

# 37MAHA Outdoor Unit Single Zone Ductless System Sizes 6K to 33K

## SERVICE MANUAL

### TABLE OF CONTENTS

|                                      | PAGE |
|--------------------------------------|------|
| SAFETY CONSIDERATIONS .....          | 1    |
| INTRODUCTION .....                   | 1    |
| MODEL NUMBER NOMENCLATURE .....      | 2    |
| WIRING .....                         | 3    |
| CONNECTION DIAGRAMS .....            | 3    |
| WIRING DIAGRAMS .....                | 4    |
| REFRIGERANT CYCLE DIAGRAM .....      | 8    |
| REFRIGERANT LINES .....              | 13   |
| SYSTEM EVACUATION AND CHARGING ..... | 14   |
| ELECTRONIC FUNCTIONS .....           | 15   |
| INQUIRY MODE .....                   | 19   |
| GENERAL TROUBLESHOOTING .....        | 23   |
| ERROR CODES .....                    | 25   |
| TROUBLESHOOTING BY ERROR CODE .....  | 27   |
| CHECK PROCEDURES .....               | 62   |
| DISASSEMBLY INSTRUCTIONS .....       | 66   |
| APPENDIX .....                       | 87   |

### SAFETY CONSIDERATIONS

Installing, starting up, and servicing air-conditioning equipment can be hazardous due to system pressures, electrical components, and equipment location (roofs, elevated structures, etc.). Only trained, qualified installers and service mechanics should install, start-up, and service this equipment.

Untrained personnel can perform basic maintenance functions such as coil cleaning. All other operations should be performed by trained service personnel.

When working on the equipment, observe precautions in the product literature and on tags, stickers, and labels attached to the equipment.

Follow all safety codes. Wear safety glasses and work gloves. Keep a quenching cloth and fire extinguisher nearby when brazing. Use care in handling, rigging, and setting bulky equipment.

Read this manual thoroughly and follow all warnings or cautions included in the literature and attached to the unit. Consult local building codes and National Electrical Code (NEC) for special requirements. Recognize safety information. This is the safety-alert symbol . When you see this symbol on the unit and in instructions or manuals, be alert to the potential for personal injury.

Understand these signal words: **DANGER**, **WARNING**, and **CAUTION**. These words are used with the safety-alert symbol. **DANGER** identifies the most serious hazards which **will** result in severe personal injury or death. **WARNING** signifies hazards which **could** result in personal injury or death. **CAUTION** is used to identify unsafe practices which **may** result in minor personal injury or product and property damage. **NOTE** is used to highlight suggestions which **will** result in enhanced installation, reliability, or operation.



### WARNING

#### ELECTRICAL SHOCK HAZARD

Failure to follow this warning could result in personal injury or death.

Before installing, modifying, or servicing system, main electrical disconnect switch must be in the **OFF** position. There may be more than 1 disconnect switch. Lock out and tag switch with a suitable warning label.



### WARNING

#### EXPLOSION HAZARD

Failure to follow this warning could result in death, serious personal injury, and/or property damage. Never use air or gases containing oxygen for leak testing or operating refrigerant compressors.

Pressurized mixtures of air or gases containing oxygen can lead to an explosion.



### CAUTION

#### EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation. Do not bury more than 36 in. (914 mm) of refrigerant pipe in the ground. If any section of pipe is buried, there must be a 6 in. (152 mm) vertical rise to the valve connections on the outdoor units.

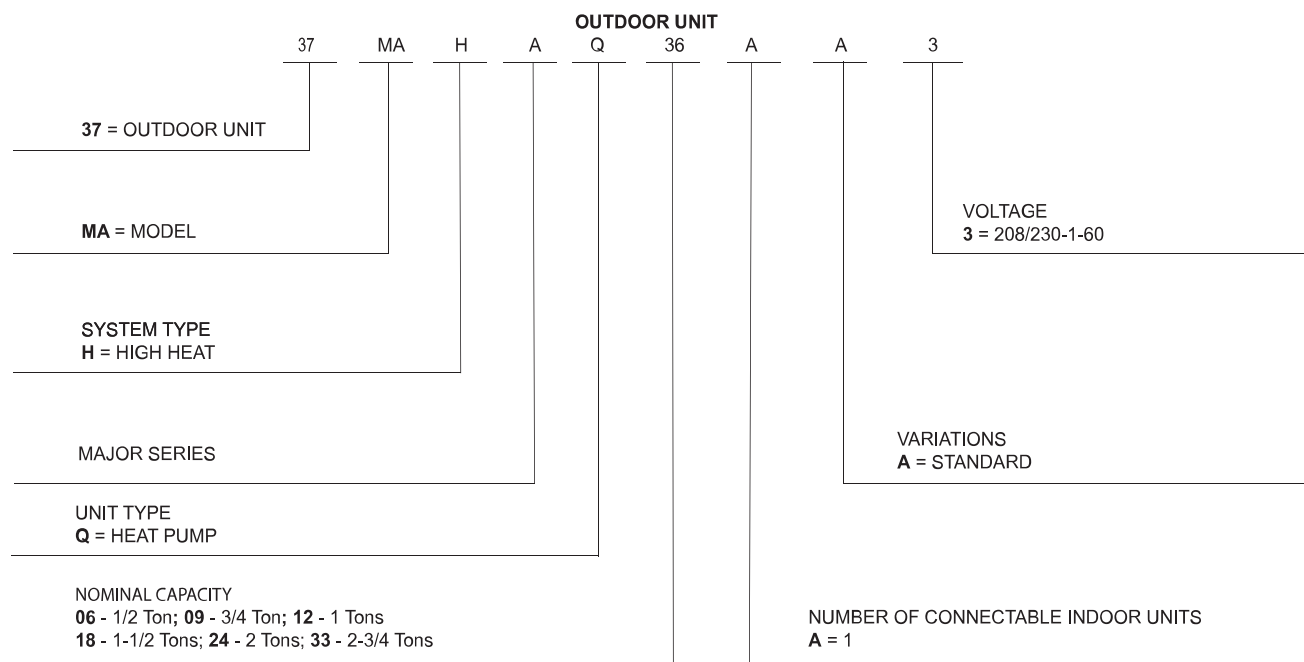
If more than the recommended length is buried, refrigerant may migrate to the cooler buried section during extended periods of system shutdown. This causes refrigerant slugging and could possibly damage the compressor at start-up.

### INTRODUCTION

This service manual provides the necessary information to service, repair, and maintain the 37MAHA family of heat pumps. This manual has an "APPENDIX" on page 87 with data required to perform troubleshooting. Use the "TABLE OF CONTENTS" on page 1 to locate a desired topic.

## MODEL NUMBER NOMENCLATURE

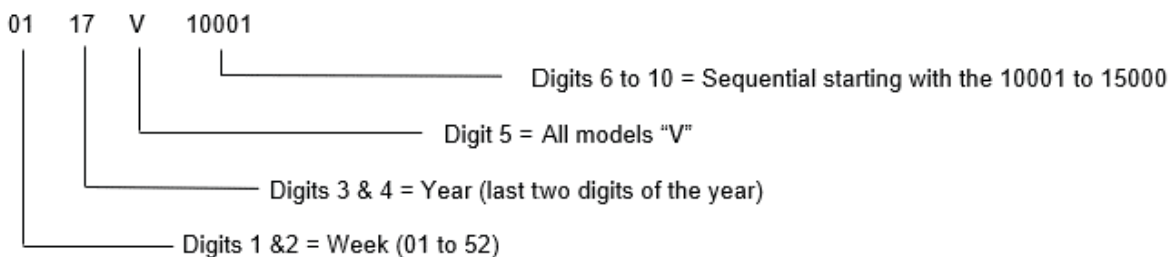
| SYSTEM TONS | BTUH   | VOLTAGE   | MODEL        |
|-------------|--------|-----------|--------------|
| .5          | 6,000  | 208/230-1 | 37MAHAQ06AA3 |
| 0.75        | 9,000  | 208/230-1 | 37MAHAQ09AA3 |
| 1           | 12,000 | 208/230-1 | 37MAHAQ12AA3 |
| 1.5         | 18,000 | 208/230-1 | 37MAHAQ18AA3 |
| 2           | 24,000 | 208/230-1 | 37MAHAQ24AA3 |
| 2.75        | 33,000 | 208/230-1 | 37MAHAQ33AA3 |



**Fig. 1 —Nomenclature**

For the Carrier/Bryant brands, the coded by a two digit week, a two followed with a 5 digit sequential number. The sequential number for MIDEA is between 10001 and 15000.

b. The SERIAL NUMBER will be as follows: SERIAL NUMBER will be date digit year followed by "V"



**Fig. 2 —Serial Number Nomenclature**



Use of the AHRI Certified TM Mark indicates a manufacturer's participation in the program. For verification of certification for individual products, go to [www.ahridirectory.org](http://www.ahridirectory.org).

WIRING

ELECTRICAL WIRING INSTALLATION

Wiring for the outdoor unit must conform to NEC (National Electrical Code) or CEC (Canadian Electrical Code) and local codes. Use Electrical Data table MCA (minimum circuit amps) and MOCP (maximum over current protection) to correctly size the wires and the disconnect fuse or breakers respectively.

All field wiring construction should be finished by a qualified electrician.

Air conditioning equipment must be grounded according to the local electrical codes.

Provide electrical disconnect per local codes.

**NOTE: DO NOT connect the power wire to the terminal of the signal wire. Connection of power to any other terminal other than L1 or L2 will cause damage to the control board.**

Any control signal cable should be run separately from the power wiring.

Use of metallic conduit or shielded cable is recommended. Maintain a distance of 12 inches(300mm) from the power wiring.

**NOTE: DO NOT run the power wiring and control wiring in the same conduit.**

Size the wiring in accordance to the NEC / CEC. Select different colors for different wire according to relevant regulations.

Ensure that the wire color of the outdoor and the terminal number are the same as those of the indoor unit.

!

CAUTION

EQUIPMENT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

Be sure to comply with local codes while running wire from the indoor unit to the outdoor unit.

Wires should be sized based on NEC and local codes.

Every wire must be connected firmly. Loose wiring may cause the terminal to overheat or result in unit malfunction. A fire hazard may also exist. Ensure all wiring is tightly connected.

No wire should touch the refrigerant tubing, compressor or any moving parts.

Disconnecting means must be provided and shall be located within sight and readily accessible from the air conditioner.

**NOTE:** Matches with multi-family and residential fan coils require separate power for the indoor and outdoor unit. A 24V interface kit is required for compatibility. Refer to the 24V Interface Kit installation manual.

CONNECTION DIAGRAMS

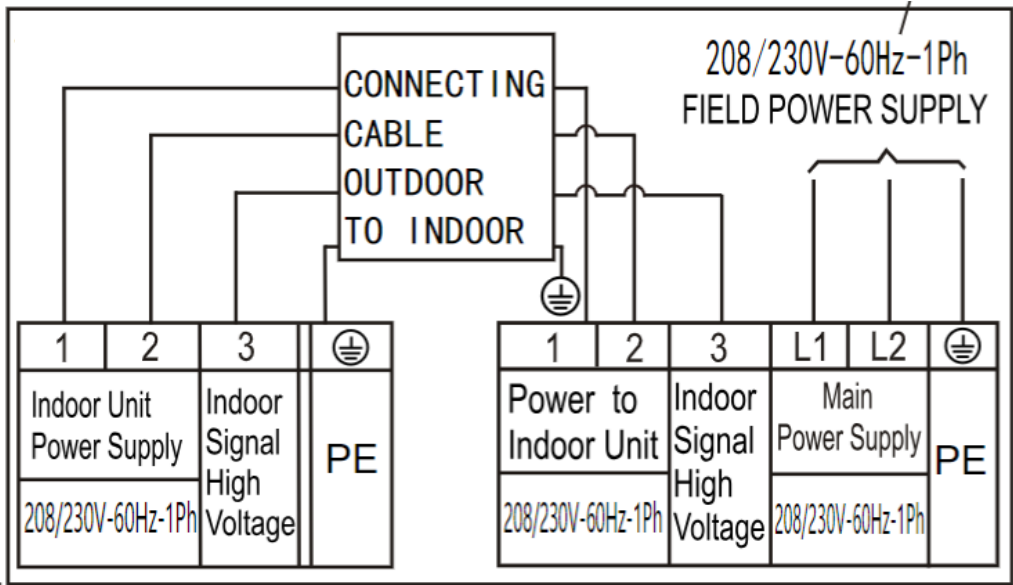


Fig. 3 —Connection Diagram Sizes 6K-33K (208/230-1-60V)

## WIRING DIAGRAMS

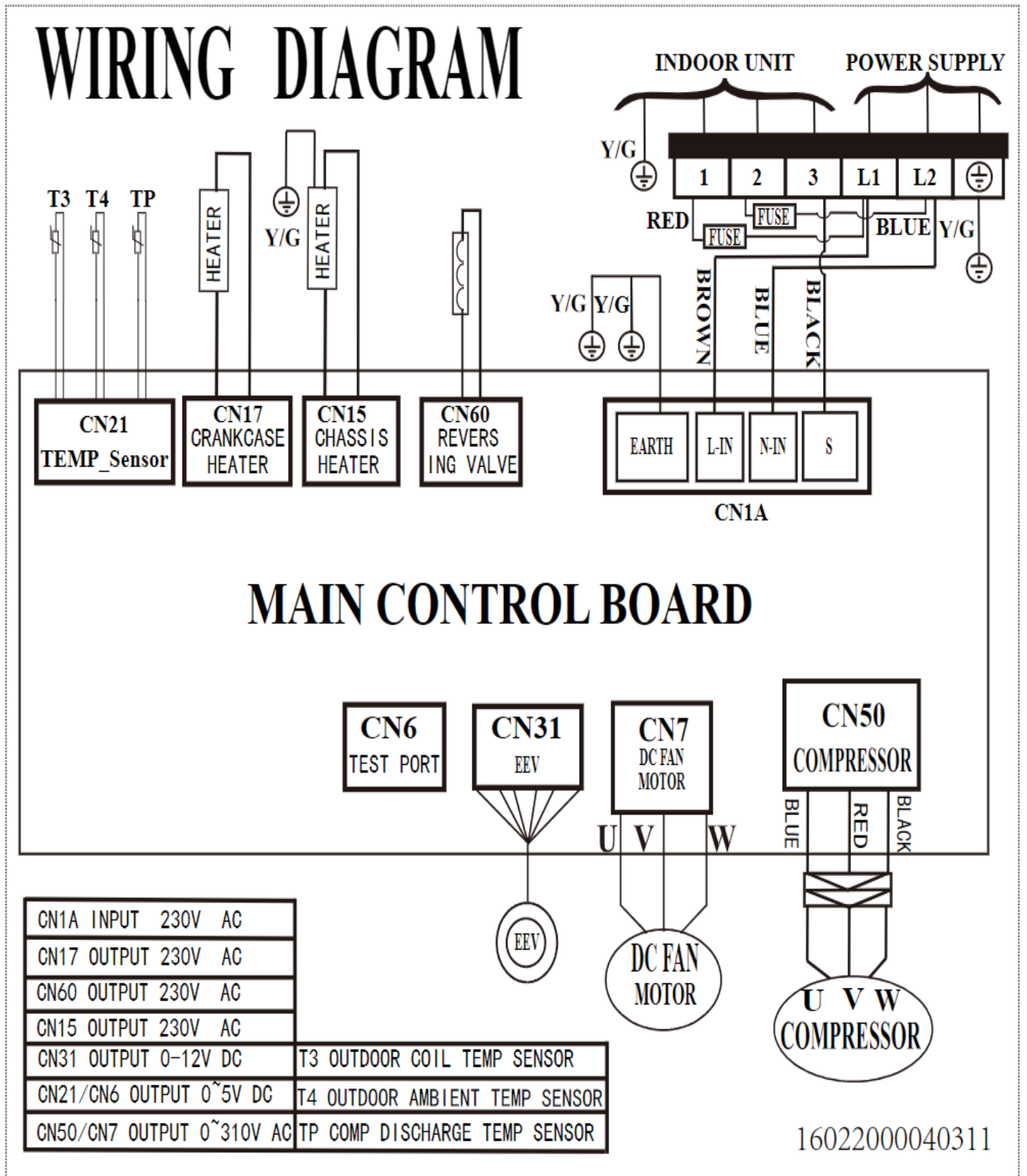


Fig. 4 —Wiring Diagram Sizes 6K/9K/12K

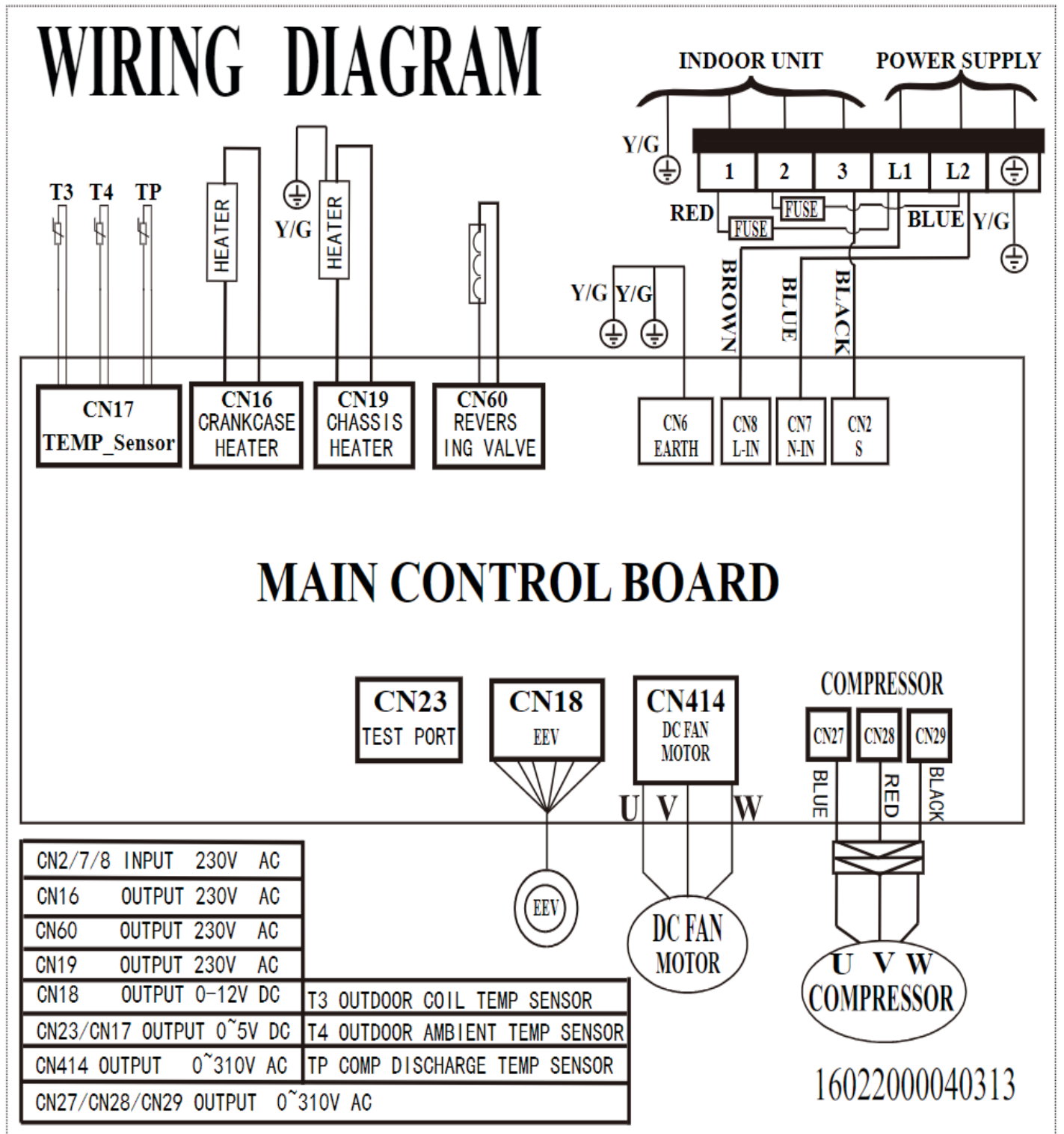


Fig. 5 —18K – 208/230V Wiring Diagram

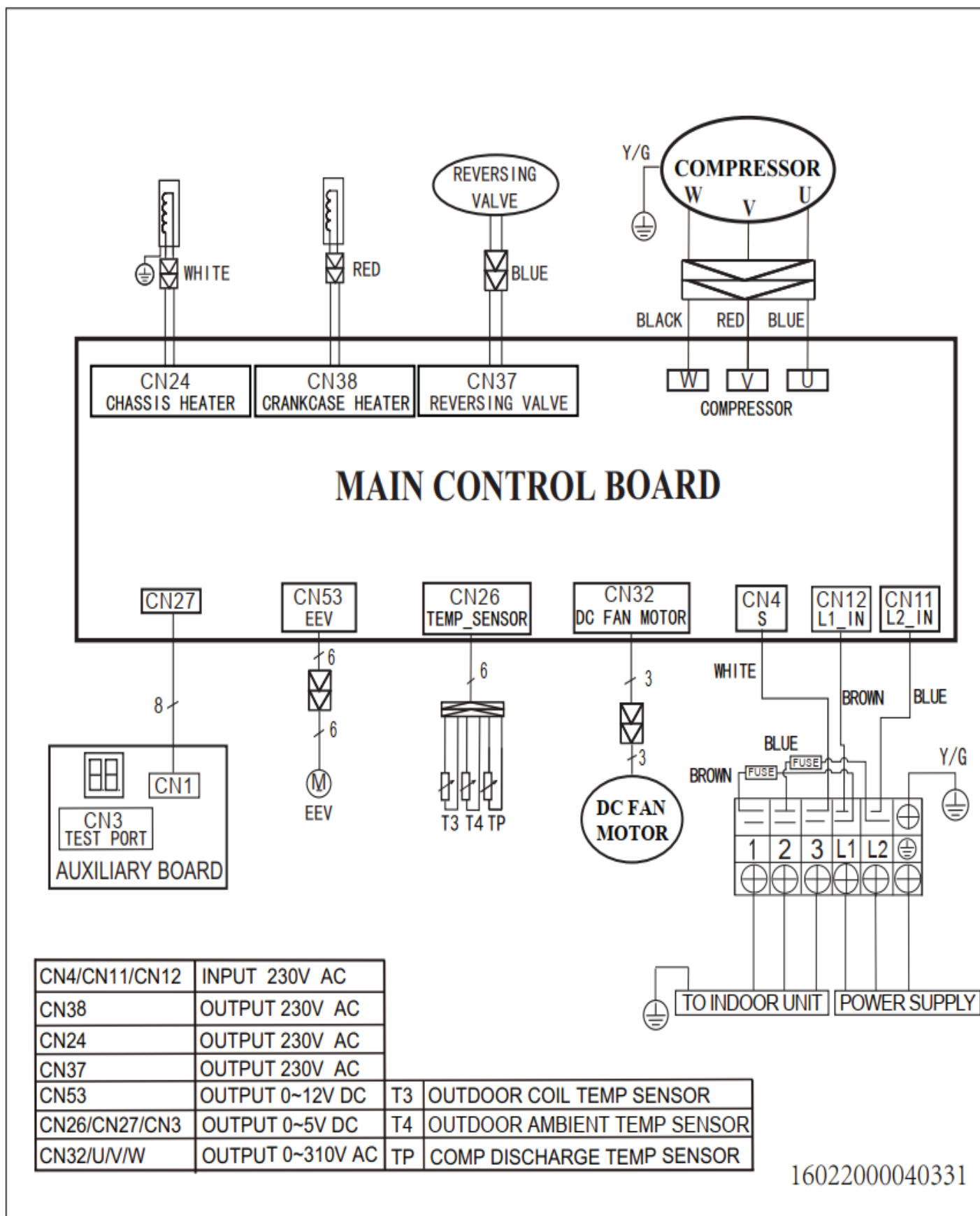


Fig. 6 —24K – 208/230V Wiring Diagram

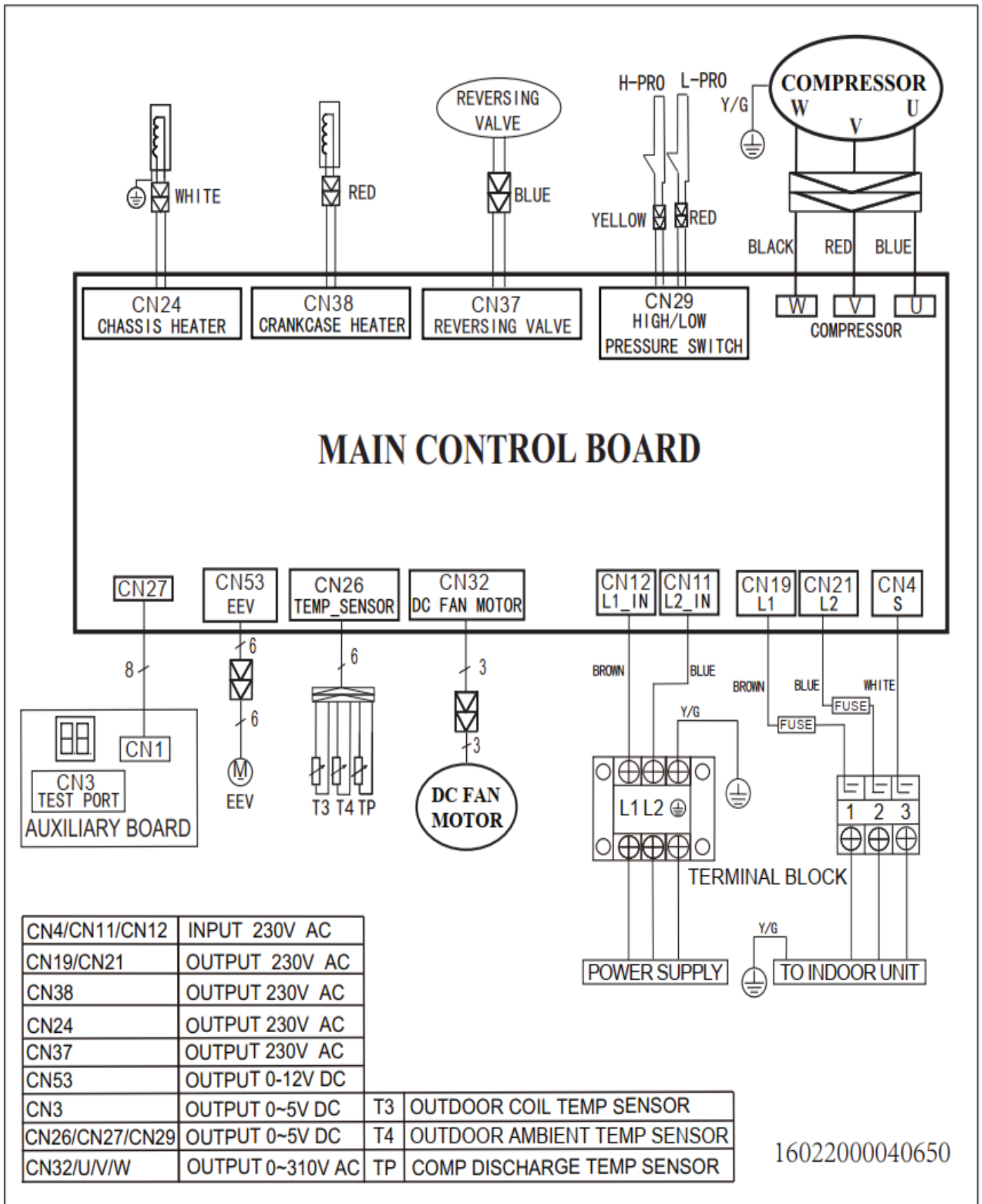


Fig. 7 —33K – 208/230V Wiring Diagram

## REFRIGERANT CYCLE DIAGRAM

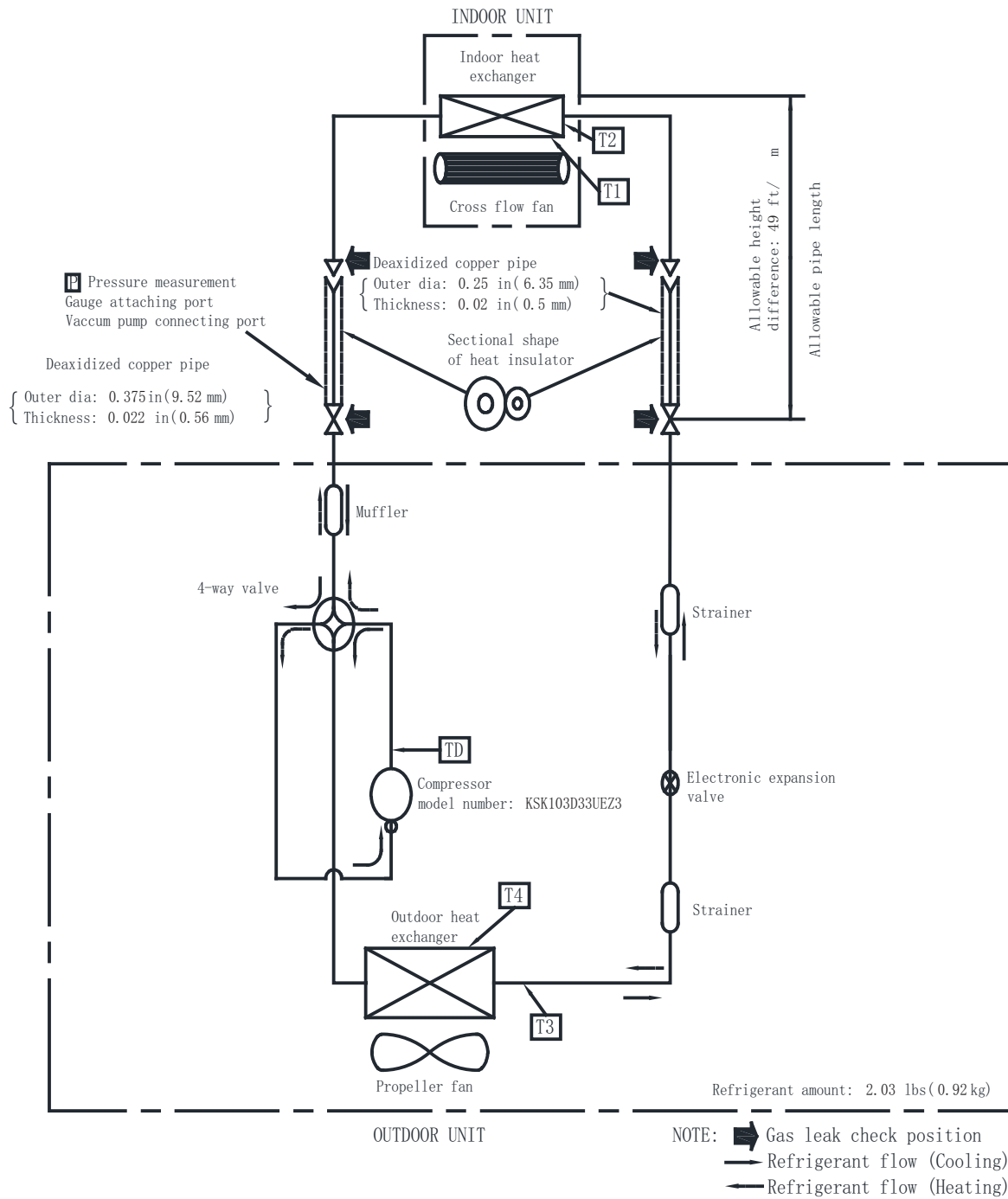


Fig. 8 —Size 6K

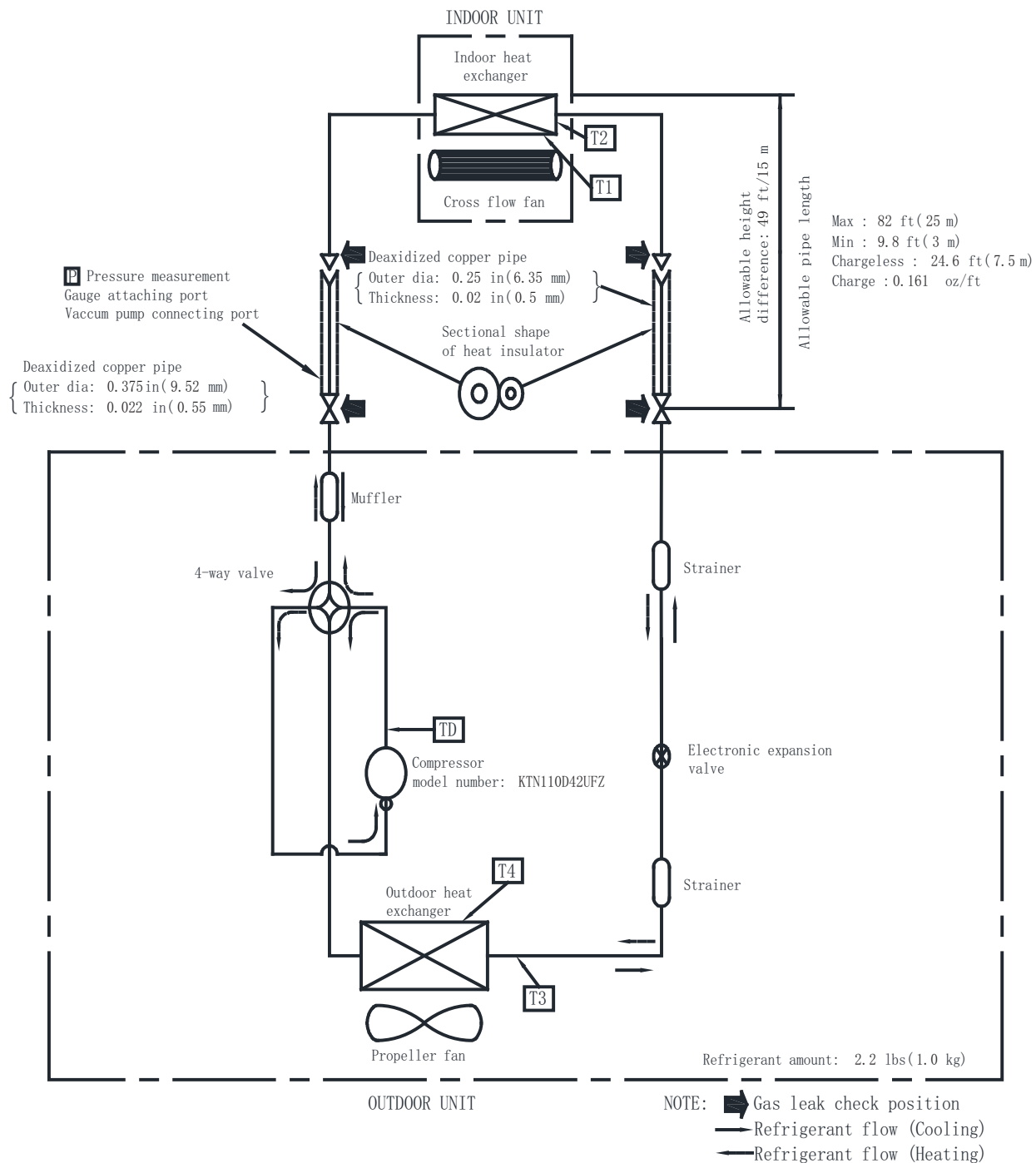


Fig. 9 —Size 9K/12K

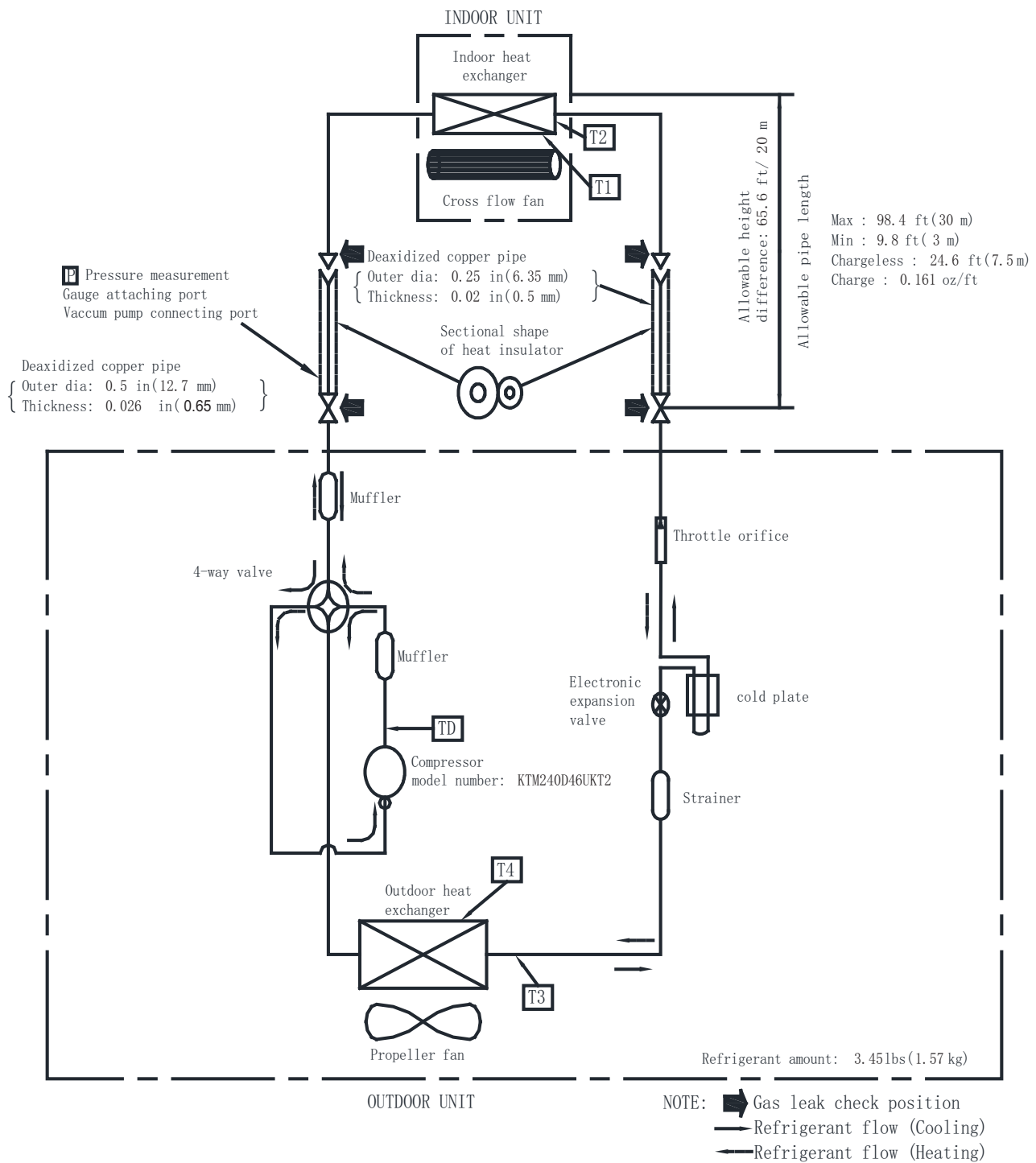


Fig. 10 —Size 18K

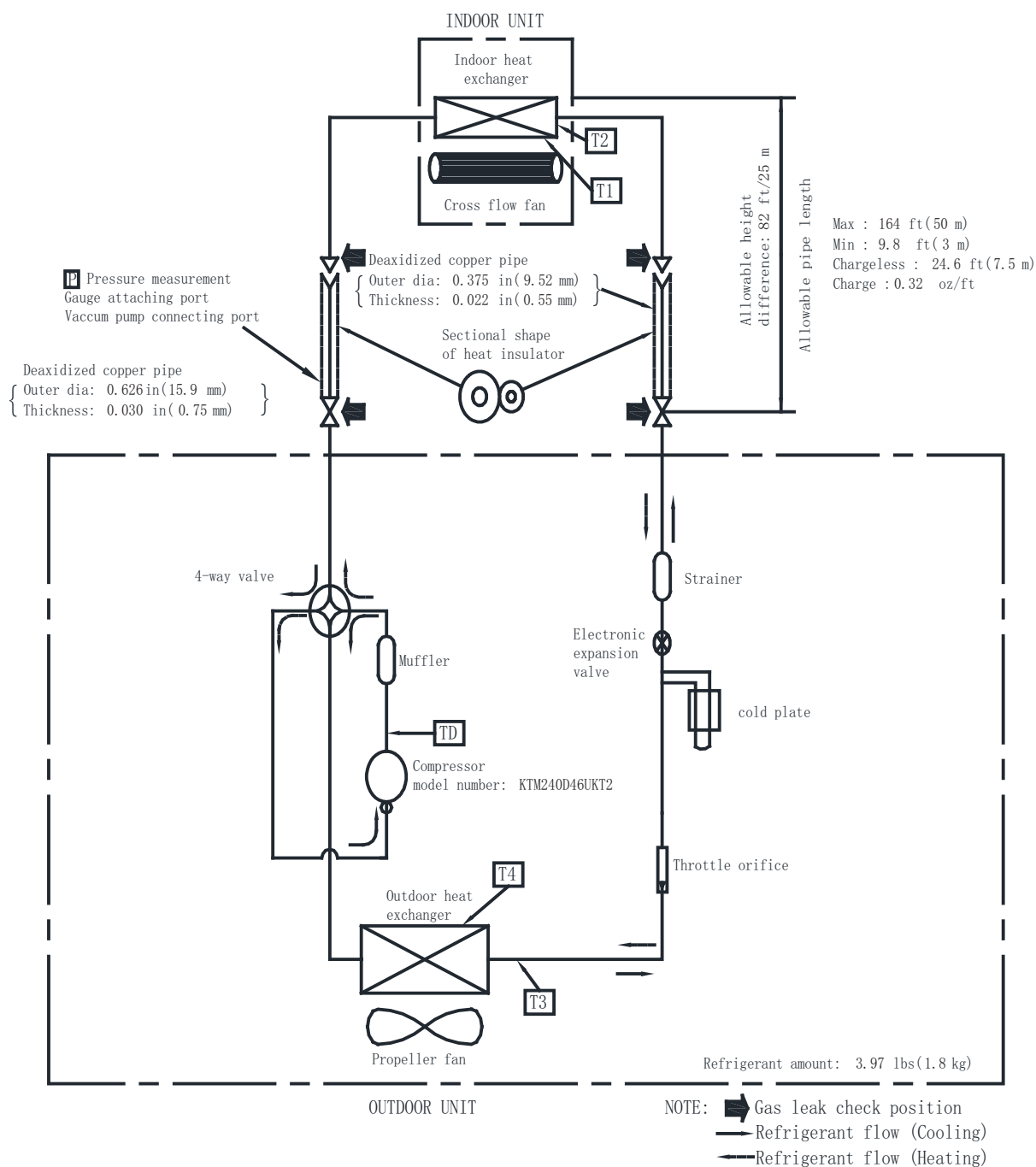


Fig. 11 —Size 24K

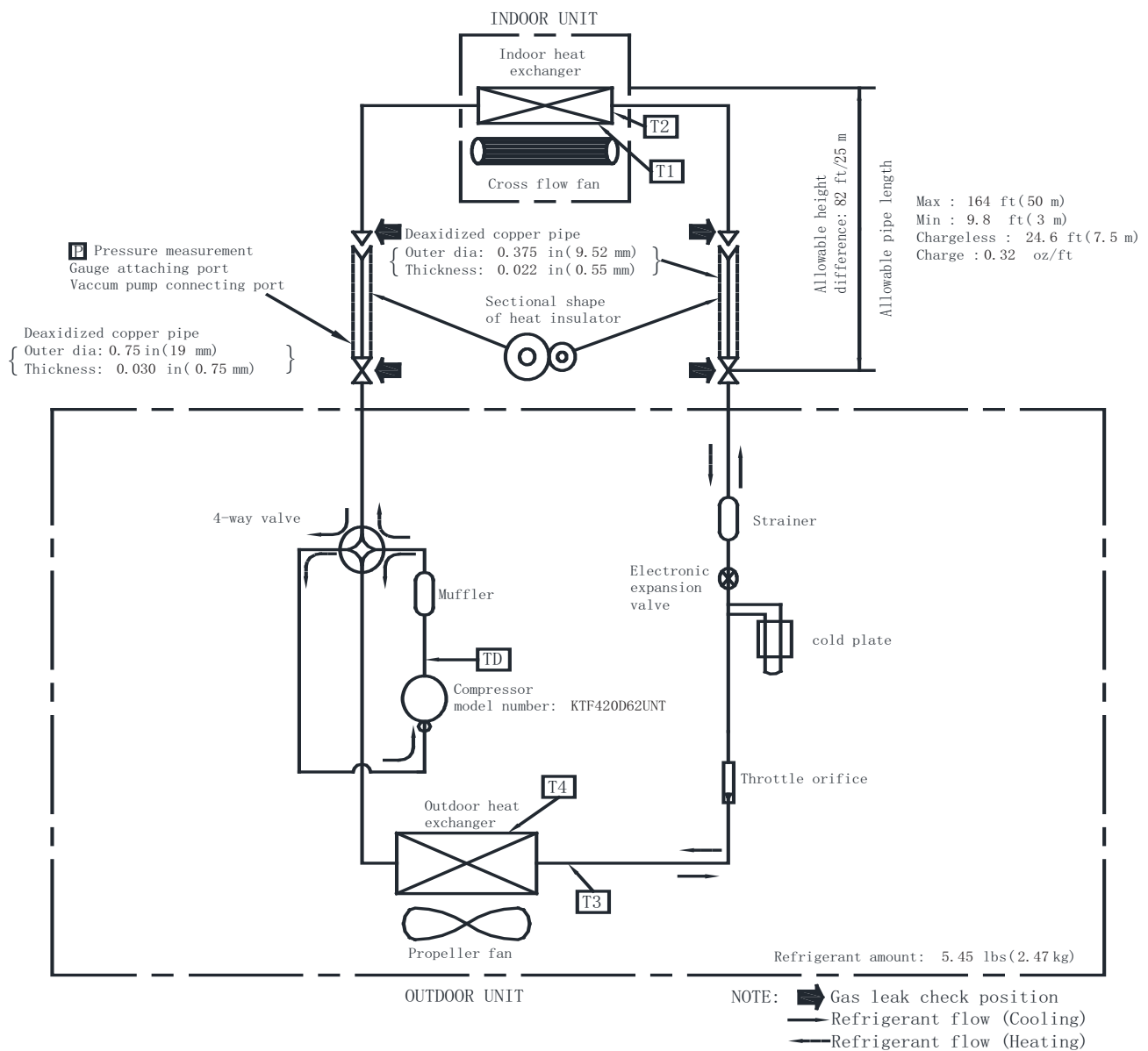


Fig. 12 —Size 33

## REFRIGERANT LINES

### General Refrigerant Line Sizing

1. The outdoor units are shipped with a full charge of R-454B refrigerant. All charges, line sizing, and capacities are based on runs of 25ft. (7.6 m). For runs over 25 ft. (7.6 m), consult [Table 1](#), below, for the proper charge adjustments.
2. The minimum refrigerant line length between the indoor and outdoor units is 10 ft. (3 m).
3. Refrigerant lines should not be buried in the ground. If it is necessary to bury the lines, not more than 36 in (914 mm) should be buried. Provide a minimum 6in (152 mm) vertical rise to the service valves to prevent refrigerant migration.
4. Suction line must be insulated. Use a minimum of 1/2in. (12.7 mm) thick insulation. Closed-cell insulation is recommended in all long-line applications.
5. Special consideration should be given to isolating interconnecting tubing from the building structure. Isolate the tubing so vibration or noise is not transmitted into the structure.

**NOTE:** [Table 1](#) displays the following maximum lengths allowed.

**Table 1 – Refrigerant Piping**

| MODEL                                                               |             | 37MAHAQ06AA3   | 37MAHAQ09AA3   | 37MAHAQ12AA3   | 37MAHAQ18AA3   | 37MAHAQ24AA3   | 37MAHAQ33AA3   |
|---------------------------------------------------------------------|-------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Refrigerant Type                                                    | Type        | R454B          | R454B          | R454B          | R454B          | R454B          | R454B          |
| Charge Amount                                                       | lb. (kg)    | 2.03 (0.92)    | 2.2 (1.00)     | 2.2 (1.00)     | 3.46 (1.57)    | 3.97 (1.80)    | 5.45 (2.47)    |
| Additional refrigerant charge                                       | Oz/ft (g/m) | 0.16 (15)      | 0.16 (15)      | 0.16 (15)      | 0.16 (15)      | 0.32 (30)      | 0.32 (30)      |
| Liquid Pipe (size - connection type)                                | In (mm)     | 1/4in (6.35mm) | 1/4in (6.35mm) | 1/4in (6.35mm) | 1/4in (6.35mm) | 3/8in (9.52mm) | 3/8in (9.52mm) |
| Suction Pipe (size - connection type)                               | In (mm)     | 3/8in (9.52mm) | 3/8in (9.52mm) | 3/8in (9.52mm) | 1/2in (12.7mm) | 5/8in (15.9)   | 3/4in (19mm)   |
| Min. Piping Length                                                  | ft. (m)     | 10 (3)         | 10 (3)         | 10 (3)         | 10 (3)         | 10 (3)         | 10 (3)         |
| Standard Piping Length                                              | ft. (m)     | 24.6 (7.5)     | 24.6 (7.5)     | 24.6 (7.5)     | 24.6 (7.5)     | 24.6 (7.5)     | 24.6 (7.5)     |
| Max. Piping Length with no additional refrigerant charge per System | ft. (m)     | 24.6 (7.5)     | 24.6 (7.5)     | 24.6 (7.5)     | 24.6 (7.5)     | 24.6 (7.5)     | 24.6 (7.5)     |
| Total Maximum Piping Length per system                              | ft. (m)     | 82.02 (25)     | 82.02 (25)     | 82.02 (25)     | 98.42 (30)     | 164.04 (50)    | 164.04 (50)    |
| Max. outdoor-indoor height difference (OU higher than IU)           | ft. (m)     | 49.21 (15)     | 49.21 (15)     | 49.21 (15)     | 65.62 (20)     | 82.02 (25)     | 82.02 (25)     |
| Max. outdoor-indoor height difference (IU higher than OU)           | ft. (m)     | 49.21 (15)     | 49.21 (15)     | 49.21 (15)     | 65.62 (20)     | 82.02 (25)     | 82.02 (25)     |

1. The charge amount listed in [Table 1](#) is for piping runs up to 25 ft. (7.6 m).

## SYSTEM EVACUATION AND CHARGING



### CAUTION

#### UNIT DAMAGE HAZARD

Failure to follow this caution may result in equipment damage or improper operation.

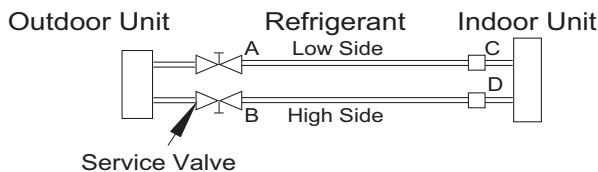
Never use the system compressor as a vacuum pump.

Refrigerant tubes and the indoor coil should be evacuated using the recommended 500 micron deep vacuum method. The alternate triple evacuation method may be used if the procedure outlined below is followed.

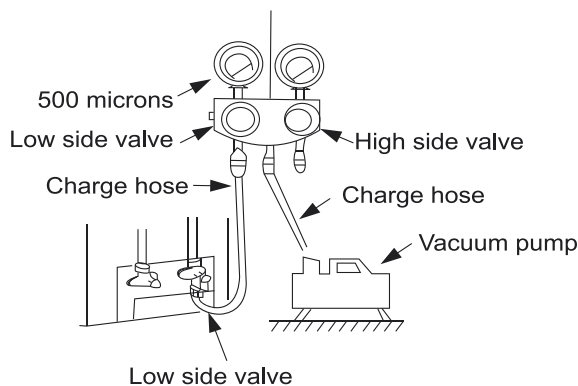
**NOTE:** Always break a vacuum with dry nitrogen.

### USING VACUUM PUMP

1. Completely tighten flare nuts A, B, C, D. Connect the manifold gage charge hose to a charge port of the low side service valve (see Fig. 24).
2. Connect the charge hose to vacuum pump.
3. Fully open the low side of manifold gage (see Fig. 13).
4. Start the vacuum pump.
5. Evacuate using either the deep vacuum or triple evacuation method.
6. After evacuation is complete, fully close the low side of manifold gage and stop the vacuum pump operation.
7. The factory charge contained in the outdoor unit is good for up to 25 ft. (8 m) of line length. For refrigerant lines longer than 25 ft. (8 m), add refrigerant, up to the allowable length.
8. Disconnect the charge hose from the charge connection of the low side service valve.
9. Fully open service valves B and A.
10. Securely tighten the service valve caps.



**Fig. 13 —Service Valve**



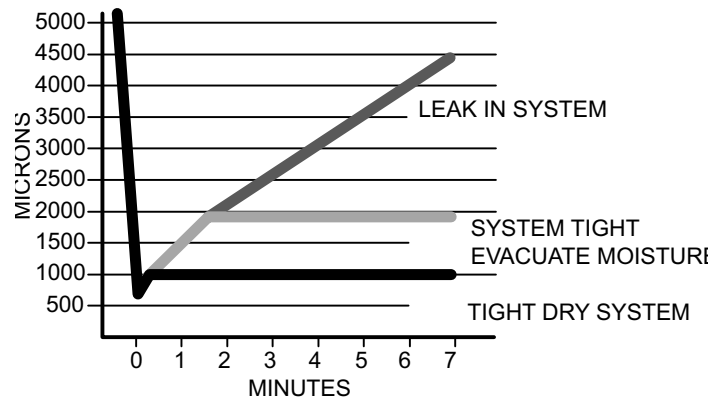
**Fig. 14 —Manifold**

### EVACUATION

Evacuation of the system will remove air or nitrogen (non-condensables) as well as moisture. A proper vacuum will assure a tight, dry system before charging with refrigerant. The two methods used to evacuate a system are the deep vacuum method and the triple vacuum method.

### DEEP VACUUM METHOD

The deep vacuum method requires a vacuum pump capable of pulling a vacuum of 500 micron and a vacuum gauge capable of accurately measuring this vacuum depth. This method is the most positive way of assuring a system is free of air and moisture (see Figure 15).

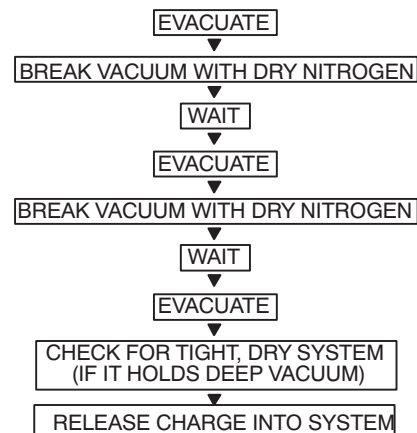


**Fig. 15 —Deep Vacuum Graph**

### TRIPLE EVACUATION METHOD

The triple evacuation method should be used when vacuum pump is not capable of pumping down to 500 microns and system does not contain any liquid water. Refer to Fig. 16 and proceed as follows:

1. Attach refrigeration gauges and evacuate system down to 28 inches of mercury and allow pump to continue operating for an additional 15 minutes.
2. Close service valves and shut off vacuum pump.
3. Connect a nitrogen cylinder and regulator to system and flow nitrogen until system pressure is 2 psig.
4. Close service valve and allow system to stand for 1 hour. During this time, dry nitrogen will be able to diffuse throughout the system absorbing moisture.
5. Repeat this procedure as indicated in Fig. 16. System is now free of any contaminants and water vapor.



**Fig. 16 —Triple Evacuation Method**

### FINAL TUBING CHECK

Check to be certain factory tubing on both the indoor and outdoor unit has not shifted during shipment. Ensure tubes are not rubbing against each other or any sheet metal. Pay close attention to the feeder tubes, making sure wire ties on feeder tubes are secure and tight.

## ELECTRONIC FUNCTIONS

### Abbreviation:

Table 2 – Unit element abbreviations

| Abbreviation                | Element                                                               |
|-----------------------------|-----------------------------------------------------------------------|
| T1                          | Indoor room temperature                                               |
| T2                          | Coil temperature of evaporator                                        |
| T3                          | Coil temperature of condenser                                         |
| T4                          | Outdoor ambient temperature                                           |
| TP                          | Compressor discharge temperature                                      |
| Tsc                         | Adjusted setting temperature                                          |
| CDIFTEMP                    | Cooling shutdown temperature                                          |
| HDIFTEMP2                   | Heating shutdown temperature                                          |
| TCDE1                       | Exit defrost temperature1                                             |
| TCDE2                       | Exit defrost temperature2<br>(maintain for a period of time)          |
| TIMING_DEFROST_<br>TIME_ADD | Enter defrost time                                                    |
| EE_TIME_DEFROST7_ST<br>RONG | Enter enhanced defrost time                                           |
| TCDE1_ADD_STRONG            | Exit enhanced defrost temperature1                                    |
| TCDE2_ADD_STRONG            | Exit enhanced defrost temperature2<br>(maintain for a period of time) |

**NOTE:** In this manual, terms such as CDIFTEMP, HDIFTEMP2, TCDE1, TCDE2, TIMING\_DEFROST\_TIME\_ADD...etc., are EEPROM parameter settings.

### FAN Mode

When fan mode is activated:

- Outdoor fan and compressor are stopped

### COOLING Mode

#### Compressor Control

Reach the configured temperature:

**When the compressor runs continuously for less than 120 minutes.**

- If the following conditions are satisfied, the compressor ceases operation.
- Calculated frequency (fb) is less than minimum limit frequency (FminC).
- Compressor runs at FminC more than ten minutes.
- Indoor room temperature(T1) is lower than or equal to (Tsc-CDIFTEMP 0.9°F/-0.5°C)

**When the compressor runs continuously for more than 120 minutes.**

If the following conditions are satisfied, the compressor ceases operation.

- Calculated frequency (fb) is less than minimum limit frequency (FminC).
- Compressor runs at FminC more than 10 minutes.
- When indoor room temperature (T1) is lower than or equal to (Tsc-CDIFTEMP).

**NOTE:** CDIFTEMP is EEPROM setting parameter. It is 4°F (2°C) usually.

**If one of the following conditions is satisfied, not judge protective time.**

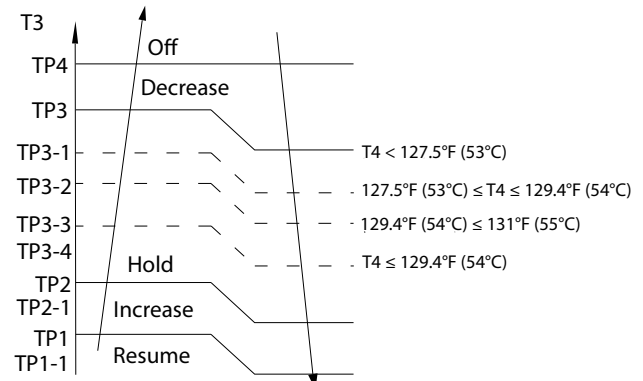
- Compressor running frequency is more than test frequency.

- When compressor running frequency is equal to test frequency, outdoor ambient temperature (T4) is more than 59°F (15°C) or outdoor ambient sensor (T4) fault.
- Change setting temperature.
- Turning on/off turbo or sleep function
- Various frequency limit shutdown occurs.

#### Outdoor Fan Control

- The outdoor unit will be run at different fan speed according to outdoor ambient temperature(T4) and compressor frequency.
- For different outdoor units, the fan speeds are different.

#### Condenser Temperature Protection



When the condenser temperature exceeds a configured value, the compressor ceases operation.

### Heating Mode (Heat Pump Modes)

#### Compressor Control

**Reach the configured temperature**

- If the following conditions are satisfied, the compressor ceases operation.
- Calculated frequency (fb) is less than minimum limit frequency (FminH).
- Compressor runs at FminH more than 10 minutes.
- T1 is higher than or equal to Tsc+ HDIFTEMP2.

**NOTE:** HDIFTEMP2 is the EEPROM parameter setting. It is normally 4°F (2°C).

- If one of the following conditions is satisfied, not judge protective time.
- Compressor running frequency is more than test frequency.
- Compressor running frequency is equal to test frequency, outdoor ambient temperature (T4) is more than 59°F (15°C) or Outdoor ambient sensor (T4) fault.
- Change setting temperature.
- Turning on/off turbo or sleep function

**When the current is higher than the predefined safe value, surge protection is activated, causing the compressor to cease operations.**

## Outdoor Fan Control

- The outdoor unit will be run at different fan speed according to outdoor ambient temperature (T4) and compressor frequency.
- For different outdoor units, the fan speeds are different.

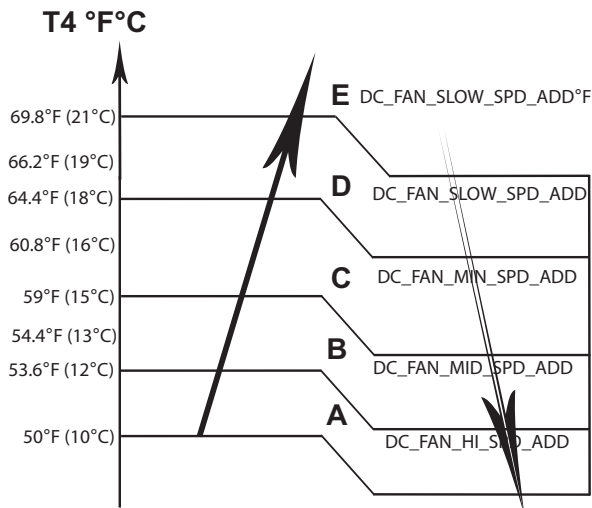


Fig. 17 —Outdoor Fan Running Rules

### Indoor Fan Running Rules:

- In the COOLING mode, the indoor fan runs continuously and the user can select any of the following speeds: HIGH, MEDIUM, LOW and AUTO.
- When the setting temperature is reached, if the compressor stops operating, the indoor fan motor runs in the minimum or setting speed (see Fig. 18).

| Setting Fan Speed | T1-Td °F (°C) |   | Actual Fan Speed |
|-------------------|---------------|---|------------------|
| H                 | 8.1°F (4.5°C) | A | H + (H+=H+G)     |
|                   | 5.4°F (3.0°C) | B | H (=H)           |
|                   | 2.7°F (1.5°C) | C | H - (H-=H-G)     |
| M                 | 8.1°F (4.5°C) | D | M + (M+=M+Z)     |
|                   | 5.4°F (3.0°C) | E | M (M=M)          |
|                   | 2.7°F (1.5°C) | F | M - (M-=M-Z)     |
| L                 | 8.1°F (4.5°C) | G | L + (L+=L+D)     |
|                   | 5.4°F (3.0°C) | H | L (L=L)          |
|                   | 2.7°F (1.5°C) | I | L - (L-=L-D)     |

Fig. 18 —Indoor Fan Running Rules

The AUTO fan adheres to the following rules (see Fig. 19:)

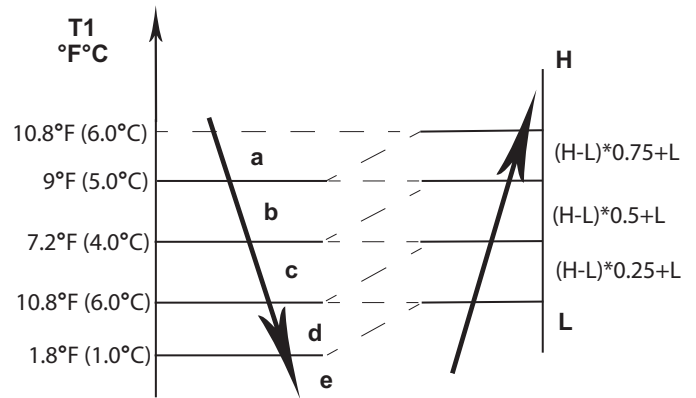


Fig. 19 —AUTO Fan Running Rules

### Compressor Temperature Protection

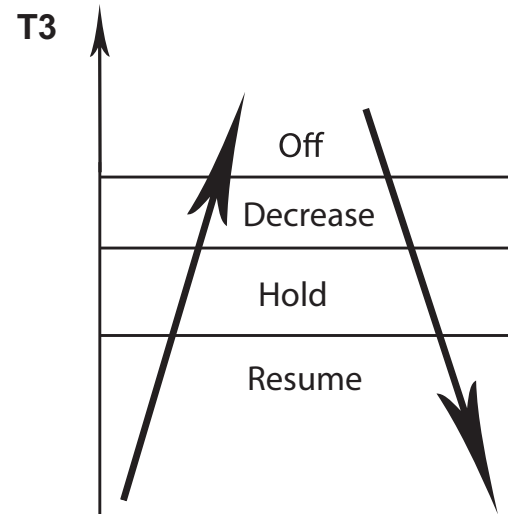


Fig. 20 —Compressor Temperature Protection

- Off: Compressor stops
- Decrease: Decrease the running frequency to the lower level
- Hold: Keep the current frequency
- Resume: No limitation for frequency

When the condenser temperature is higher than the setting value, the compressor stops.

### Evaporator Temperature Protection

When the evaporator temperature is lower than the setting value the compressor stops.

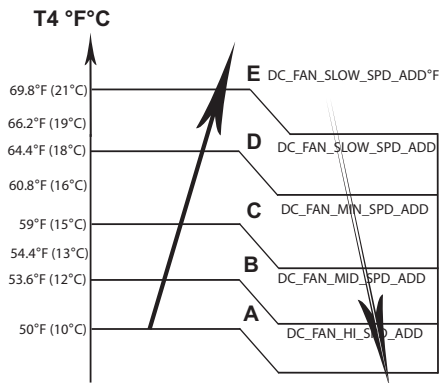
## HEATING Mode

### Compressor Running Rules:

- When  $T1-Ts > f'T$ , the compressor stops.
- When  $T1-Ts < f'T-1.5$ , the compressor is on.  $f'T$  is the programmed parameter for temperature compensation.
- When the AC runs in the MUTE mode, the compressor runs with a low frequency.
- When the current is more than the setting value, the current protection function activates and the compressor stops.

### Outdoor Fan Running Rules:

The outdoor unit runs at a different fan speed according to T4. For different outdoor units, the fan speeds differ.



**Fig. 21 —Outdoor Fan Running Rules**

### Defrosting mode

The unit enters defrosting mode according to the temperature value of condenser temperature (T3) and outdoor ambient temperature (T4) as well as the compressor running time.

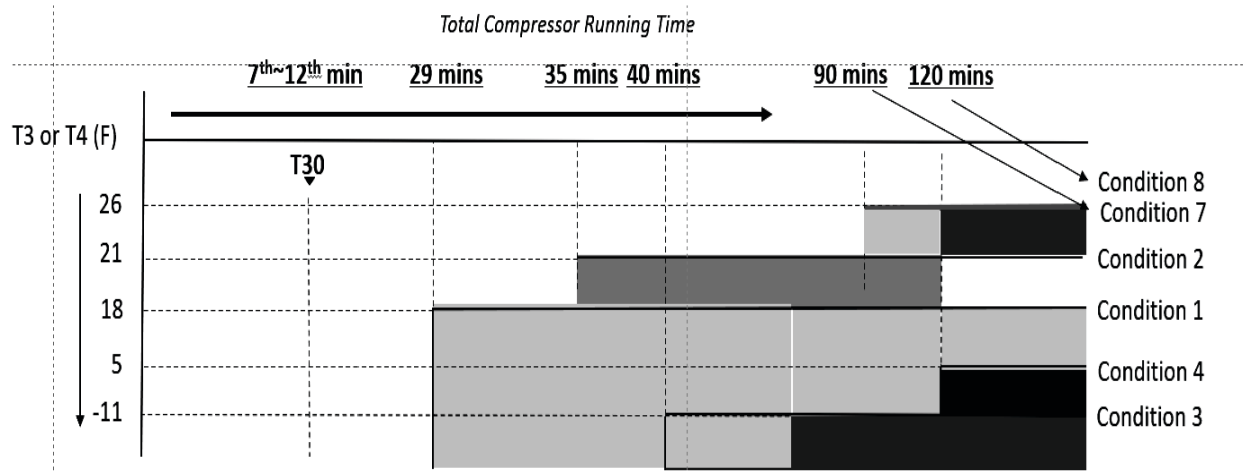
- In defrosting mode, the compressor continues to run, the indoor and outdoor motor will cease operation, the defrost light of the indoor unit will turn on, and the “dF” symbol is displayed.
- If any one of the following conditions is satisfied, defrosting ends and the machine switches to normal heating mode:
- Condenser temperature (T3) rises above TCDE1.
- Condenser temperature (T3) maintained above TCDE2 for 80 seconds.
- Unit runs for 15 minutes consecutively in defrosting mode.
- If Outdoor ambient temperature (T4) is lower than or equal to -7.6°F (-22°C) and compressor running time is more than

TIMING\_DEFROST\_TIME, if any one of the following conditions is satisfied, defrosting ends and the machine switches to normal heating mode:

- Unit runs for 10 minutes consecutively in defrosting mode.
- Condenser temperature (T3) rises above 50°F/ 10°C.
- If any one of the following conditions is satisfied, the unit enters defrosting mode
- If condenser temperature (T3) or outdoor ambient temperature (T4) is lower than -3°C for 30 seconds, Ts-T1 is lower than 5°C and compressor running time is more than EE\_TIME\_DEFROST7\_ADD.
- If condenser temperature (T3) or outdoor ambient temperature (T4) is lower than -3°C for 30 seconds and compressor running time is more than EE\_TIME\_DEFROST7\_ADD+30 minutes.
- If any one of the following conditions is satisfied, defrosting ends and the machine switches to normal heating mode:
  - Condenser temperature (T3) rises above TCDE1 7.2°F/+4°C.
  - Condenser temperature (T3) maintained above TCDE2 7.2°F/+4°C for 80 seconds.
  - Unit runs for 15 minutes consecutively in defrosting mode.

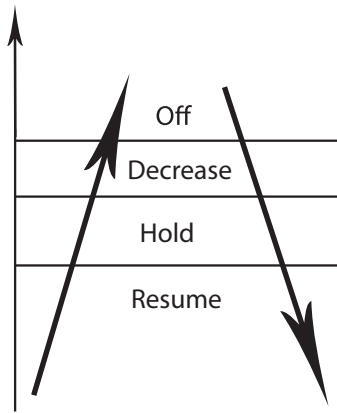
### Enhanced Defrost

Enhanced Defrost is a feature that is enabled by turning on dipswitch SW4 on the 24 volt ODU board. This feature is designed to be used for problem areas of defrosting (near water sources and areas of high humidity during cold outdoor conditions). When enabled, the total heating cumulative run time will be reduced to 40 minutes, allowing for defrosting to occur at the 40th minute providing the T3 or T4 temperature is 26°F or cooler at that time. The defrost termination temperature at T3 will also increase to 77°F to ensure better completion of defrost prior to terminating defrost.



|              |                                                                                                                                                                                       |                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| Condition 1  | Total compressor running time is 29 mins                                                                                                                                              | T3 ≤ 18F, T3 is less than/equal to T30 - 4.5F (2.5C) and T4 > -8F |
| Condition 2  | Total compressor running time is 35 mins                                                                                                                                              | T3 ≤ 21F, T3 less than/equal to T30 - 5.4F (3C) and T4 > -8F      |
| Condition 3  | Total compressor running time is 29 mins                                                                                                                                              | T3 ≤ -11F and last for 3 mins, and T4 > -8F                       |
| Condition 4  | Total compressor running time is 120 mins                                                                                                                                             | T3 ≤ 5F and T4 > -8F                                              |
| Condition 5  | Cumulative running 30 mins                                                                                                                                                            | T4 - T3 > (0.5T4 + 5F) and T3 ≤ 10F, T4 > -8F                     |
| Condition 6  | Cumulative running 8 hours                                                                                                                                                            | T4 ≤ -8F, with T4 operating without malfunction                   |
| Condition 7  | Cumulative running time 90 mins and diff. of Ts-T1 ≤ 9F<br>Cumulative running time 120 minutes (if Ts-T1 diff. above not applicable)                                                  | T3 or T4 ≤ 26F last for 30 seconds                                |
| *Condition 8 | If enhanced defrost is ON, cumulative running time is 40 minutes (T30 does not apply) *Only applies to Crossover ODU's. Enable enhanced defrost by turning on dipswitch 4 in the ODU. | T3 or T4 ≤ 26F last for 30 seconds                                |

## Evaporator Coil Temperature Protection



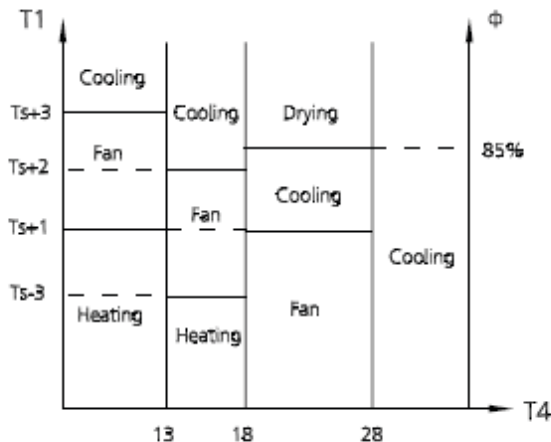
**Fig. 22 —Evaporator Coil Temperature Protection**

When the evaporator temperature is higher than the setting protection value, the compressor stops.

## AUTO Mode

AUTO mode can be selected with the remote controller and the setting temperature can be changed between 60.0°F~86°F (16°C~30°C).

In the AUTO mode, the unit chooses either the COOLING, HEATING or the FAN-ONLY mode accT2, T4 and relative humidity.



**Fig. 23 —AUTO Mode**

**Heating\*:** COOLING ONLY models run at fan speed. The indoor fan runs in the AUTO fan speed for the relevant mode. The louver operates the same as in the relevant mode.

If the unit switches mode between HEATING and COOLING, the compressor repeatedly stops for a certain time and then chooses the mode according to T1-Ts. If the setting temperature is modified, the unit selects a running function again.

## DRYING mode

The indoor fan speed is fixed at BREEZE and can not be changed. The louver angle is the same as in the COOLING mode.

**Low Indoor Room Temperature Protection** In the DRYING mode, if the room temperature is lower than 50°F(10°C),

the compressor stops and does not resume until the room temperature exceeds 53.6°F (12°C).

Evaporator anti-freezing protection, condenser high temperature protection and outdoor unit frequency limit are active and are the same as that in the COOLING mode. The outdoor fan operates the same as in COOLING mode.

## FORCED OPERATION Function

### Enter FORCED OPERATION function:

When the machine is off, press TOUCH to engage the Forced Auto Mode. Press TOUCH again, within 5 seconds, to engage the FORCED COOLING mode. In FORCED AUTO, FORCED COOLING or any other operation mode, press TOUCH to turn off the unit.

In the FORCED OPERATION mode, all general protections and the remote controller are available.

### Operation Rules:

#### FORCED COOLING mode:

The compressor runs at the F2 frequency and the indoor fan runs as a breeze. After running for 30 minutes, the unit enters the AUTO mode at a 75.2°F (24°C) setting temperature.

#### FORCED AUTO mode:

The FORCED AUTO mode is the same as the normal AUTO mode with a 75.2°F (24°C) setting temperature.

#### AUTO-RESTART function

The indoor unit is equipped with an AUTO-RESTART function, which is carried out through an auto-restart module. In case of a sudden power failure, the module memorizes the setting conditions before the power failure. The unit resumes the previous operation setting (not including the swing function) automatically 3 minutes after the power returns.

If the memorization condition is the FORCED COOLING mode, the unit runs in the COOLING mode for 30 minutes and enters the AUTO mode as 75.2°F (24°C) setting temp.

If the air conditioner turns off before the unit powers off and the air conditioner is required to restart immediately, the compressor delays for 1 minute when the power is on. Under other conditions, the compressor has a 3 minute delay when it restarts.

#### Refrigerant Leakage Detection

With this new technology, the display area displays EC when the outdoor unit detects a refrigerant leak.

#### 46°F (8°C) Heating

When the compressor is running, the indoor fan motor runs without the anti-cold air function. When the compressor is off, the indoor fan motor is off.

## INQUIRY MODE

### Accessing the INQUIRY Mode

#### CAUTION

Read and understand the function changes you wish to make in advance. Neither the indoor unit nor the remote control displays the new level of any of the changes made while in the INQUIRY mode. Be sure to document the changes you've made to the system's programming using the INQUIRY mode. Once you complete the changes and exit the INQUIRY mode, if additional changes are made to the programming, the system will not show the new previously set level(s).

For example, when you first access CODE 22, Heating Temperature Compensation, the remote control display defaults to 0. If you change it to -2, then save and exit out of the INQUIRY mode, the next time someone goes back in and accesses CODE 22, the remote's display will not display -2.

Instead it will show 0 because that's the default. If you are unsure of the previous changes, due to a lack of documentation, you could press the DOWN symbol to the maximum change range of -6, then press the UP symbol until you are back to 0, and make the new adjustments accordingly. Be sure to document the changes when you are done.

### Remote Controller Service Mode Functions

**NOTE:** While in the INQUIRY mode, refer to the following instructions to enter SERVICE mode for the applicable codes.

Below is a list of INQUIRY modes and serviceable functions.

- a. Before using the remote's service functions, turn OFF the indoor unit with the remote.
- b. Turn OFF the power to the outdoor unit for 2 minutes. Turn the power back ON.
- c. Remove the batteries from the remote and wait for the remote screen to clear. Within 30 seconds of replacing the batteries, use UP or DOWN to scroll through the INQUIRY modes.
- d. To enter the SERVICE mode for an applicable INQUIRY mode, press ON/OFF for 2 seconds.
- e. After SERVICE adjustments have been made, press ON/OFF for 2 seconds to exit the SERVICE mode and return to the INQUIRY mode.
- f. Once operations in the INQUIRY mode are complete, press ON/OFF and FAN SPEED for 2 seconds to exit. All buttons on the remote controller are disabled for 60 seconds
- g. To ensure changes are locked, power down the outdoor unit for three (3) minutes after all the service mode changes are made.

1. Simultaneously press ON/OFF   and FAN SPEED  for 8 seconds.
  - a. The remote is now in the INQUIRY mode.
  - b. The remote control remains in the INQUIRY mode for 1 minute if no other button is pressed.
  - c. While in the INQUIRY Mode, the remote display cancels all icons except AUTO, COOL, DRY, HEAT and Battery Strength.
  - d. The remote control digital display defaults to 0 upon entering the INQUIRY mode.
  - e. In the INQUIRY mode, each digital code (from 0 to 30) is accessed by pressing the UP or DOWN arrows .
  - f. The INQUIRY information appears on the indoor unit display in approximately 1 second after accessing the digital code. Press OK to send as well.
  - g. In the INQUIRY mode, all other buttons and operations are invalid except for UP, DOWN and OK or the operation to exit the INQUIRY mode.

## Service Inquiry Codes

| CODE | INQUIRY | INQUIRY DESCRIPTION              | SERVICE/ INQUIRY    | FOR SERVICE, PRESS ON/OFF FOR 2 SECONDS TO:                                                                                                                                                                                                                                              | SELECTION GUIDE/NOTES                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|---------|----------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0    |         | Error Code Check                 | SERVICE AND INQUIRY | Review error memory function. Displays "Ch". Press <b>OK</b> to send the query error code memory.                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1    | T1      | Indoor Ambient Temperature       | SERVICE AND INQUIRY | Change the power off memory selection. This feature determines whether the unit memorizes the set conditions prior to a power failure.<br>Displays "Ch". Press <b>OK</b> to return the current setting. Press <b>UP</b> or <b>DOWN</b> to cycle through settings <b>1</b> and <b>0</b> . | Memory settings are off<br>Memory settings are on                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2    | T2      | Indoor Coil Temperature          | SERVICE AND INQUIRY | Change the option to control the indoor fan operation after reaching the set temperature.<br><br>Displays "Ch". Press <b>OK</b> to return the current setting. Press <b>UP</b> or <b>DOWN</b> to cycle through settings 1 through 11. Next, press <b>OK</b> to confirm the selection.    | Stop the fan<br>Minimum fan speed<br>Set speed - intermittent<br>fan-off 4 minutes/on 1 min<br>Terminate after run time of 10 mins<br>Terminate after run time of 15 mins<br>Terminate after run time of 20 mins<br>Terminate after run time of 30 mins<br>Terminate after run time of 40 mins<br>Terminate after run time of 50 mins<br>Terminate after run time of 60 mins                                                                                                                  |
| 3    | T3      | Outdoor Coil Temperature         | SERVICE AND INQUIRY | Change the option to control the <b>COOLING</b> and <b>HEATING</b> modes available for use on the unit. Press <b>UP</b> or <b>DOWN</b> to cycle through the settings <b>CH</b> , <b>HH</b> , <b>CC</b> or <b>nU</b> . Press <b>OK</b> to confirm.                                        | <b>CH</b> - <b>COOLING</b> and <b>HEATING</b> : <b>AUTO</b> , <b>COOLING</b> , <b>DRY</b> , <b>HEATING</b> and <b>FAN</b> modes available<br><b>HH</b> - <b>HEATING</b> Only: <b>HEATING</b> and <b>FAN</b> modes available<br><b>CC</b> - <b>COOLING</b> without <b>AUTO</b> : <b>COOLING</b> , <b>DRY</b> and <b>FAN</b> modes available<br><b>nU</b> - <b>COOLING</b> and <b>HEATING</b> without <b>AUTO</b> : <b>COOLING</b> , <b>DRY</b> , <b>HEATING</b> and <b>FAN</b> modes available |
| 4    | T4      | Outdoor Ambient Temperature      | SERVICE AND INQUIRY | Change the selection of the lowest set temperature.<br><br>NOTE: Temperature range is 60°F ~ 75°F (16°C ~ 24°C). Press <b>UP</b> or <b>DOWN</b> to select temperature setting. Press <b>OK</b> to confirm.                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5    | TP (T5) | Compressor Discharge Temperature | SERVICE AND INQUIRY | Change the selection of the highest set temperature.<br><br>NOTE: Temperature range is 77°F ~ 86°F (25°C ~ 30°C). Press <b>UP</b> or <b>DOWN</b> to select the temperature setting. Press <b>OK</b> to confirm.                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 6    | FT      | Compressor target frequency      | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 7    | Fr      | Compressor run frequency         | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8    | dL      | Unit amperage                    | SERVICE AND INQUIRY | Change the static pressure selection.<br><br>Displays "Ch". Press <b>OK</b> to return the current setting. Press <b>UP</b> or <b>DOWN</b> to cycle through settings <b>0</b> through <b>4</b> or <b>AF</b> (constant air volume test). Press <b>OK</b> to confirm.                       | Only available on ducted/AHU units.<br><br>Refer to the ducted/AHU installation manuals for Fan performances at varying static pressures for airflow settings.                                                                                                                                                                                                                                                                                                                                |
| 9    | Uo      | Unit voltage                     | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 10   | Sn      | Capacity test (special usage)    | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 11   | ----    | Not available                    | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 12   | Pr      | Indoor fan speed                 | SERVICE AND INQUIRY | Change the heating frequency lower limit selection. Displays "Ch". Press <b>OK</b> to return the current heating minimum frequency limit selection code. Press <b>UP</b> and <b>DOWN</b> to select the minimum heating frequency limit value. Press <b>OK</b> to confirm.                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| CODE | INQUIRY | INQUIRY DESCRIPTION                      | SERVICE/ INQUIRY    | FOR SERVICE, PRESS ON/OFF FOR 2 SECONDS TO:                                                                                                                                                                                                                                                                                                                                       | SELECTION GUIDE/NOTES                                                                                                                                     |
|------|---------|------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13   | Lr      | Electronic Expansion Valve (EEV) opening | SERVICE AND INQUIRY | Change the maximum operating frequency of T4 Cooling Only intervals.<br><br>Displays " <b>Ch</b> ". Press <b>OK</b> to return the current operating frequency code of the T4 Cooling Only intervals. Press <b>UP</b> or <b>DOWN</b> to select the limit value and then press <b>OK</b> .                                                                                          |                                                                                                                                                           |
| 14   | ir      | Indoor fan speed                         | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                                                                                                                   | Multiple the display number by 8 to calculate the actual RPM                                                                                              |
| 15   | HU      | Relative Humidity                        | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                                                                                                                   | Available in INQUIRY mode for the high tier/new mid tier units that have an RH sensor.                                                                    |
| 16   | TT      | Setpoint compensation temperature        | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                           |
| 17   | dT      | Dust concentration (not used)            | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                           |
| 18   | WIFI    | Wi-Fi signal strength                    | INQUIRY ONLY        |                                                                                                                                                                                                                                                                                                                                                                                   | The value is measured in <b>dBm</b> . The display values are 0, 1, 2, 3 and 4 (4 is the highest and 0 is the lowest)                                      |
| 19   | ----    | Not available                            | SERVICE ONLY        | Change the cooling frequency upper limit selection in Hz.<br><br>Displays " <b>Ch</b> ". Press <b>OK</b> to return the current frequency limit. Press <b>UP</b> or <b>DOWN</b> to select the preferred frequency upper limit value (in Hz). Press <b>OK</b> to confirm.                                                                                                           | For example, the unit may be factory set to fluctuate between 40 and 84 Hz. If set to 50, the unit will now be limited to operating between 40 and 50 Hz. |
| 20   | oT      | Indoor fan target frequency              | SERVICE AND INQUIRY | Change the heating frequency upper limit selection in Hz.<br><br>Displays " <b>Ch</b> "; press <b>OK</b> to return the current frequency limit. Press <b>UP</b> or <b>DOWN</b> to select the preferred frequency upper limit value (in Hz). Press <b>OK</b> to confirm.                                                                                                           | For example, the unit may be factory set to fluctuate between 40 and 84 Hz. If set to 50, the unit is limited to operating between 40 and 50 Hz.          |
| 21   | ----    | Cooling Temperature Compensation         | SERVICE ONLY        | Change the cooling temperature compensation value.<br><br>Displays " <b>Ch</b> ". Press <b>OK</b> to return the current temperature compensation value code. Press <b>UP</b> or <b>DOWN</b> to select the cooling temperature difference compensation value. Press <b>OK</b> to confirm.                                                                                          | This setting is used to adjust for temperature differences due to the height of the unit install. The offset value can be set at a range of -6° to +6°.   |
| 22   | ----    | Heating Temperature Compensation         | SERVICE ONLY        | Change the heating temperature compensation value.<br><br>Displays " <b>Ch</b> ". Press <b>OK</b> to return the current temperature compensation value code. Press <b>UP</b> or <b>DOWN</b> to select the heating temperature difference compensation value. Press <b>OK</b> to confirm.                                                                                          | This setting is used to adjust for temperature differences due to the height of unit installation. The offset value can be set at a range of -6° to +6°.  |
| 23   | ----    | Maximum Cooling Fan Speed                | SERVICE ONLY        | Change the maximum cooling fan speed setting as it relates to RPM.<br><br>Displays " <b>Ch</b> ". Press <b>OK</b> to return the current maximum cooling fan speed setting. Press <b>UP</b> or <b>DOWN</b> to select the maximum cooling fan speed. Press <b>OK</b> to confirm.                                                                                                    | For example, the unit may be factory set to fluctuate between 300 and 1000 RPM. If set to 800, the unit is limited to operating between 300 and 800 RPM.  |
| 24   | ----    | Minimum Cooling Fan Speed                | SERVICE ONLY        | Change the minimum cooling fan speed setting as it relates to RPM.<br><br>NOTE: Changing this setting is not recommended as it may trigger unit protection protocols.<br><br>Displays " <b>Ch</b> ". Press <b>OK</b> to return the current minimum cooling fan speed setting. Press <b>UP</b> or <b>DOWN</b> to select the minimum cooling fan speed. Press <b>OK</b> to confirm. | For example, the unit may be factory set to fluctuate between 300 and 1000 RPM. If set to 500, the unit is limited to operating between 500 and 1000 RPM. |

| CODE | INQUIRY | INQUIRY DESCRIPTION       | SERVICE/ INQUIRY | FOR SERVICE,<br>PRESS ON/OFF FOR 2 SECONDS TO:                                                                                                                                                                                                                                                                                                                                         | SELECTION GUIDE/NOTES                                                                                                                                             |
|------|---------|---------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25   | ----    | Maximum Heating Fan Speed | SERVICE ONLY     | <p>Change the maximum heating fan speed setting as it relates to RPM.</p> <p>Displays "<b>Ch</b>". Press <b>OK</b> to return the current maximum heating fan speed setting. Press <b>UP</b> or <b>DOWN</b> to select the maximum heating fan speed. Press <b>OK</b> to confirm.</p>                                                                                                    | For example, the unit may be factory set to fluctuate between 300 and 1000 RPM. If set to 800, the unit will now be limited to operating between 300 and 800 RPM. |
| 26   | ----    | Minimum Heating Fan Speed | SERVICE ONLY     | <p>Change the minimum heating fan speed setting as it relates to RPM.</p> <p>Note: Changing this setting is not recommended as it may trigger unit protection protocols.</p> <p>Displays "<b>Ch</b>". Press <b>OK</b> to return the current minimum heating fan speed setting. Press <b>UP</b> or <b>DOWN</b> to select the minimum heating fan speed. Press <b>OK</b> to confirm.</p> | For example, the unit may be factory set to fluctuate between 300 and 1000 RPM. If set to 500, the unit is limited to operating between 500 and 1000 RPM.         |
| 27   | ----    | Not available             |                  |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                   |
| 28   | ----    | Not available             |                  |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                   |
| 29   | ----    | Not available             |                  |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                   |
| 30   | ----    | Not available             |                  |                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                   |

**To exit the Inquiry Mode:**

Press and hold together the On/Off and Fan buttons   for 2 seconds.

## GENERAL TROUBLESHOOTING

### SAFETY CAUTION



### WARNING

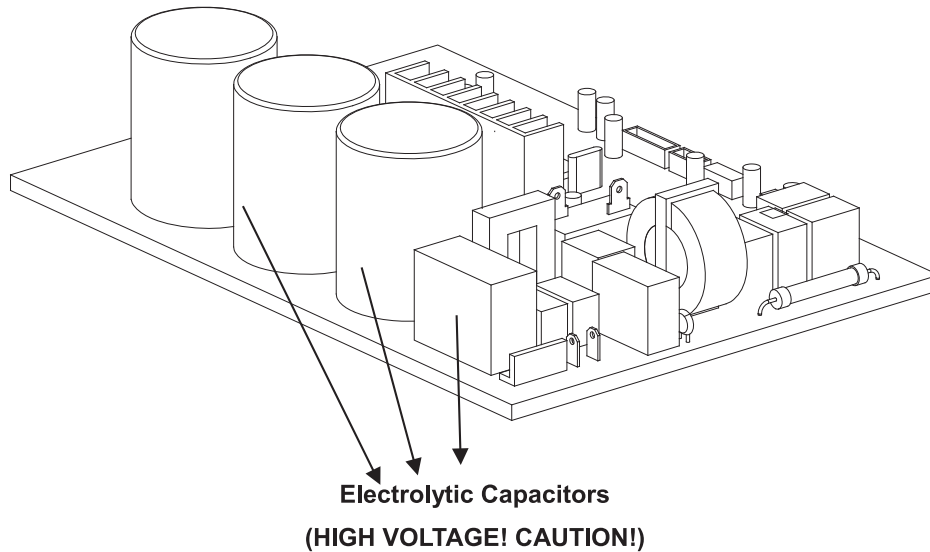
Be sure to turn off all power supplies or disconnect all wires to avoid electric shock. While checking indoor/outdoor PCB, equip yourself with anti-static gloves or wrist strap to avoid damage to the board.



### WARNING

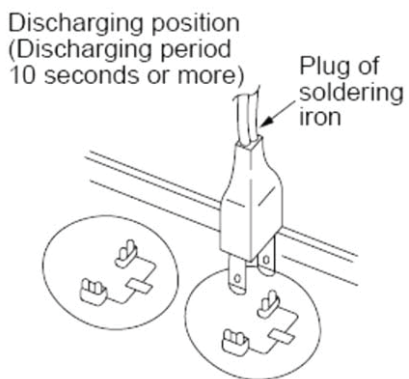
Electricity remains in capacitors even when the power supply is off. Ensure the capacitors are fully discharged before troubleshooting.

**NOTE: Remember to discharge the electrical power in capacitor.**



**Fig. 24 —Electrolytic Capacitors**

For other models, please connect discharge resistance (approximately 100 $\Omega$  40W) or a soldering iron (plug) between the +, - terminals of the electrolytic capacitor on the contrary side of the outdoor PCB.

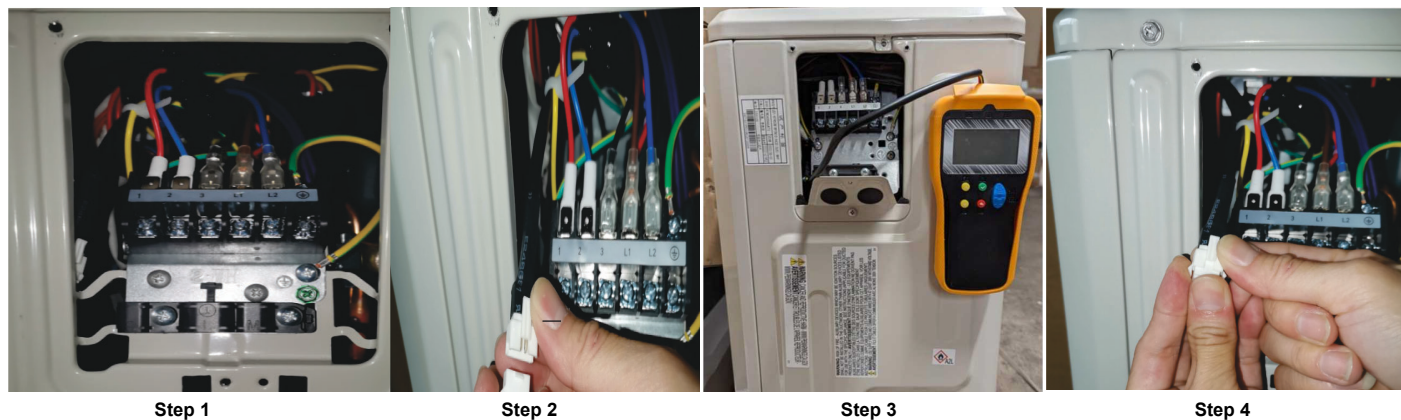


**Fig. 25 —Discharge Position**

**NOTE:** Figure is for reference only. The plug on your unit may differ.

**NOTE:** If using the inverter test tool for troubleshooting, shut off power, remove the electrical panel and locate the cable that is already connected to the test port on the outdoor unit. Connect the test tool to the cable with the connector provided with the test tool. After the maintenance is completed, insert the female end back into the port.

For the R454B single zone ODU with capacity less than 24K, there will be the test tool connector. For 24K HH and 30K-60K single zone ODU, there is a diagnosis/check board which has digital display on it, you can read the parameters from it directly.



**Fig. 26 —Inverter Test Tool Maintenance**

Connect the Dr. SMART tool to the white terminal as shown in Step 3 above.



**Fig. 27 —Dr. SMART Tool**

**NOTE:** These pictures is for reference only. Actual appearance may vary.

## ERROR CODES

### Error Display

| Display               | Indoor/Outdoor Code | Malfunction or Protection                                                           | Solution<br>(Refer to Page)    |
|-----------------------|---------------------|-------------------------------------------------------------------------------------|--------------------------------|
| d <sup>f</sup>        | N/A                 | Defrosting                                                                          | Normal Display, not error code |
| FC                    | N/A                 | Forced cooling                                                                      |                                |
| FH CC                 | IDU                 | Refrigerant Sensor Error                                                            | page 56                        |
| EC 03 / EC 07 / EC 71 | ODU                 | Fan speed out of control                                                            | page 31                        |
| EC 51                 | ODU                 | EEPROM parameter error                                                              | page 27                        |
| EC 52                 | ODU                 | Coil temp. sensor(T3) error                                                         | page 34                        |
| EC 53                 | ODU                 | Ambient temp. sensor(T4) error                                                      | page 34                        |
| EC 54                 | ODU                 | COMP. discharge temp. sensor (TP) error                                             | page 34                        |
| EC 55                 | ODU                 | IPM module temperature sensor malfunction                                           | page 36                        |
| EL 0C                 | IDU & ODU           | System Lacks Refrigerant Diagnosis and Solution                                     | page 37                        |
| EH 00 / EH0A          | IDU & ODU           | EEPROM Malfunction Error Diagnosis and Solution                                     | page 27                        |
| EH02                  | IDU                 | Zero Crossing Detection Error                                                       | page 30                        |
| EH b3                 | IDU                 | Communication Malfunction Between Wire and Master Control                           | page 55                        |
| EH bA                 | IDU & ODU           | Communication Malfunction Between Wire and Master Control                           | page 55                        |
| EH C1                 | IDU                 | Refrigerant Sensor Detects Leakage                                                  | page 57                        |
| EH C2                 | IDU                 | Refrigerant Sensor is Out of Range and Leakage is Detected                          | page 57                        |
| EH CC / FH CC         | IDU                 | Refrigerant sensor error or Refrigerant sensor is out of range                      | page 56                        |
| EH 3b                 | IDU & ODU           | External Fan DC Bus Voltage is Too High                                             | page 55                        |
| EH 06                 | IDU                 | IDU Main Control Board and Display Board Communication Error Diagnosis and Solution | page 58                        |
| EH 0E                 | IDU                 | Water-Level Alarm Malfunction Diagnosis and Solution                                | page 38                        |
| EH 60 / EH 61         | IDU & ODU           | Open circuit or Short Circuit Of Temperature Sensor                                 | page 34                        |
| EC 57                 | ODU                 | Refrigerant pipe temperature sensor error                                           | page 34                        |
| EC 5C                 | ODU                 | Suction Transducer Failure                                                          | page 33                        |
| EL 01                 | IDU & ODU           | Communication error                                                                 | page 28                        |
| EL 16                 | ODU                 | Communication malfunction between adapter board and ODU main board                  | page 59                        |
| FL 09                 | IDU & ODU           | Mismatch between the new and old platforms diagnosis and solution                   | page 59                        |
| PC 00                 | ODU                 | IPM module protection                                                               | page 39                        |
| PC 02                 | ODU                 | Compressor top (or IPM) temp. protection                                            | page 41                        |
| PC 04                 | ODU                 | Inverter Compressor Drive Error Diagnosis and Solution                              | page 44                        |
| PC 06                 | ODU                 | Discharge temperature protection of compressor                                      | page 41                        |
| PC 08                 | ODU                 | Outdoor overcurrent protection                                                      | page 50                        |
| PC 0A                 | ODU                 | High temperature protection of condenser                                            | page 52                        |
| PC 0F                 | ODU                 | PFC module protection                                                               | page 53                        |
| PC 0L                 | ODU                 | Low Ambient Temperature Protection                                                  | page 55                        |
| PC 10                 | ODU                 | Low AC voltage protection                                                           | page 54                        |
| PC 11                 | ODU                 | Main control board DC bus high voltage protection                                   | page 54                        |
| PC 12                 | ODU                 | Main control board DC bus high voltage protection /341 MCE error                    | page 54                        |
| PC 03                 | IDU                 | * Pressure Protection (low or high pressure)                                        | page 42                        |
| PC 30                 | ODU                 | System high pressure protection                                                     | page 42                        |
| PC 31                 | ODU                 | Low Pressure Protection                                                             | page 42                        |

| Display | Indoor/Outdoor Code | Malfunction or Protection                                            | Solution<br>(Refer to Page) |
|---------|---------------------|----------------------------------------------------------------------|-----------------------------|
| PC 40   | ODU                 | Communication error between ODU main chip and compressor driven chip | page 45                     |
| PC 41   | ODU                 | Compressor current sampling failure                                  | page 46                     |
| PC 42   | ODU                 | Compressor start failure of outdoor unit                             | page 50                     |
| PC 43   | ODU                 | Compressor lack phase protection                                     | page 47                     |
| PC 44   | ODU                 | Zero speed protection                                                | page 50                     |
| PC 45   | ODU                 | IR chip drive failure                                                | page 48                     |
| PC 46   | ODU                 | Compressor speed has been out of control                             | page 50                     |
| PC 49   | ODU                 | Compressor overcurrent failure                                       | page 50                     |
| LC 06   | ODU                 | High temperature protection of Inverter module (IPM)                 | page 41                     |
| PH 90   | IDU & ODU           | High temperature protection of evaporator                            | page 60                     |
| PH 91   | IDU & ODU           | Low temperature protection of evaporator                             | page 61                     |

**NOTE: PC03 Low pressure protection switch is open. Check the switch and repair or leak check the unit and recharge.**

## TROUBLESHOOTING BY ERROR CODE

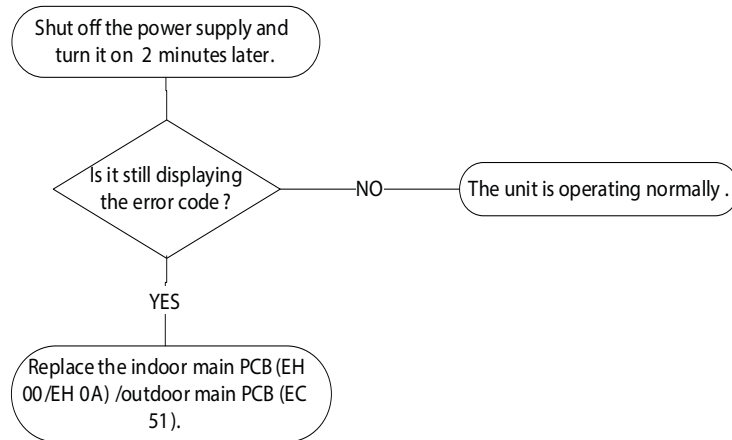
### EH 00/ EH 0A / EC 51 (EEPROM Malfunction Error Diagnosis and Solution)

**Description:** Indoor or outdoor PCB main chip does not receive feedback from EEPROM chip.

**Recommended parts to prepare:** Indoor PCB, Outdoor PCB

**Troubleshooting and repair:**

Troubleshooting and repair:

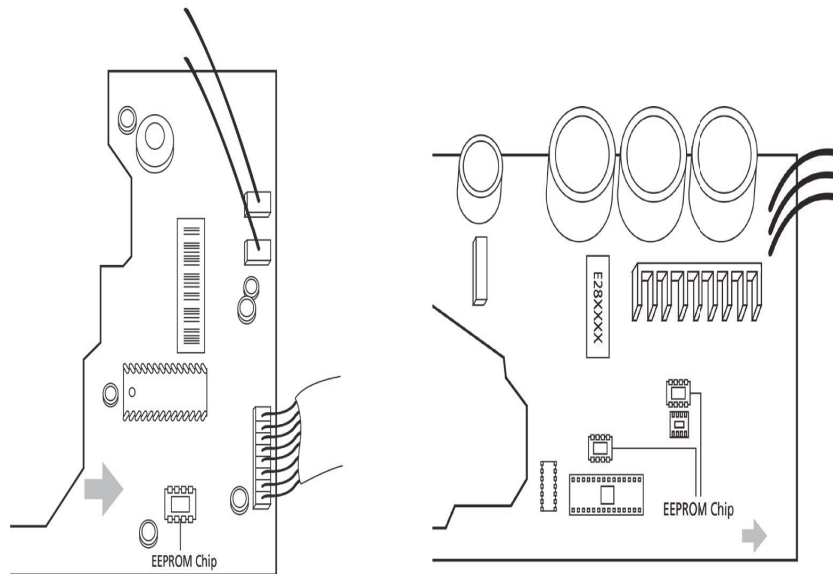


**Fig. 28 —EC 51**

**Remarks:**

**EEPROM:** A read-only memory whose contents can be erased and reprogrammed using a pulsed voltage.

The location of the EEPROM chip on the outdoor PCB is shown in the following image:



**Fig. 29 —Location of EPROM Chip**

**NOTE:** This picture is only for reference, actual appearance may vary.

**IMPORTANT:** Troubleshooting and repair of compressor driven chip EEPROM parameter error sand communication errors between outdoor main chip and compressor driven chip are same as EC 51.

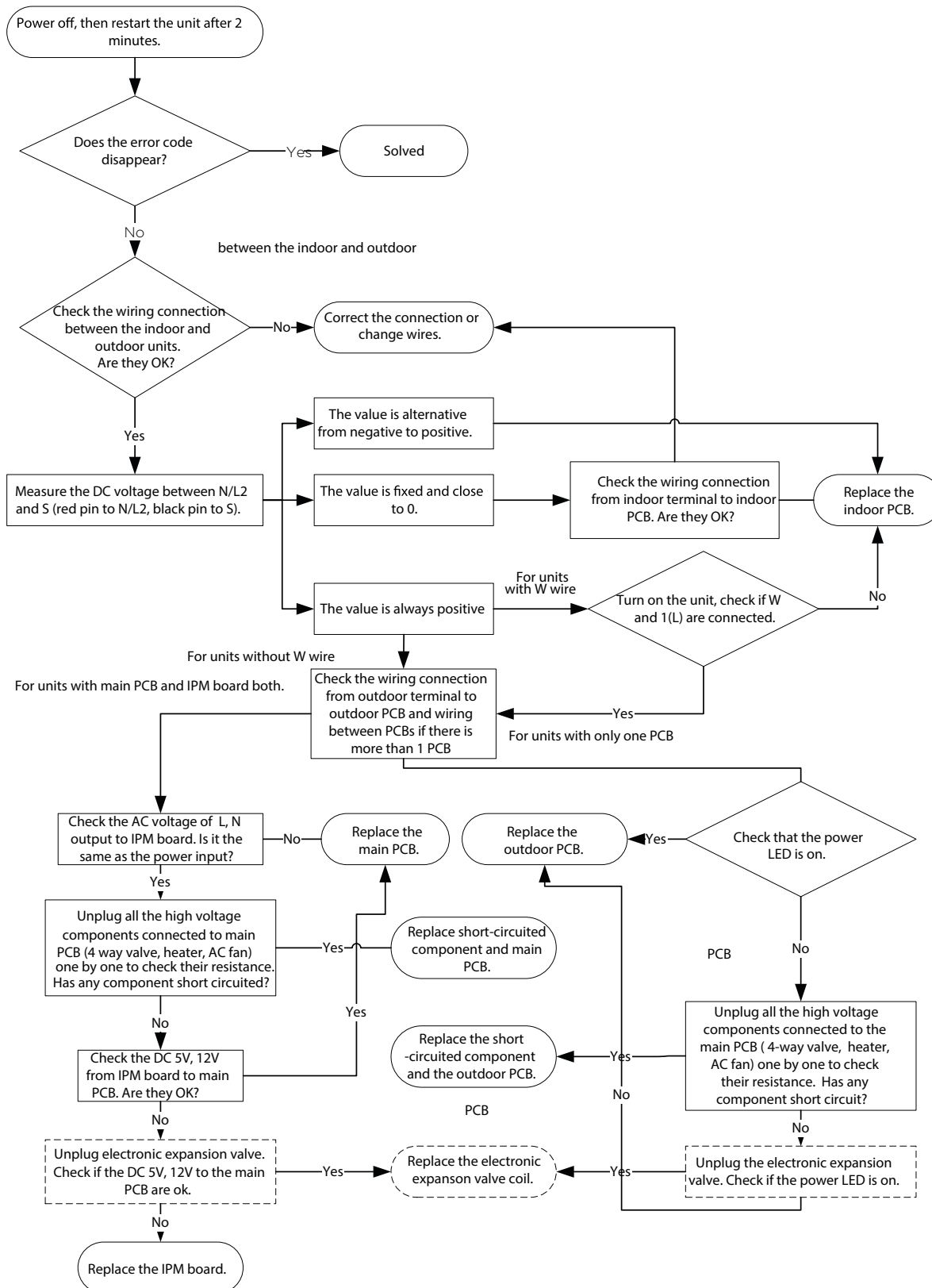
## EL 01 (Indoor and Outdoor Unit Communication Error Diagnosis and Solution)

**Description:** Indoor unit can not communicate with outdoor unit.

**Recommended parts to prepare:** Indoor PCB, Outdoor PCB

**Troubleshooting and repair:**

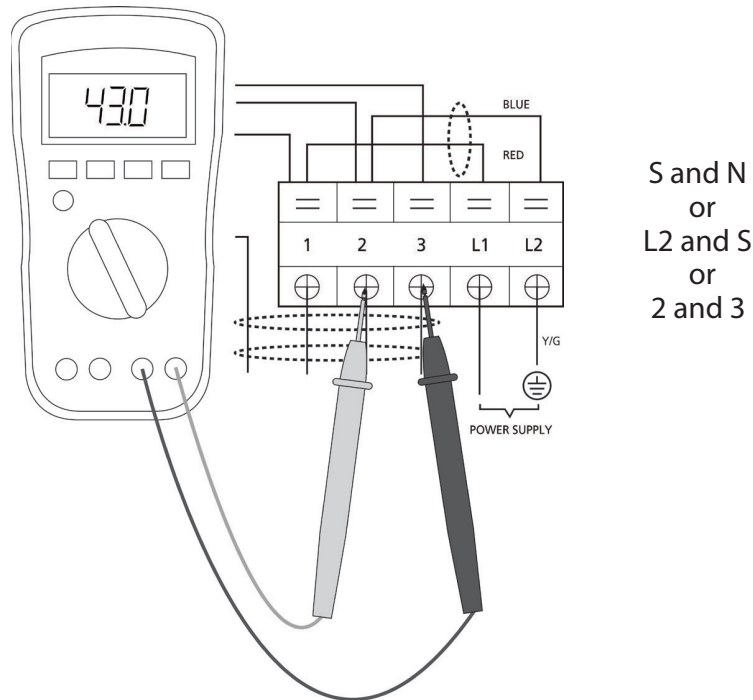
:



For certain models, outdoor PCB could not be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

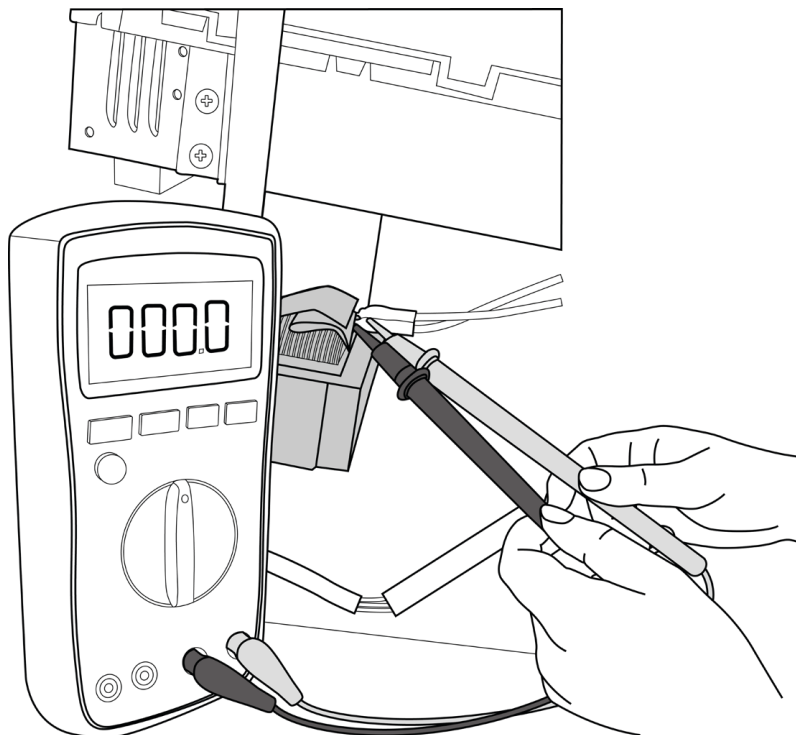
**EL 01 (Continued)****Remarks:**

- Use a multimeter to test the DC voltage between the 2 port (or S or L2 port) and 3 port (or N or S port) of the outdoor unit.
- The multimeter's red pin connects with the 2 port (or S or L2 port) while the black pin is for the 3 port (or N or S port). If the unit is running normally, the voltage moves alternately as positive values and negative values.
- If the outdoor unit malfunctions, the voltage remains a narrow positive value.
- If the indoor unit malfunctions, the voltage value will be fixed.

**Fig. 30 —Measure Voltage Between Ports**

Use a multimeter to test the reactor's resistance which does not connect with the capacitor.

- The normal value should be around zero ohm. Otherwise, the reactor has malfunctioned. Check the reactor to ensure it is not shorted to ground.

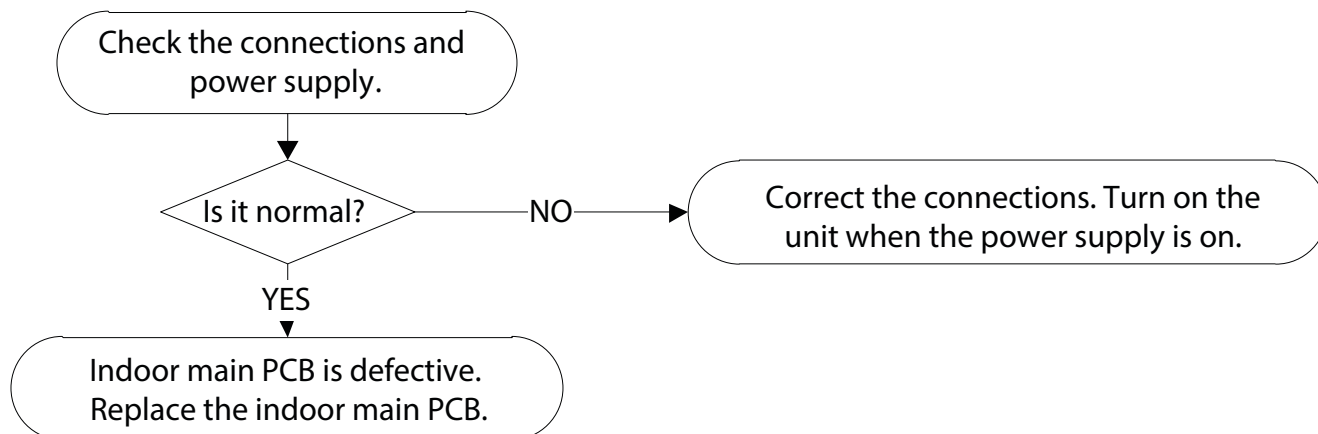
**Fig. 31 —Test Reactor Resistance**

## EH 02 Zero Crossing Detection Error Diagnosis and Solution

**Description:** When the PCB does not receive a zero crossing signal feedback for 4 minutes or the zero crossing signal time interval is abnormal.

**Recommended parts to repair:** Connection wires, Indoor main PCB

**Troubleshooting:**



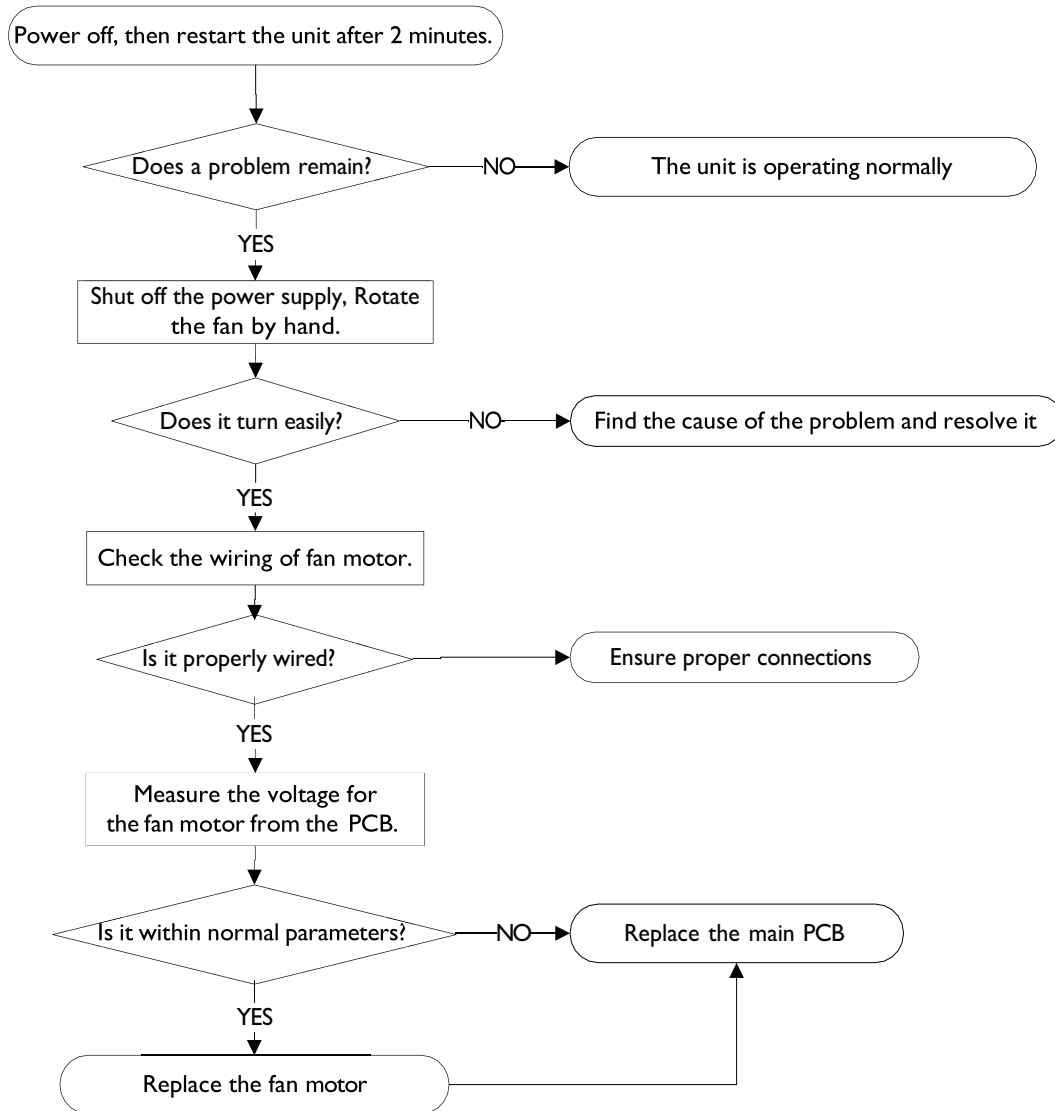
**NOTE:** A zero crossing detection error is only valid for a unit with an AC fan motor. For other models, this error does not apply.

**EH03 / EC 07/ EC 71(Fan Speed Is Operating Outside of Normal Range Diagnosis and Solution)**

**Description:** When indoor / outdoor fan speed keeps too low or too high for a certain time, the LED displays the failure code and the AC turns off.

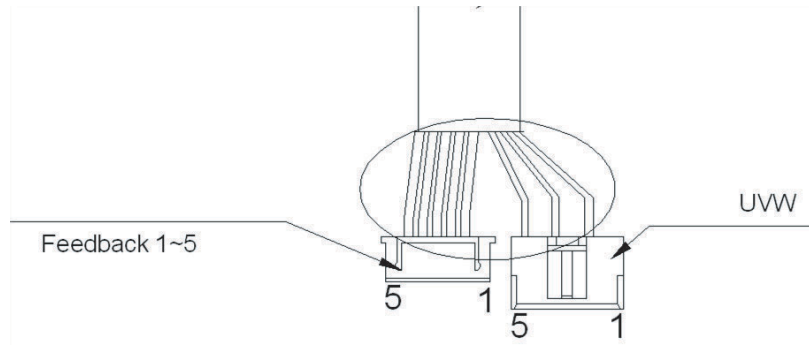
**Recommended parts to prepare:** Connection wires, Fan assembly, Fan motor, PCB

**Troubleshooting and repair:**



**Outdoor DC Fan Motor (DC motor that controls the chip on the PCB)**

1. Release the UVW connector. Measure the resistance of U-V, U-W, V-W. If the resistance is not equal to each other, the fan motor is faulty and must be replaced. Otherwise, proceed to step 2.
2. Power on the unit and when the unit is in standby, measure the pin4-5 voltage in the feedback signal connector. If the value is not 5V, change the PCB. Otherwise, proceed to step 3.
3. Rotate the fan by hand, measure the pin1-5, pin 2-5 and pin 3-5 voltage levels in the feedback signal connector. If any voltage is not in the positive voltage fluctuation, the fan motor is faulty and must be replaced.

**Fig. 32 —Outdoor DC Fan Motor (DC motor that controls the chip on the PCB)**

| NO.    | 1      | 2    | 3     | 4    | 5     |
|--------|--------|------|-------|------|-------|
| COLOR  | Orange | Grey | White | Pink | Black |
| SIGNAL | Hu     | Hv   | Hw    | Vcc  | GND   |

|        |     |      |        |
|--------|-----|------|--------|
| COLOR  | Red | Blue | Yellow |
| SIGNAL | W   | V    | U      |

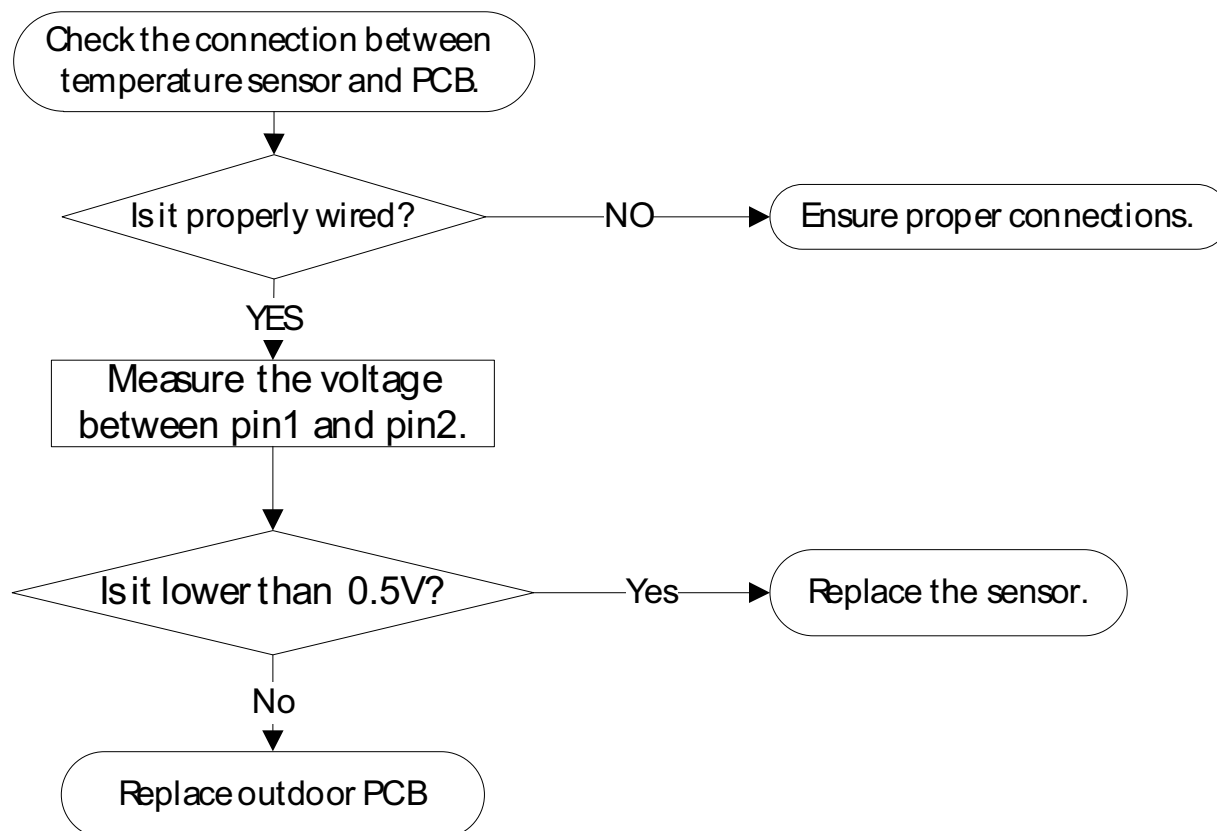
**EC 5C (Suction Transducer is in open circuit or has short circuited) (For Crossover units) diagnosis and solution**

Description: If the sampling voltage is lower than 2V or higher than 254V, the LED displays the failure code.

Recommended parts to prepare:

- Connection wires
- Sensor
- Outdoor PCB

Troubleshooting and repair:



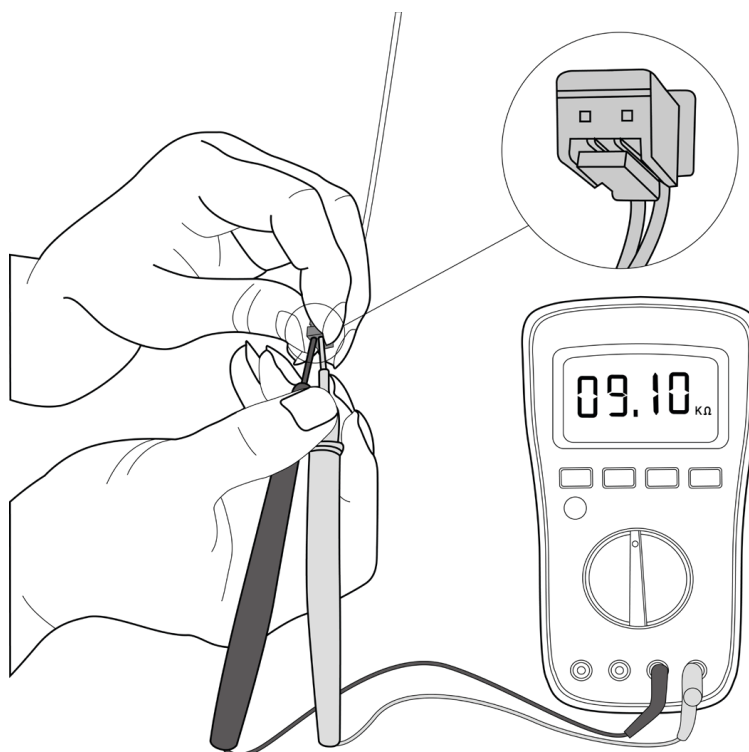
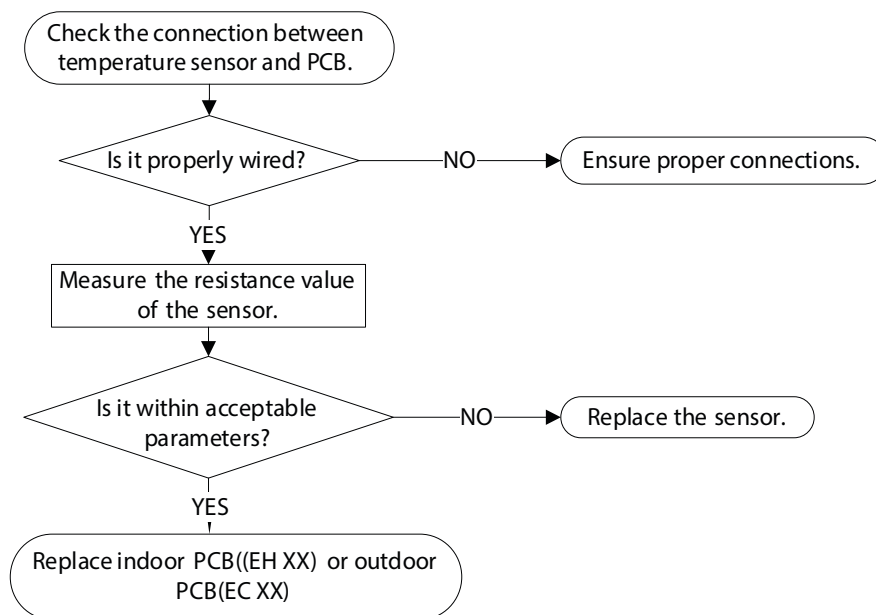
## EC 53/ EC 52/ EC 54/ EC 56/ EC 57/ EC 50/ EH 60/ EH 61(Open Circuit or Short Circuit of Temperature Sensor Diagnosis and Solution)

**Description:** If the sampling voltage is lower than 0.06V or higher than 4.94V, the LED displays the failure.

**Recommended parts to prepare:** Connection wires, Sensors, PCB

**Troubleshooting and repair:**

Refer to Appendix, page 88.



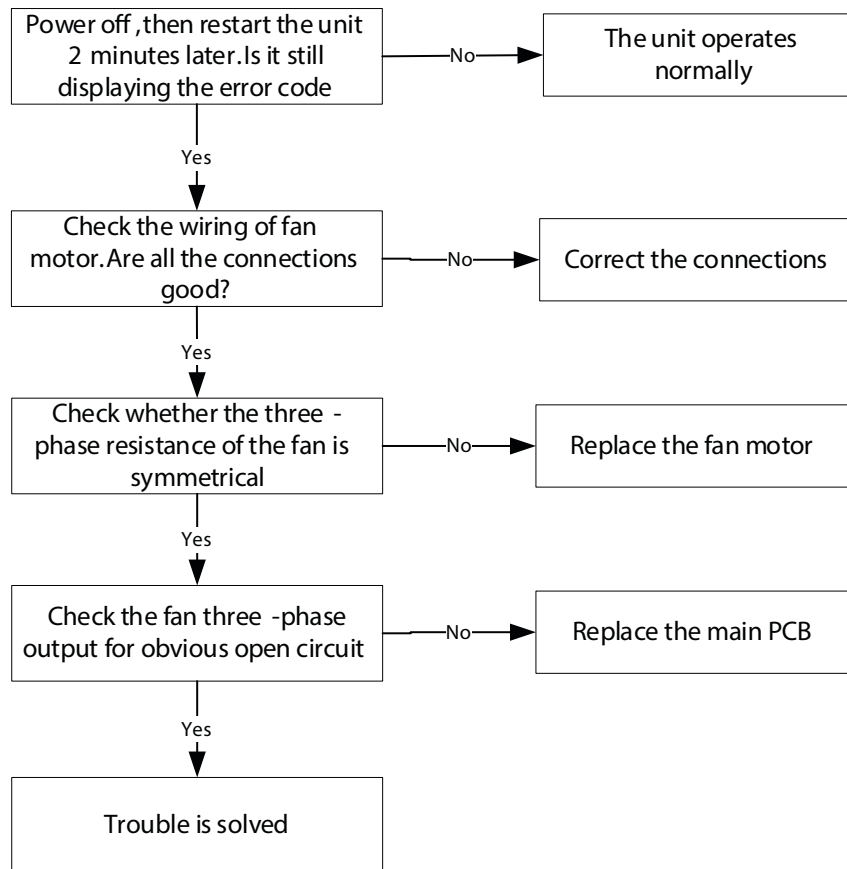
**NOTE:** This picture and the value are only for reference, actual appearance and value may vary.

## EC 72 Lack phase failure of ODU DC fan motor diagnosis and solution

**Description:** When the three-phase sampling current of the DC motor is abnormal, especially when the current of one or more phases is always small and almost 0, the LED displays the failure code.

**Recommended parts to prepare:** Connection wire, Fan motor, Outdoor PCB

**Troubleshooting and repair:**



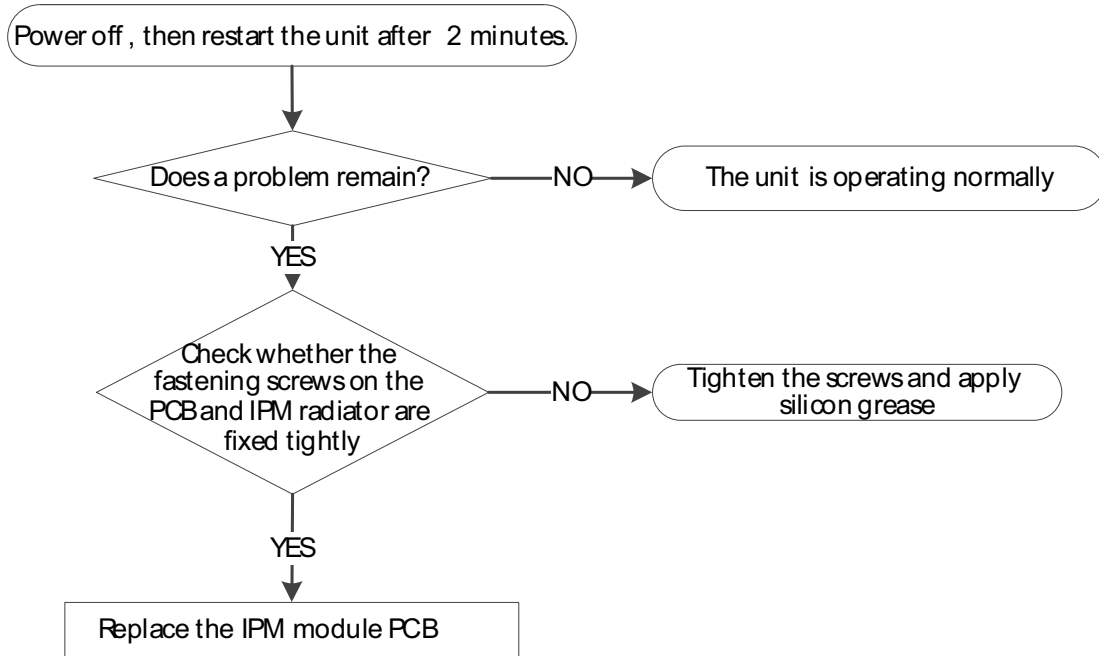
**EC 55 (ODU IPM module temperature sensor malfunction diagnosis and solution)**

**Description:** If the sampling voltage is 0V or 5V, the LED displays the failure code.

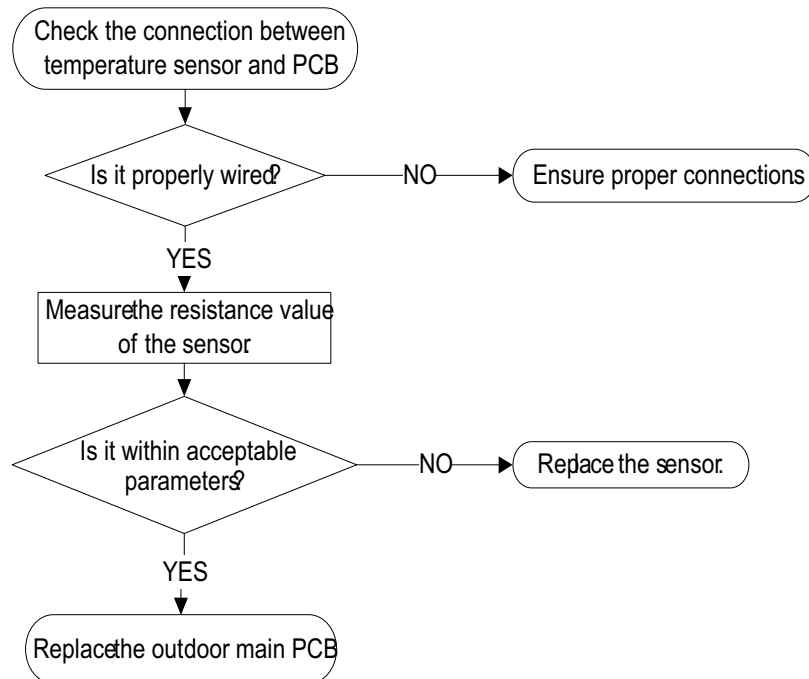
**Recommended parts to prepare:**

- IPM module PCB
- Connection wires
- Sensors
- Outdoor main PCB

**Troubleshooting and repair:** If the radiator has no sensor, follow the steps below to resolve:



If the radiator has a sensor(TH), follow the steps below:

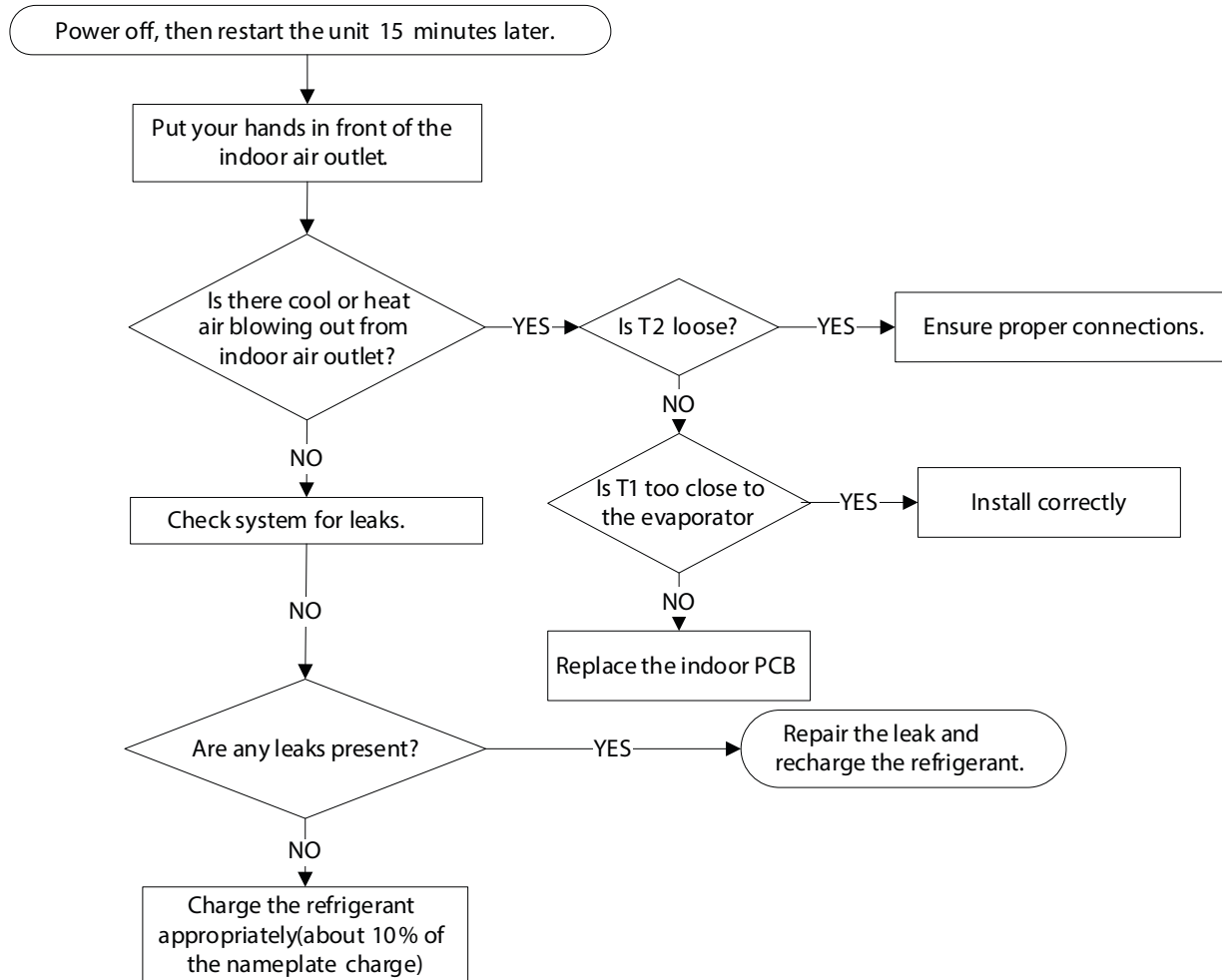


**EL 0C (System Lacks Refrigerant Diagnosis and Solution)**

**Description:** Judging the abnormality of the refrigeration system according to the number of compressor stops and the changes in operating parameters caused by excessive exhaust temperature.

**Recommended parts to prepare:** Indoor PCB, Additional refrigerant

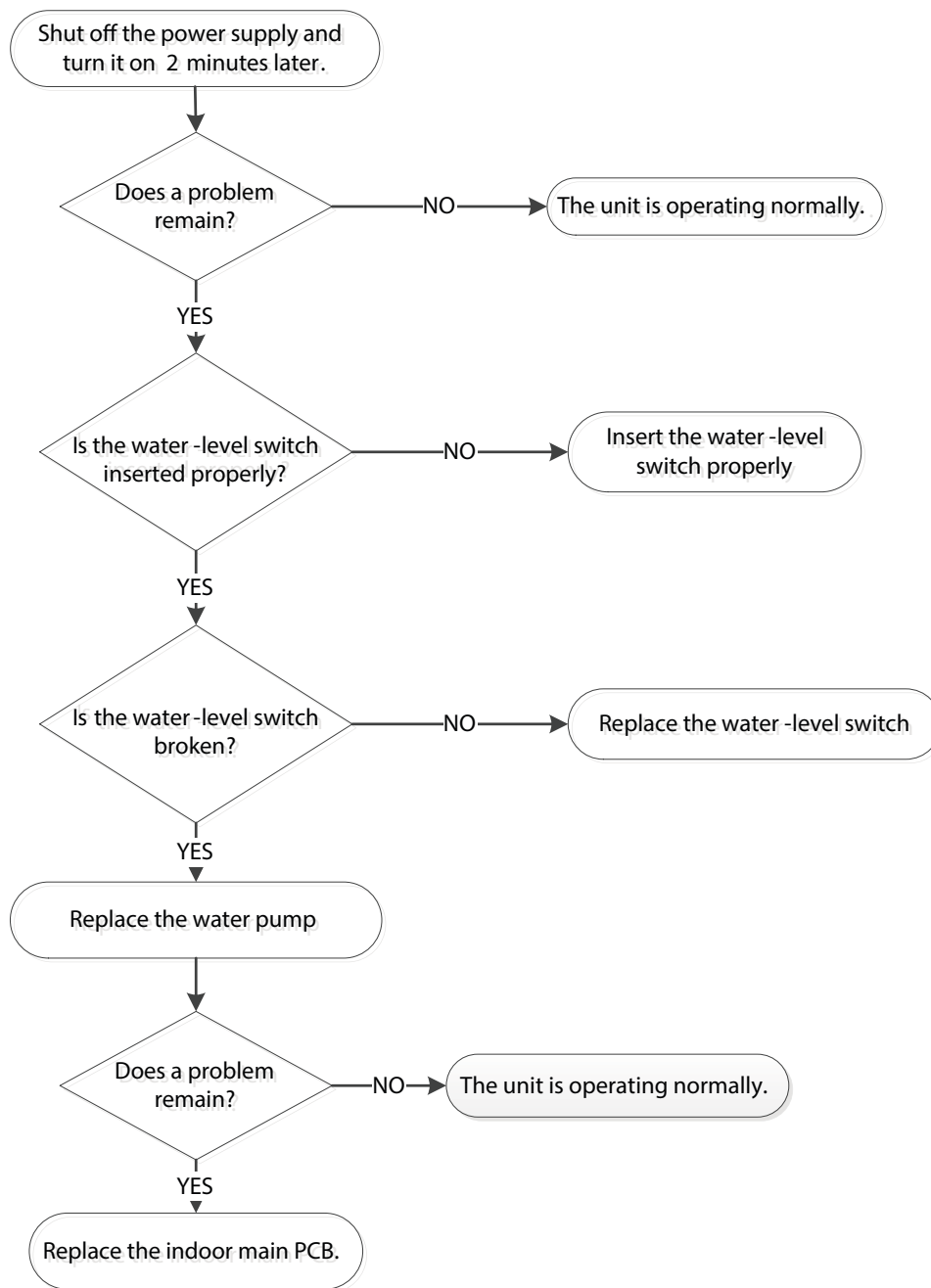
**Troubleshooting and repair:**



**EH 0E (Water-Level Alarm Malfunction Diagnosis and Solution)**

**Description:** If the sampling voltage is not 5V, the LED displays the failure code.

**Recommended parts to prepare:** Connection wires, Water-level switch, Water pump, Indoor PCB

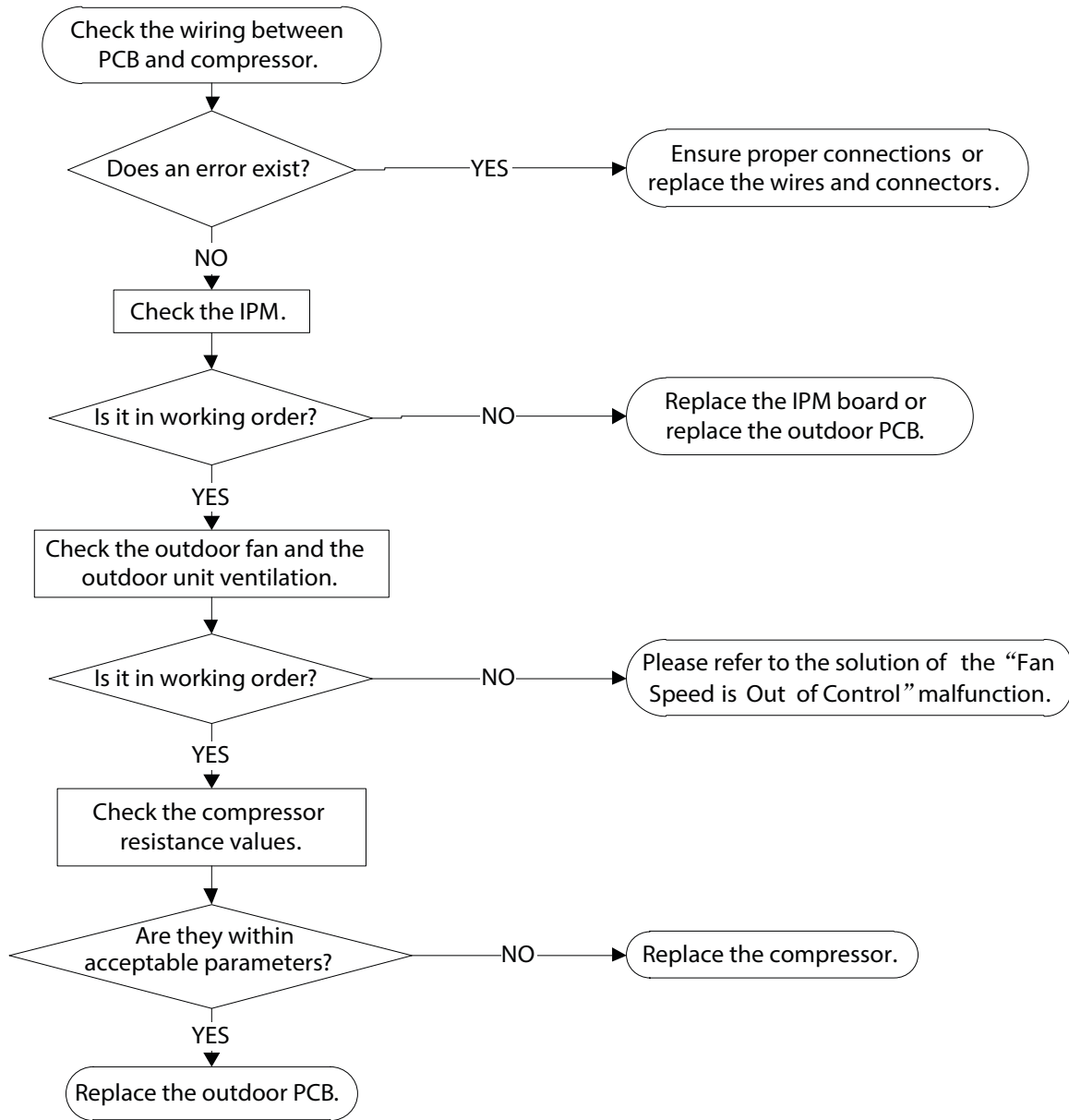


## PC 00 (ODU IPM Module Protection Diagnosis and Solution)

**Description:** When the voltage signal the IPM sends to the compressor drive chip is abnormal, the display LED shows "PC 00" and the AC turn off.

**Recommended parts to prepare:** Connection wires, IPM module board, Outdoor fan assembly, Compressor, Outdoor PCB

**Troubleshooting and repair:**



**NOTE:** For certain models, the outdoor PCB can not be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

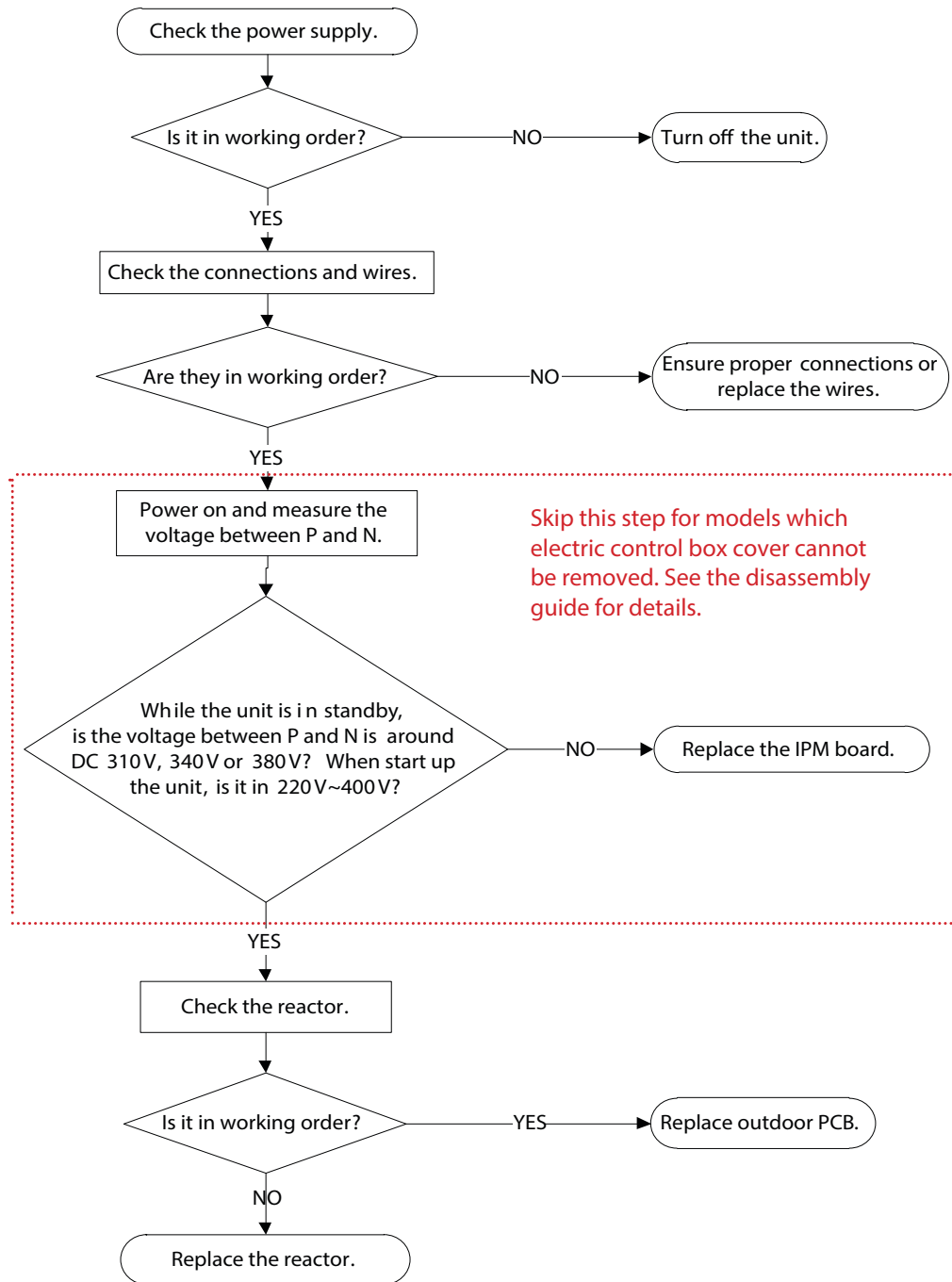
**PC 10 / PC 11 / PC 12 (ODU Voltage Protection Diagnosis and Solution)**

**PC 10 (Outdoor unit low AC voltage protection) / PC 11 (Outdoor unit main control board DC bus high voltage protection) / PC 12 (Outdoor unit main control board DC bus high voltage protection/341 MCE error) Diagnosis and Solution**

**Description:** Abnormal increases or decreases in voltage are detected by checking the specified voltage detection circuit.

**Recommended parts to prepare:** Power supply wires, IPM module board, PCB, Reactor

**Troubleshooting and repair:**



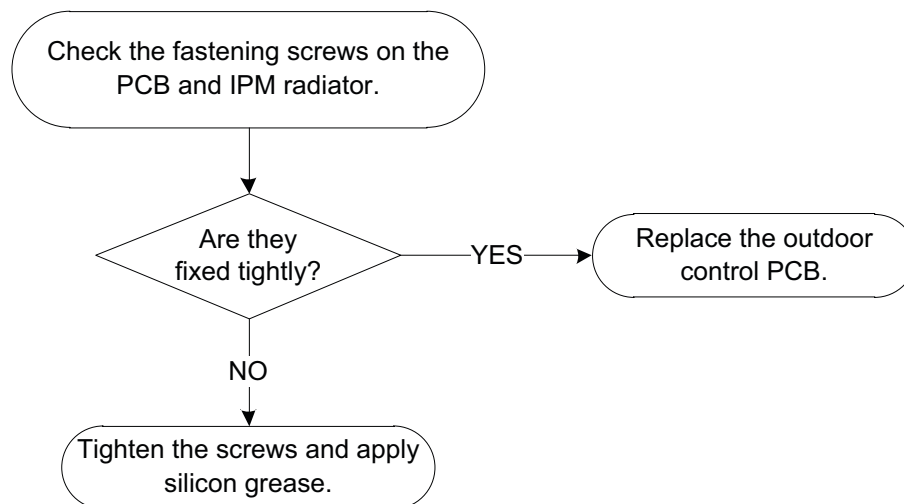
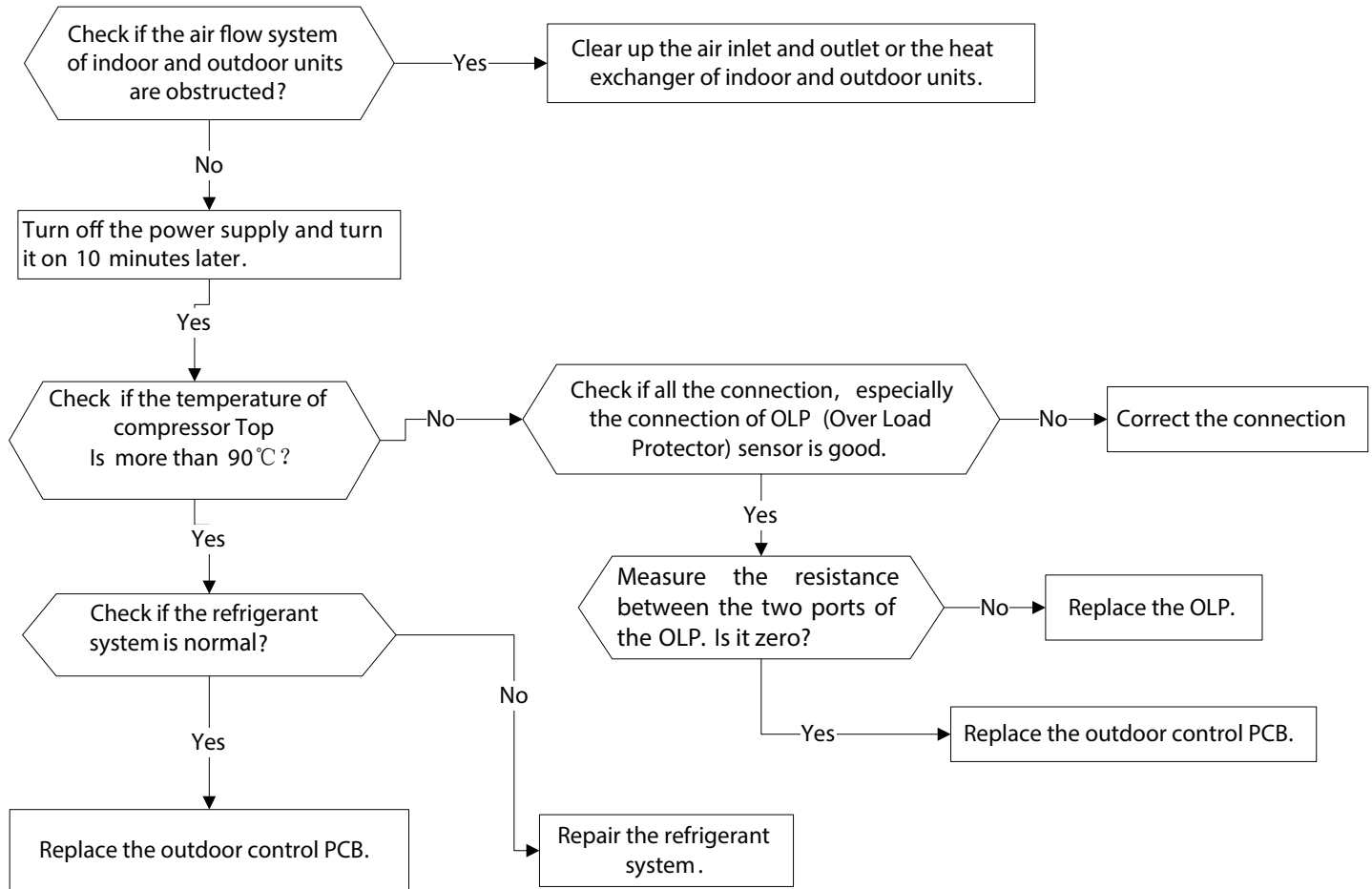
**NOTE:** For certain models, the outdoor PCB cannot be removed separately. In this case, the outdoor electric control box should be replaced as a whole.

**PC 02 / LC 06 (Compressor Top (or IPM) Temp. Protection Diagnosis and Solution)**

**Description:** For some models with overload protection, If the sampling voltage is not 5V, the LED will display the failure. If the temperature of IPM module is higher than a certain value, the LED displays the failure code.

**Recommended parts to prepare:** Connection wires, Outdoor PCB, IPM module board, High pressure protector, System blockages

**Troubleshooting and repair:**

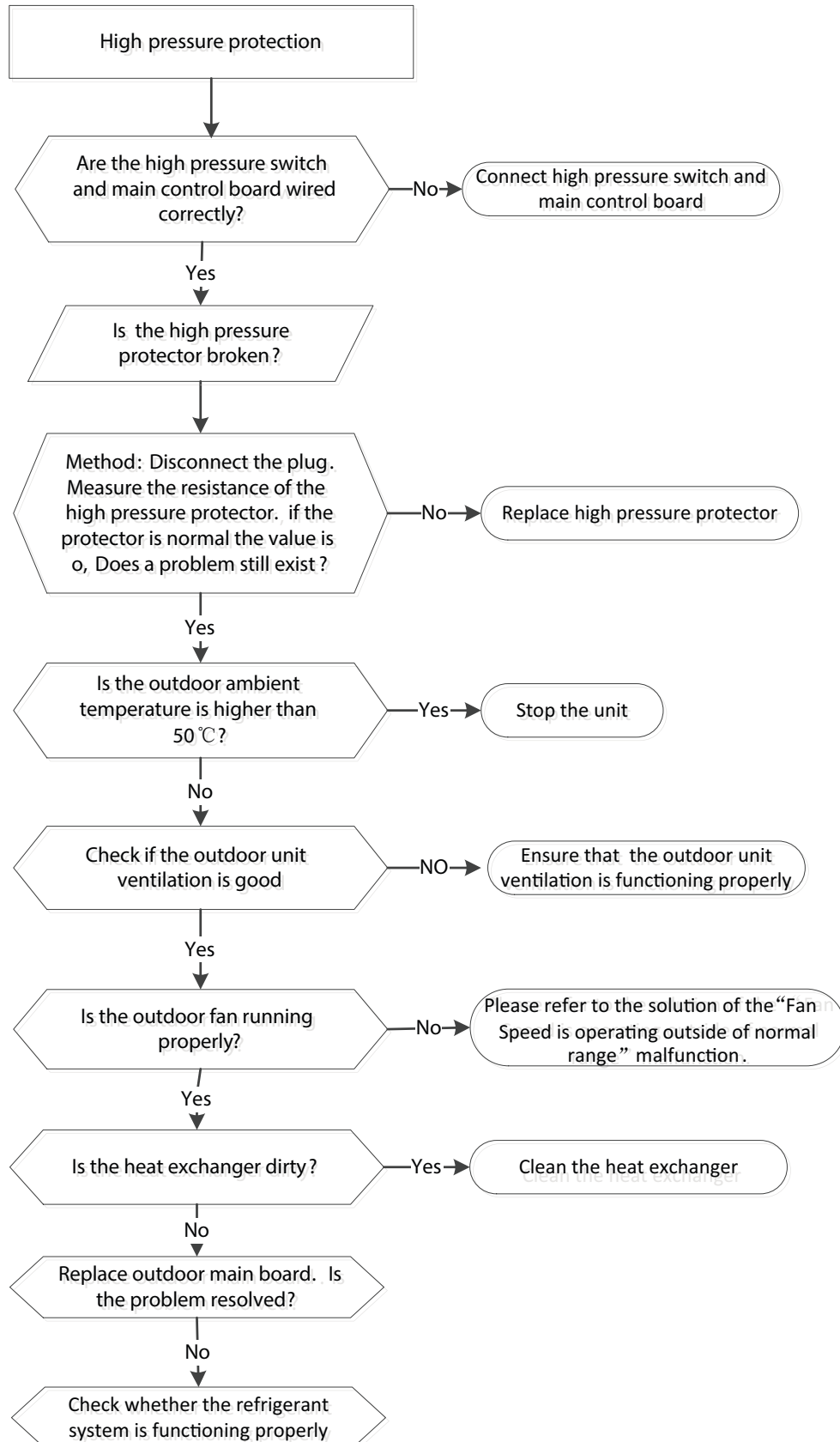


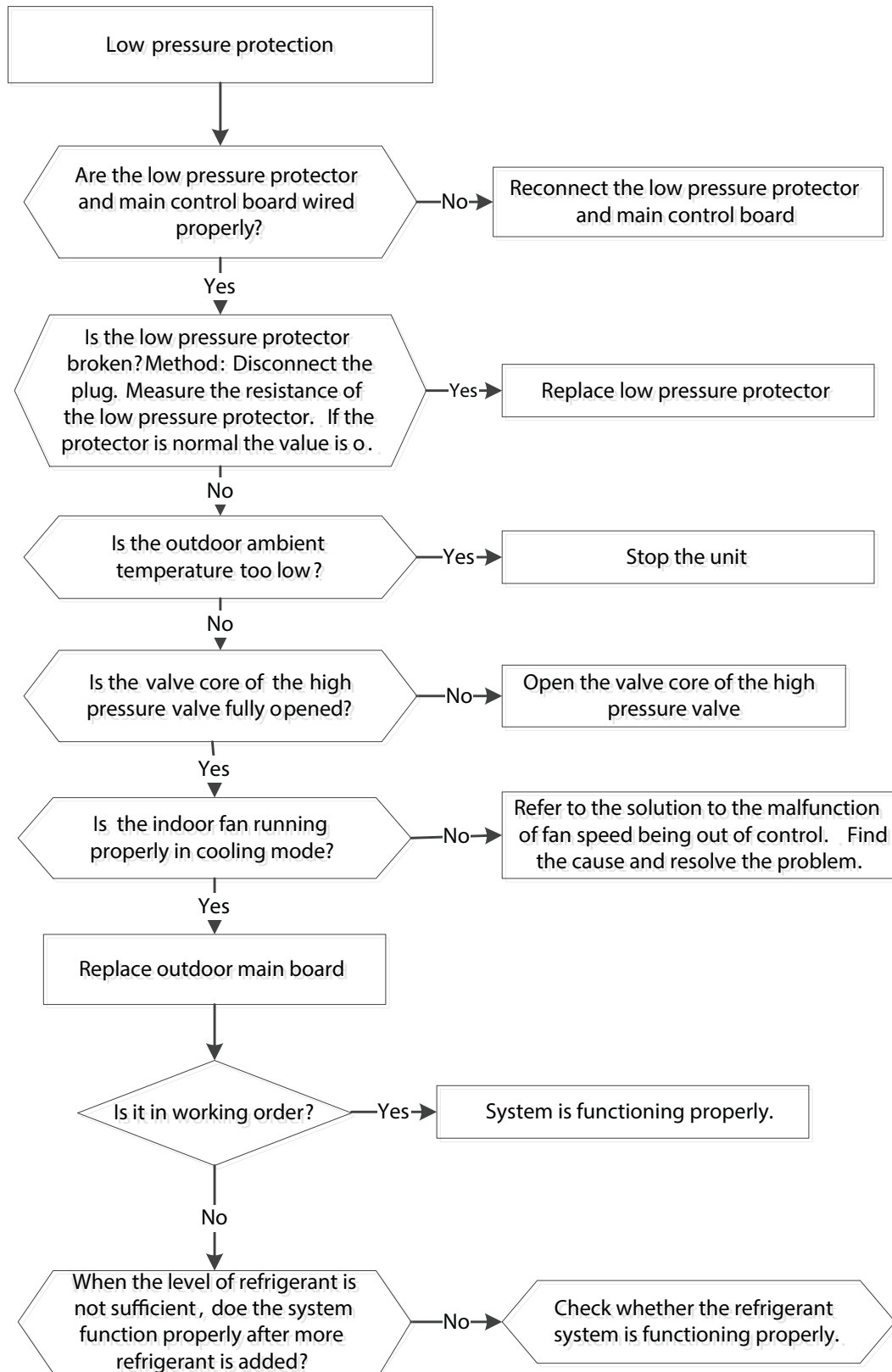
## PC 03 Pressure Protection (low or high pressure), PC 30 High Pressure Protection, PC 31 Low Pressure Protection (Diagnosis and Solution)

**Description:** Outdoor pressure switch cut off the system because high pressure is higher than 4.4 MPa or outdoor pressure switch cut off the system because low pressure is lower than 0.13 MPa, the LED displays the failure code.

**Recommended parts to prepare:** Connection wires, Pressure switch, Outdoor fan, Outdoor main PCB, Refrigerant

**Troubleshooting and repair:**



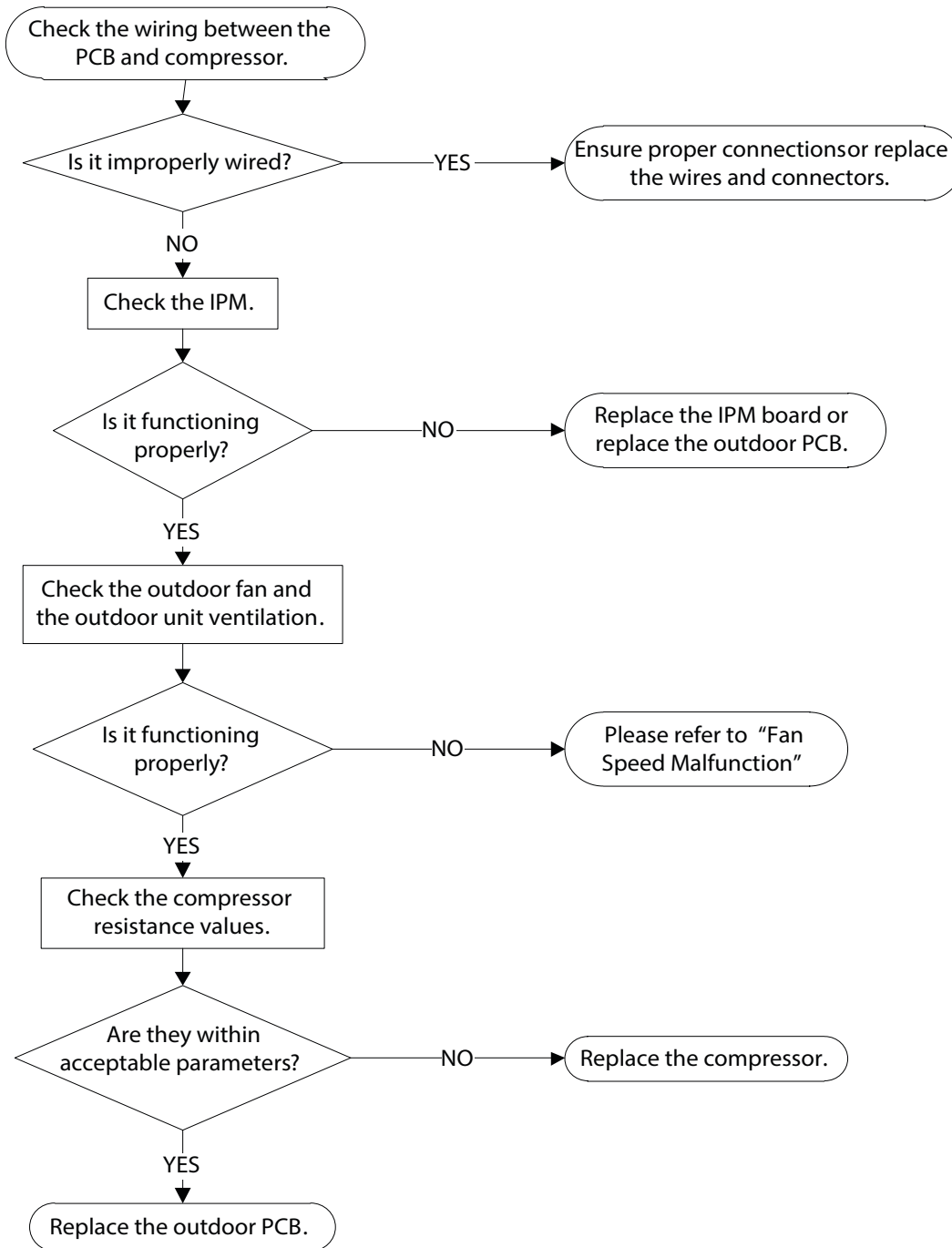


## PC 04 (Inverter Compressor Drive Error Diagnosis and Solution)

**Description:** An abnormal inverter compressor drive is detected by a special detection circuit, including communication signal detection, voltage detection, compressor rotation speed signal detection and so on.

**Recommended parts to prepare:** Connection wires, IPM module board, Outdoor fan assembly, Compressor, Outdoor PCB

**Troubleshooting and repair:**

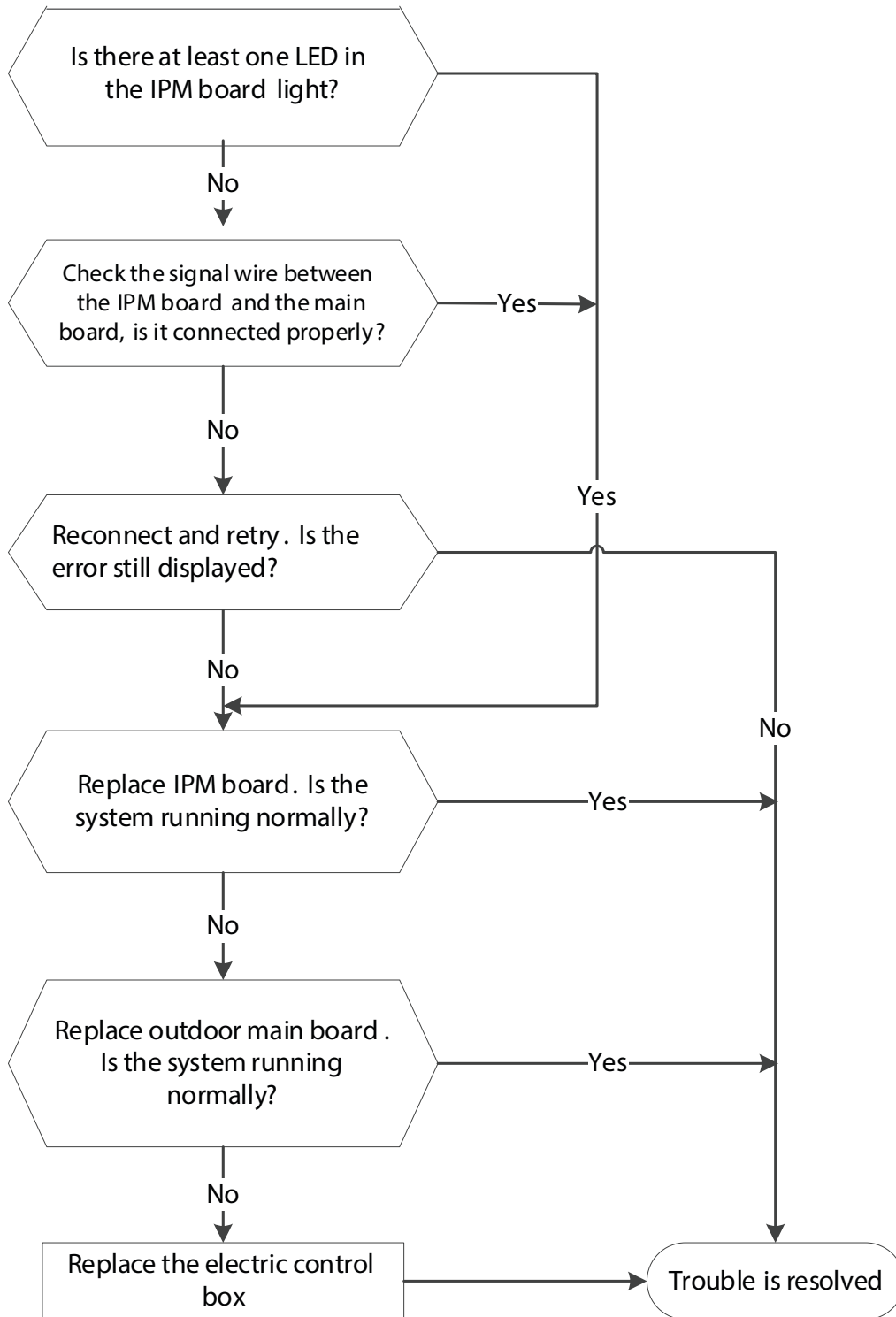


## PC 40 (Communication error between ODU main chip and compressor driven chip diagnosis and solution)

**Description:** The main PCB cannot detect the IPM board.

**Recommended parts to prepare:** Connection wires, Outdoor PCB, IPM module board, Electric control box

**Troubleshooting and repair:**

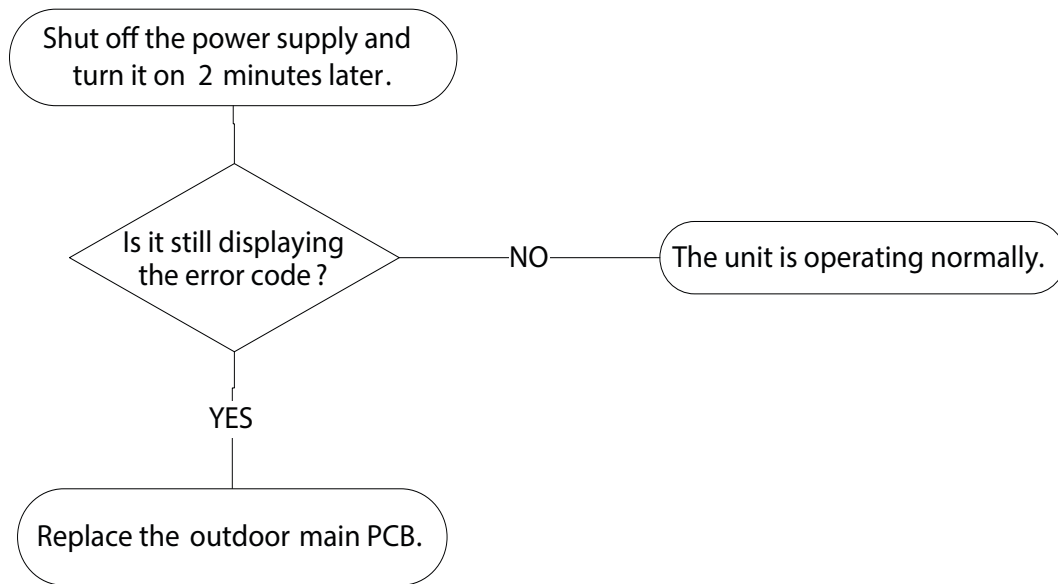


## PC 41 (Outdoor compressor current sampling circuit failure diagnosis and solution)

**Description:** Three-phase sampling offset voltage error, the static bias voltage is normally 2.5V.

**Recommended parts to prepare:** Outdoor main PCB

**Troubleshooting and repair:**

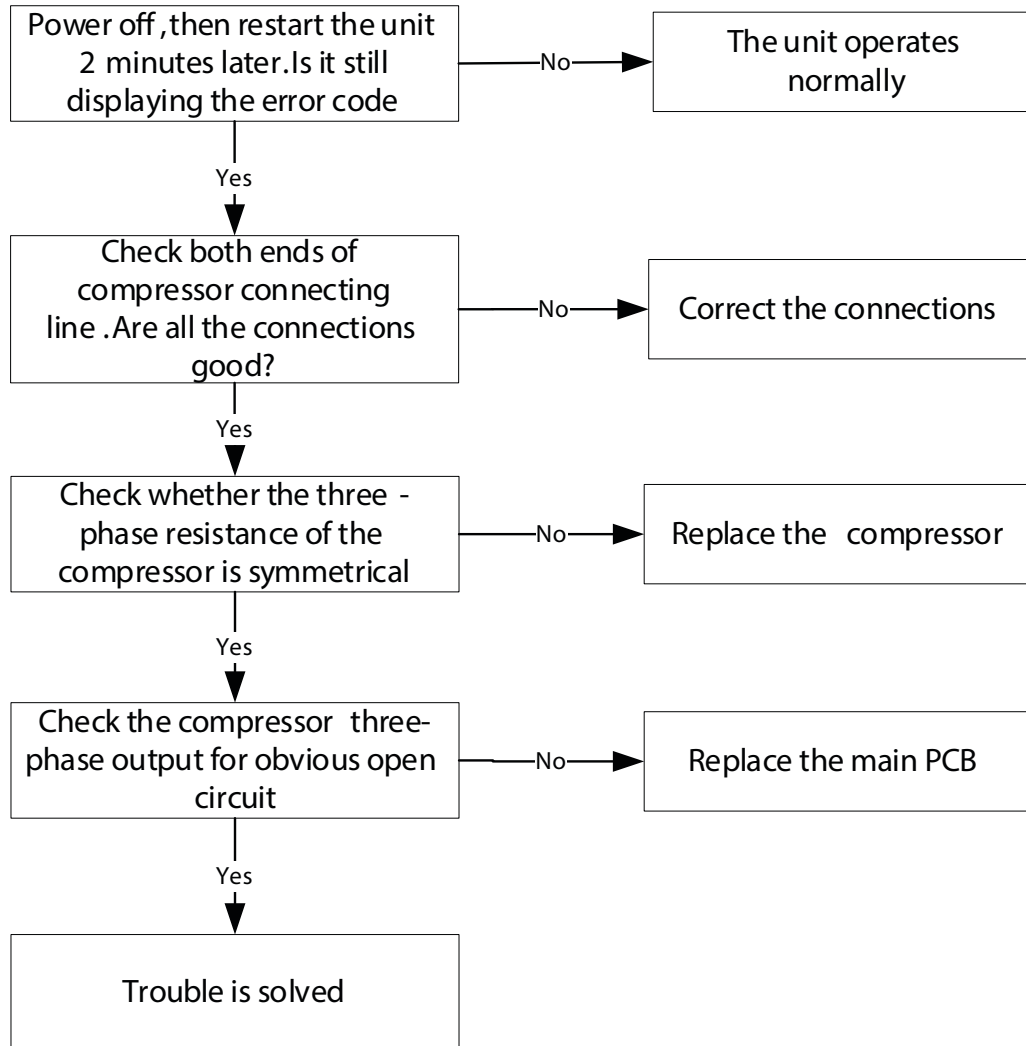


**PC 43 (ODU compressor lack phase protection diagnosis and solution)**

**Description:** When the three-phase sampling current of the compressor is abnormal, especially when the current of one or more phases is always small and almost 0, the LED displays the failure code.

**Recommended parts to prepare:** Connection wire, Compressor, Outdoor PCB

**Troubleshooting and repair:**

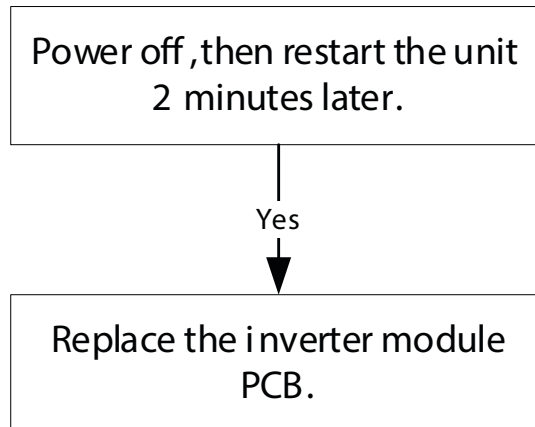


### PC 45 (ODU IR chip drive failure diagnosis and solution)

**Description:** When the IR chip detects its own parameter error, the LED displays the failure code when power on.

**Recommended parts to prepare:** Inverter module PCB.

**Troubleshooting and repair:**

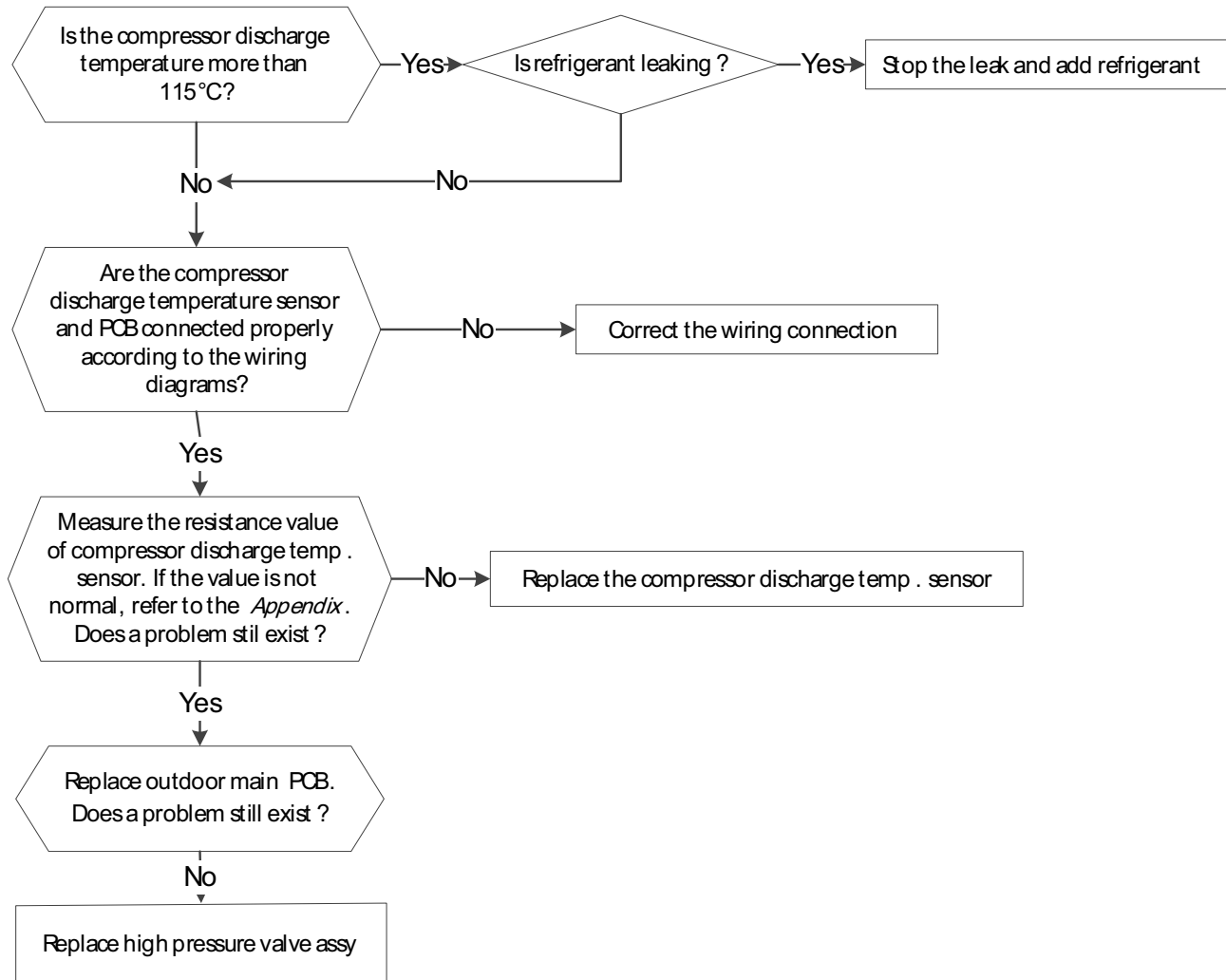


**PC 06 (Discharge temperature protection of compressor diagnosis and solution)**

**Description:** When the compressor discharge temperature (TP) is more than 115°C for 10 seconds, the compressor ceases operation and does not restart until TP is less than 90°C

**Recommended parts to prepare:** Connection wires, Outdoor PCB, Discharge temperature sensor, Refrigerant

**Troubleshooting and repair:**

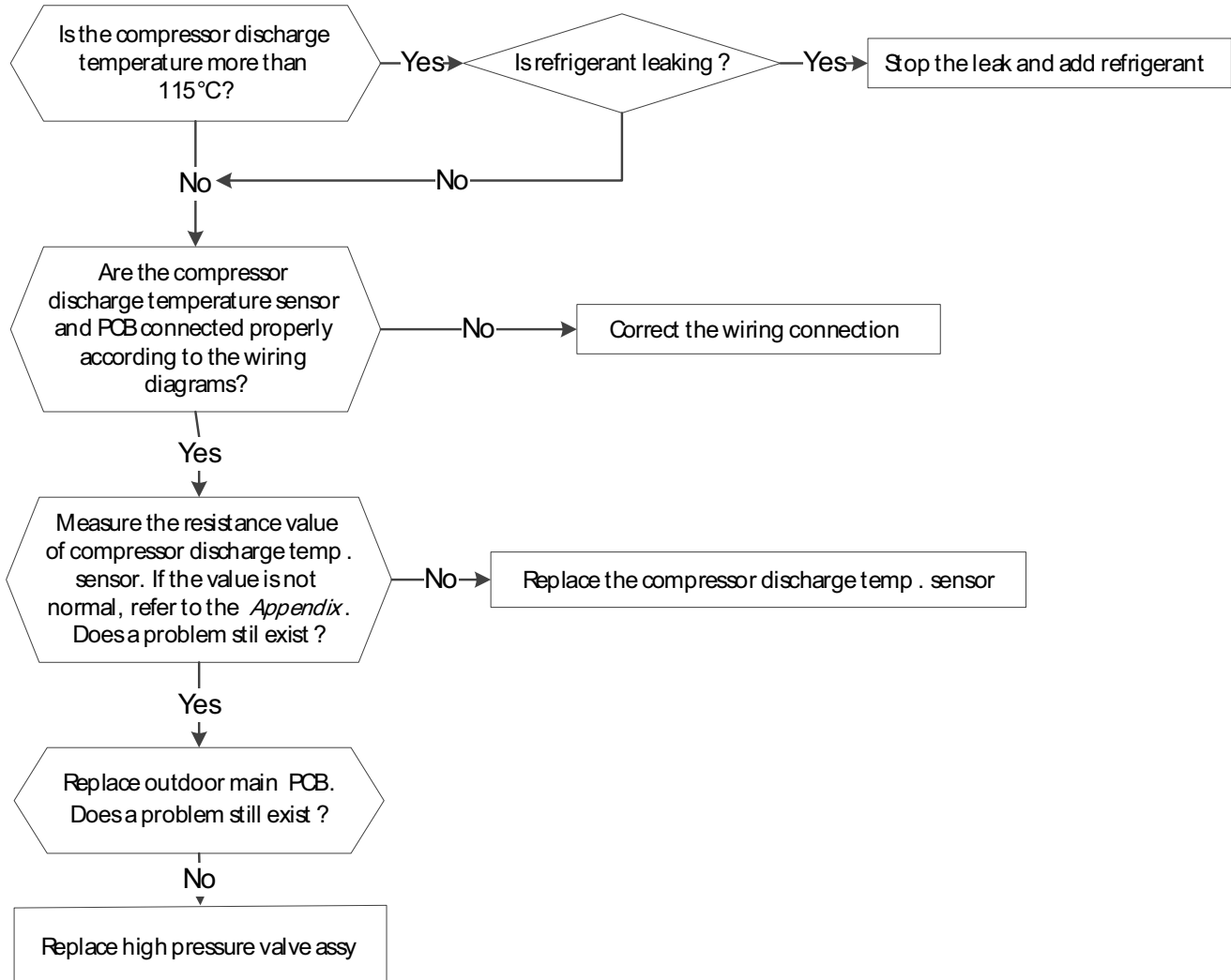


**PC 08 (Current overload protection)/ PC 42 (Compressor start failure of outdoor unit)/ PC 44 (ODU zero speed protection) / PC 46 (Compressor speed has been out of control)/ PC 49 (Compressor overcurrent failure)**

**Description:** An abnormal current rise is detected by checking the specified current detection circuit.

**Recommended parts to prepare:** Outdoor PCB, Connection wires, Bridge rectifier, PFC circuit or reactor, Refrigeration piping system, Pressure switch, Outdoor fan, PM module board

**Troubleshooting and repair:**

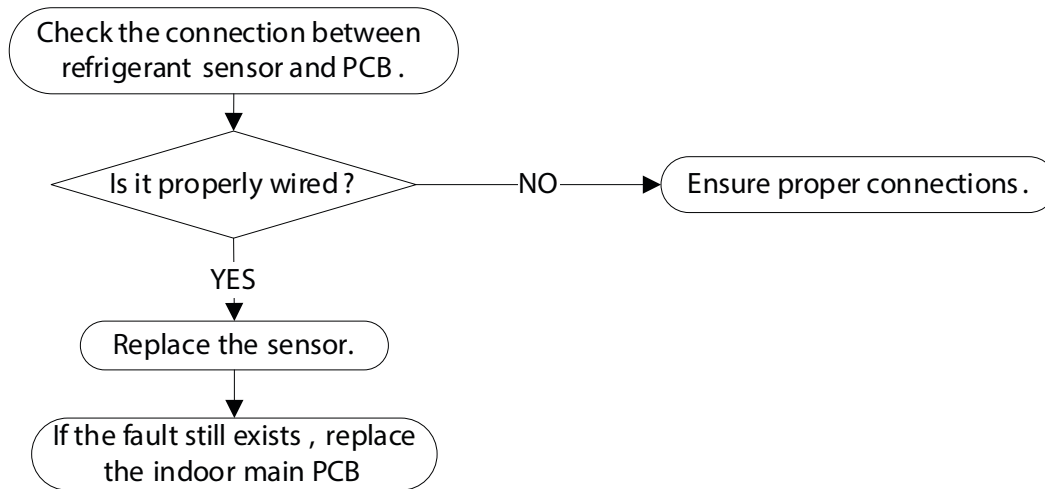


**FH CC/ EH C3 Refrigerant sensor error or Refrigerant sensor is out of range diagnosis and solution**

**Description:** Indoor unit receives fault signal for 10s or indoor unit does not receive feedback from refrigerant sensor for 150s.

**Recommended parts to prepare:** Connection wires, Sensors, Indoor main PCB

**Troubleshooting and repair:**

**EH C1/ EH C2/ EC C1 Refrigerant sensor detects leakage or Refrigerant sensor is out of range and leakage is detected diagnosis and solution**

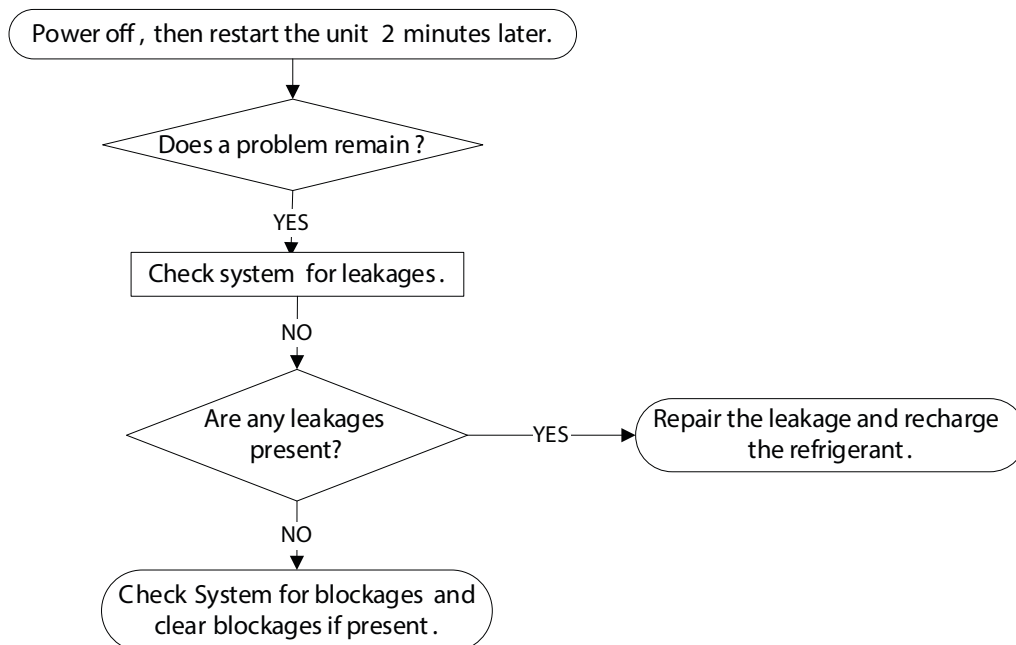
**Description:**

The refrigerant sensor detects a concentration higher than or equal to 10%\*LFL for 10 seconds or the refrigerant sensor detects a concentration higher than or equal to 20%\*LFL or the multi model receives the refrigerant leakage protection fault sent by the outdoor unit.

Multi-zone: Only the buzzer of the indoor unit that detects refrigerant leakage continues to sound the alarm, the shortest sound is 10 seconds, and the longest sound is 5 minutes (you can press any key such as remote control or wire control, APP and so on to eliminate the alarm), and the other non-refrigerant leakage fault indoor unit only displays ECC1, but the buzzer does not sound.

**Recommended parts to prepare:** Additional refrigerant

**Troubleshooting and repair:**

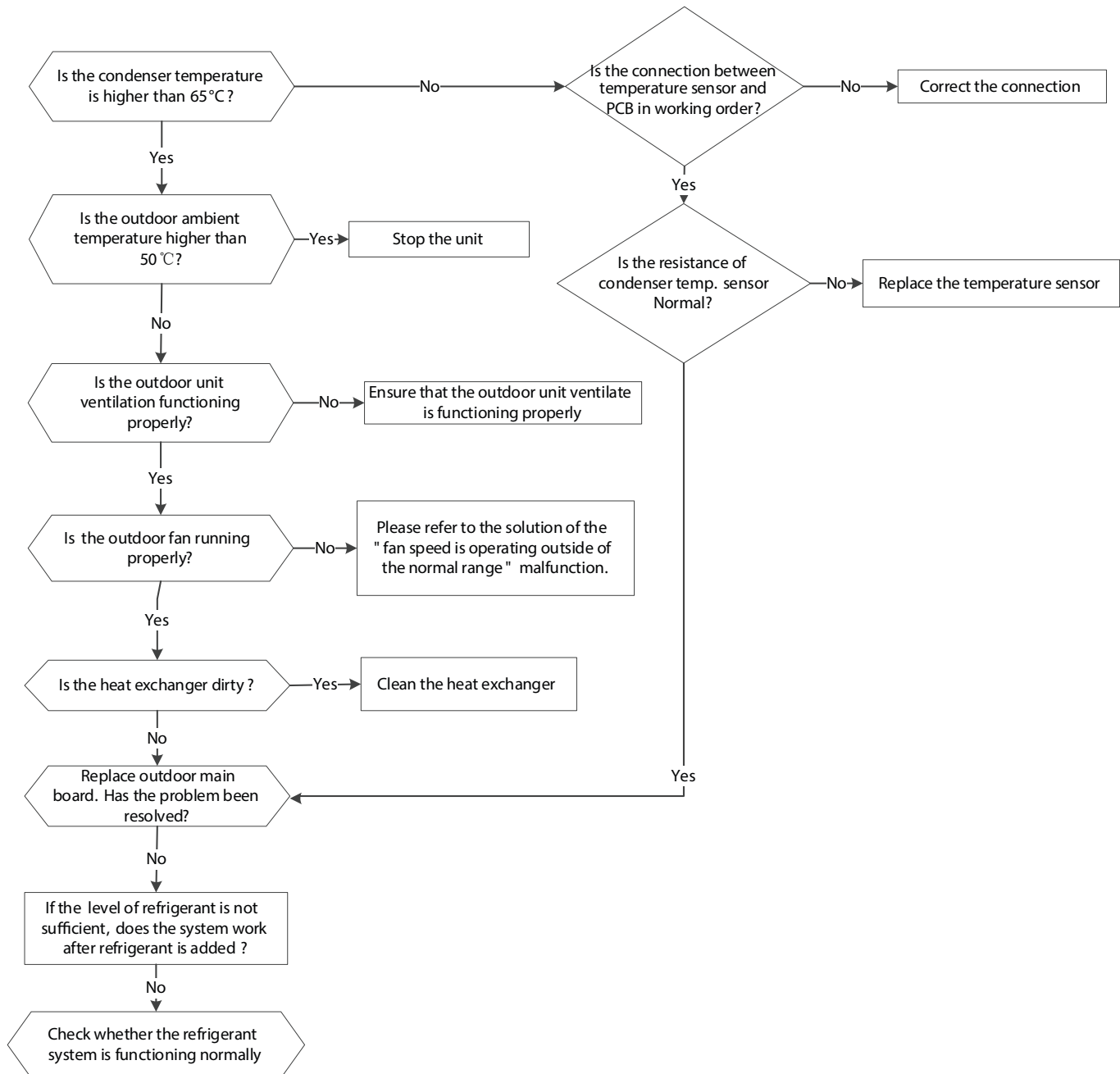


## PC 0A (High temperature protection of condenser diagnosis and solution)

**Description:** The unit will stop when condenser temperature is higher than 65°C, and runs again when it is less than 52°C

**Recommended parts to prepare:** Connection wires, Condenser temperature sensor, Outdoor fan, Outdoor main PCB, Refrigerant

**Troubleshooting and repair:**

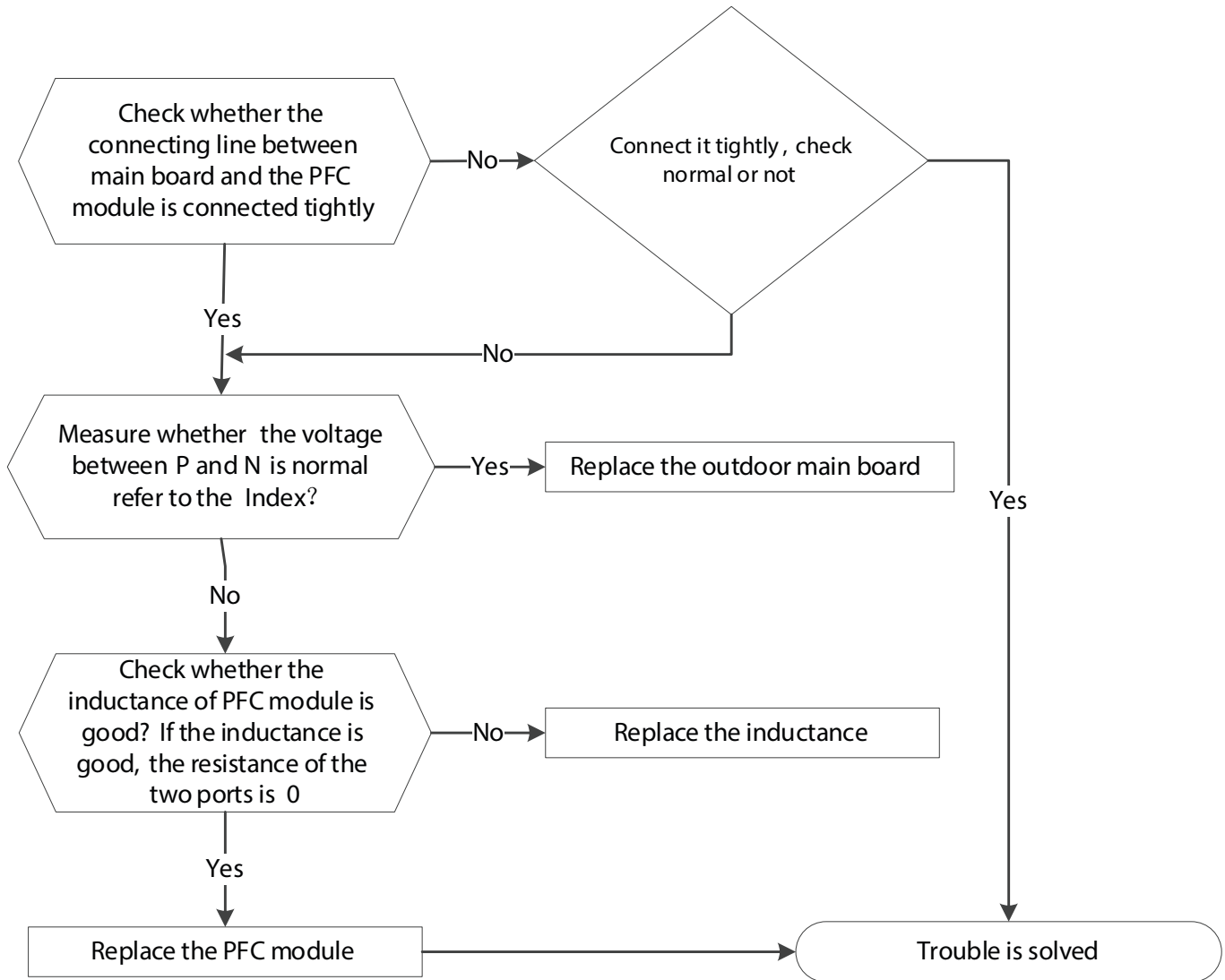


**PC 0F (PFC module protection diagnosis and solution)**

**Description:** Outdoor PCB detects PFC signal is low voltage or DC voltage is lower than 340V for 6s when quick check.

**Recommended parts to prepare:** Connection wires Outdoor PCB, Inductance, PFC circuit or IPM module board

**Troubleshooting and repair:**

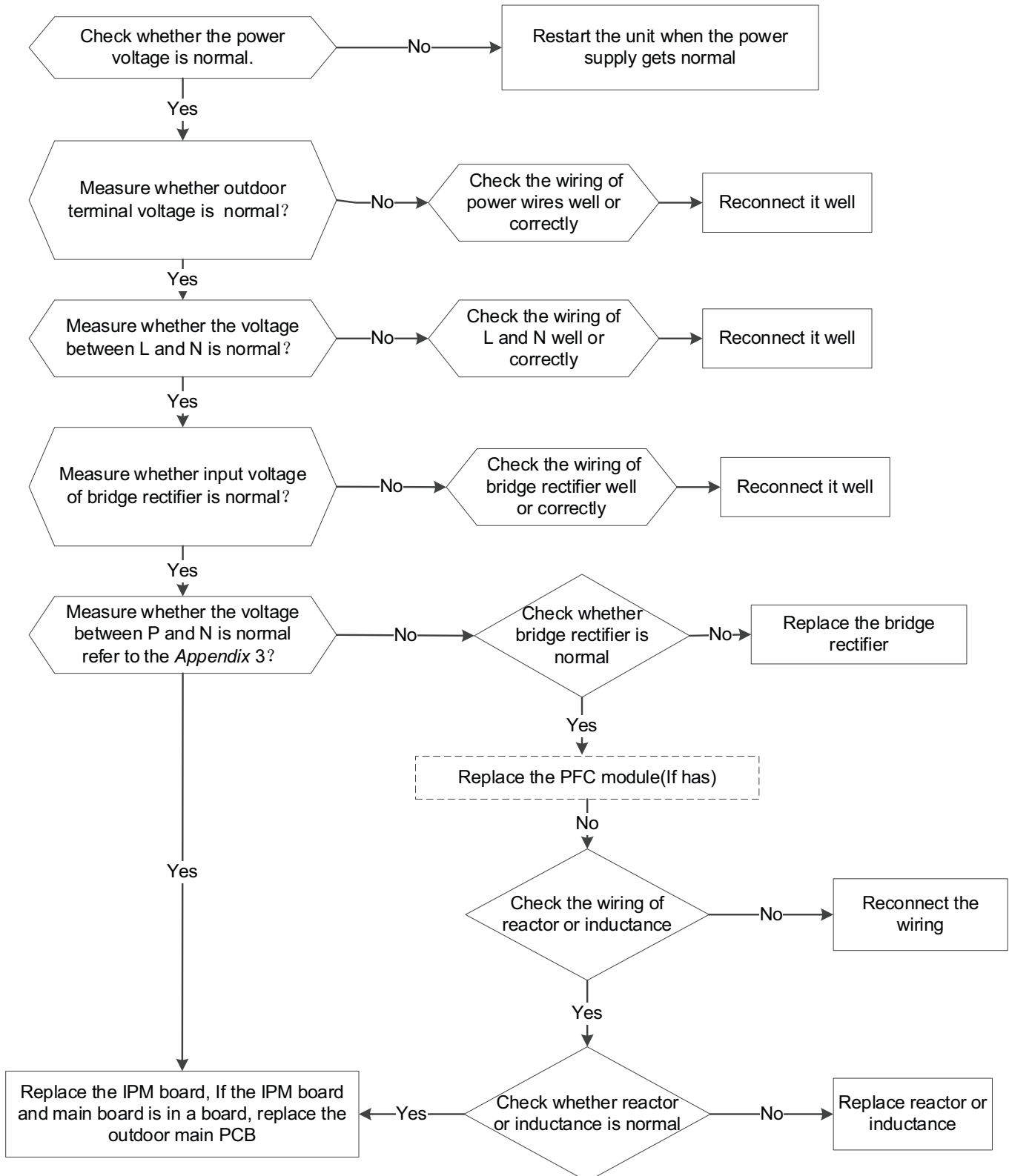


## PC 10 (ODU low AC voltage protection)/ PC 11 (ODU main control board DC bus high voltage protection)/ PC 12 (ODU main control board DC bus high voltage protection /341 MCE error) Diagnosis and Solution

**Description:** Abnormal increases or decreases in voltage are detected by checking the specified voltage detection circuit.

**Recommended parts to prepare:** Power supply wires, IPM module board, Outdoor PCB, Bridge rectifier, PFC circuit or reactor

**Troubleshooting and repair:**



## PC 0L (Low Ambient Temperature Protection)

**Description:** It is a protection function. When compressor is off, outdoor ambient temperature(T4) is lower than -35oC. for 10s, the AC will stop and display the failure code.

When compressor is on, outdoor ambient temperature(T4) is lower than -40oC.for 10s, the AC will stop and display the failure code.

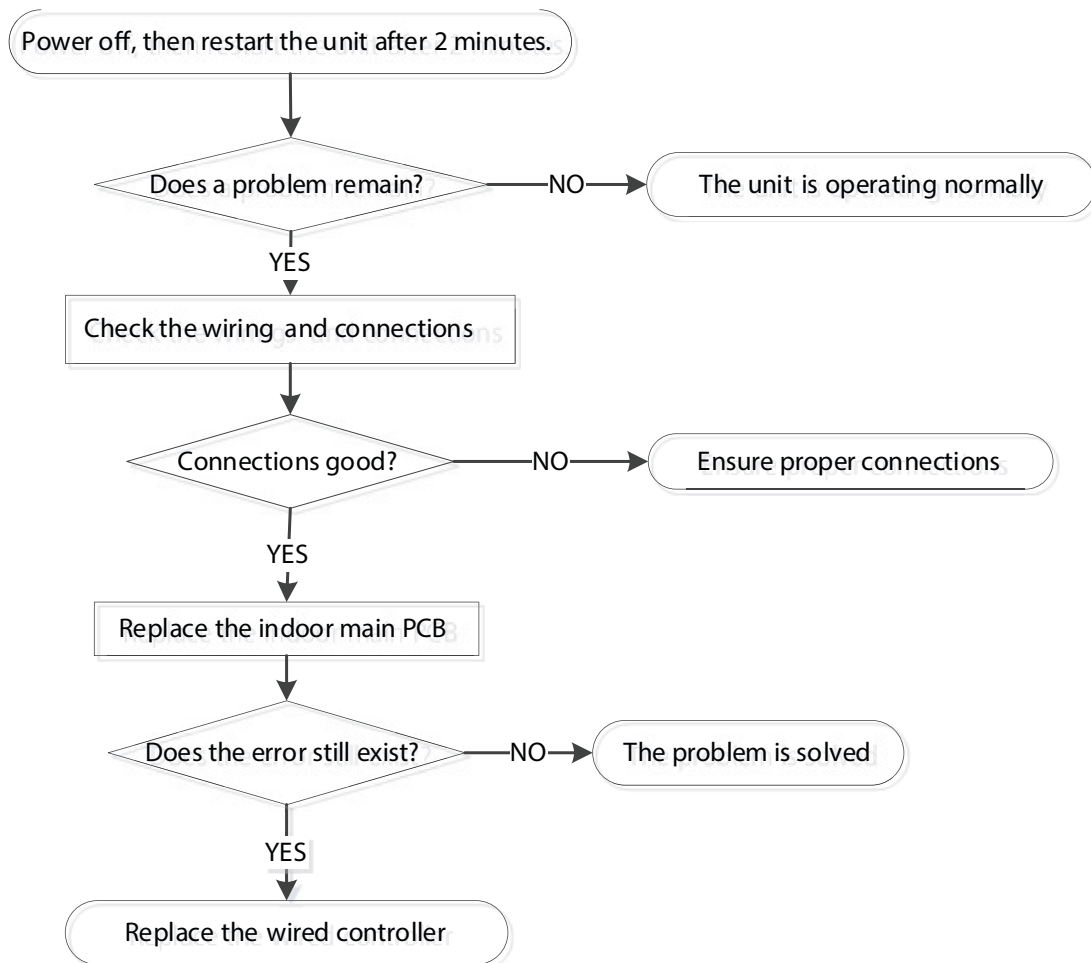
When outdoor ambient temperature(T4) is no lower than -32oC.for 10s, the unit will exit protection

## EH b3 (Communication Malfunction Between Wire and Master Control) Diagnosis and Solution

**Description:** If Indoor PCB does not receive feedback from wired controller, the error displays on the wired controller

**Recommended parts to prepare:** Connection wires, Indoor PCB, Wired controller

**Troubleshooting and repair:**

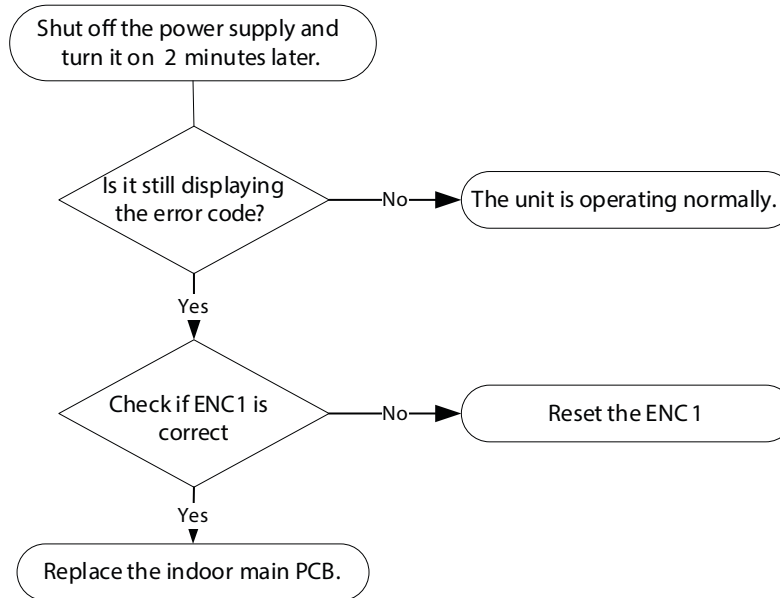


## EH bA (Communication Malfunction Between Indoor Unit and External Fan Module)/ EH 3A(External Fan DC Bus Voltage Is Too Low Protection)/ EH 3b (External Fan DC Bus Voltage is Too High) Fault) Diagnosis and Solution

**Description:** Indoor unit does not receive the feedback from external fan module during 150 seconds. or Indoor unit receives abnormal increases or decreases in voltage from external fan module.

**Recommended parts to prepare:** Indoor main PCB

**Troubleshooting and repair:**

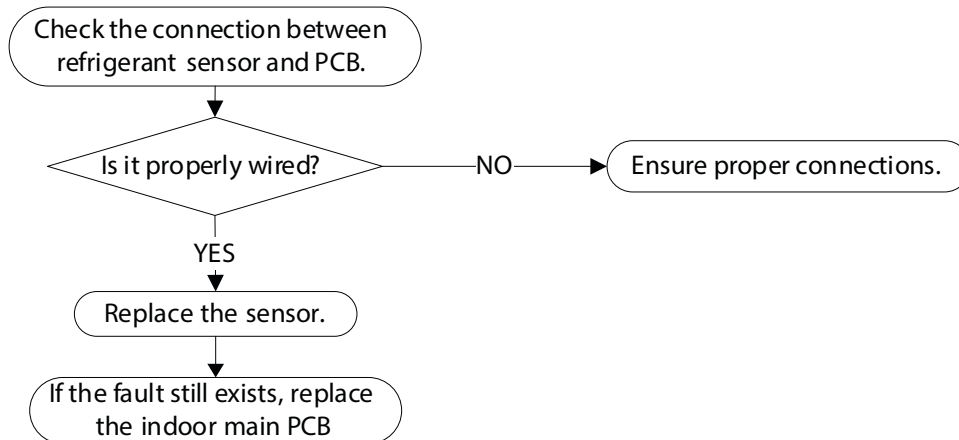


## FH CC (Refrigerant Sensor Error) or EH C3(Refrigerant Sensor is Out of Range) Diagnosis and Solution

**Description:** Indoor unit receives fault signal for 10s or indoor unit does not receive feedback from refrigerant sensor for 150s.

**Recommended parts to prepare:** Connection wires, Sensors, Indoor main PCB

**Troubleshooting and repair:**



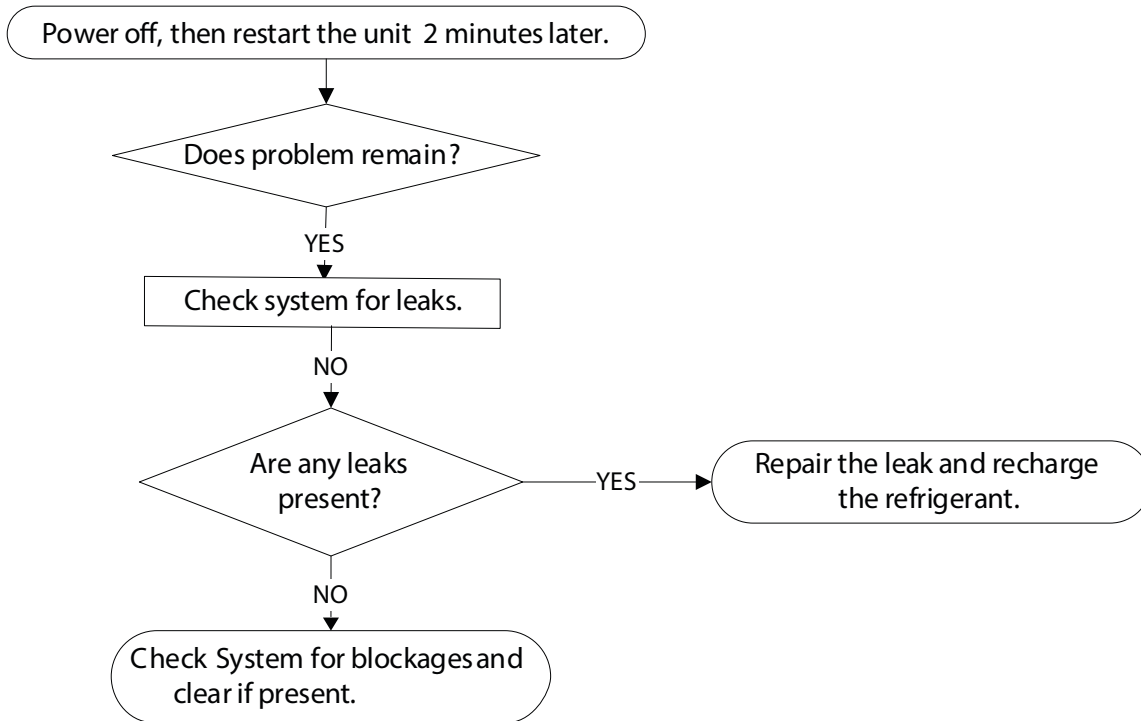
## EH C1 (Refrigerant Sensor Detects Leakage) or EH C2 (Refrigerant Sensor is Out of Range and Leakage is Detected) Diagnosis and Solution

**Description:** The refrigerant sensor detects a concentration higher than or equal to 10%\*LFL for 10 seconds or the refrigerant sensor detects a concentration higher than or equal to 20%\*LFL or the multi model receives the refrigerant leakage protection fault sent by the outdoor unit.

**Multi-zone:** Only the buzzer of the indoor unit that detects refrigerant leakage continues to sound the alarm, the shortest sound is 10 seconds, and the longest sound is 5 minutes (you can press any key such as remote control or wire control, APP and so on to eliminate the alarm), and the other non-refrigerant leakage fault indoor unit only displays "ECC1", but the buzzer does not sound.

**Recommended parts to prepare:** Additional refrigerant

**Troubleshooting and repair:**

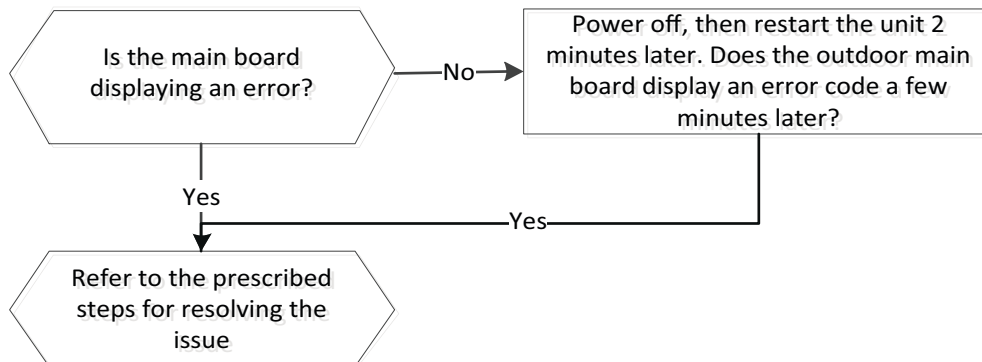


## EC 0d (ODU Malfunction Diagnosis and Solution)

**Description:** The indoor unit detects the outdoor unit in error.

**Recommended parts to prepare:** Outdoor unit

**Troubleshooting and repair:**

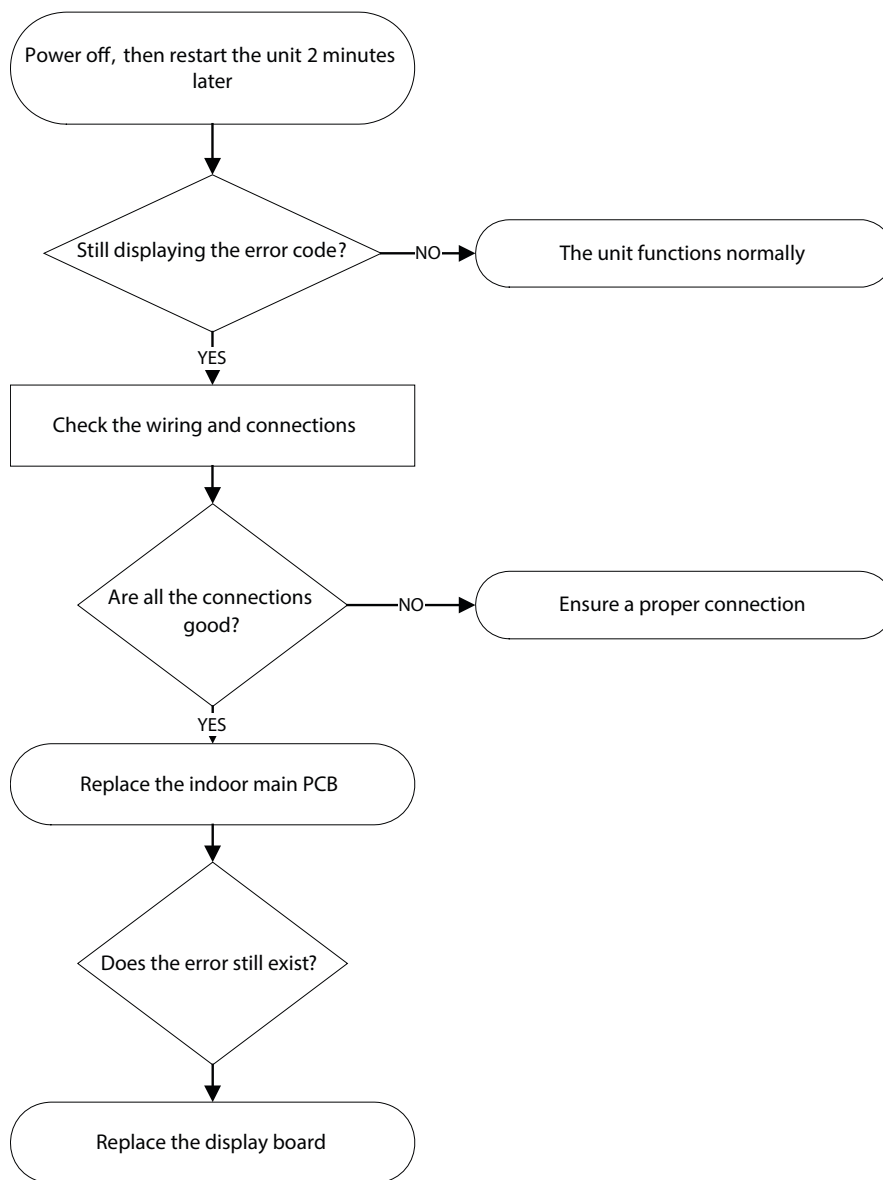


**EH 06 (IDU Main Control Board and Display Board Communication Error Diagnosis and Solution)**

**Description:** Indoor PCB does not receive feedback from the display board.

**Recommended parts to prepare:** Communication wire, Indoor PCB, Display board

**Troubleshooting and repair:**

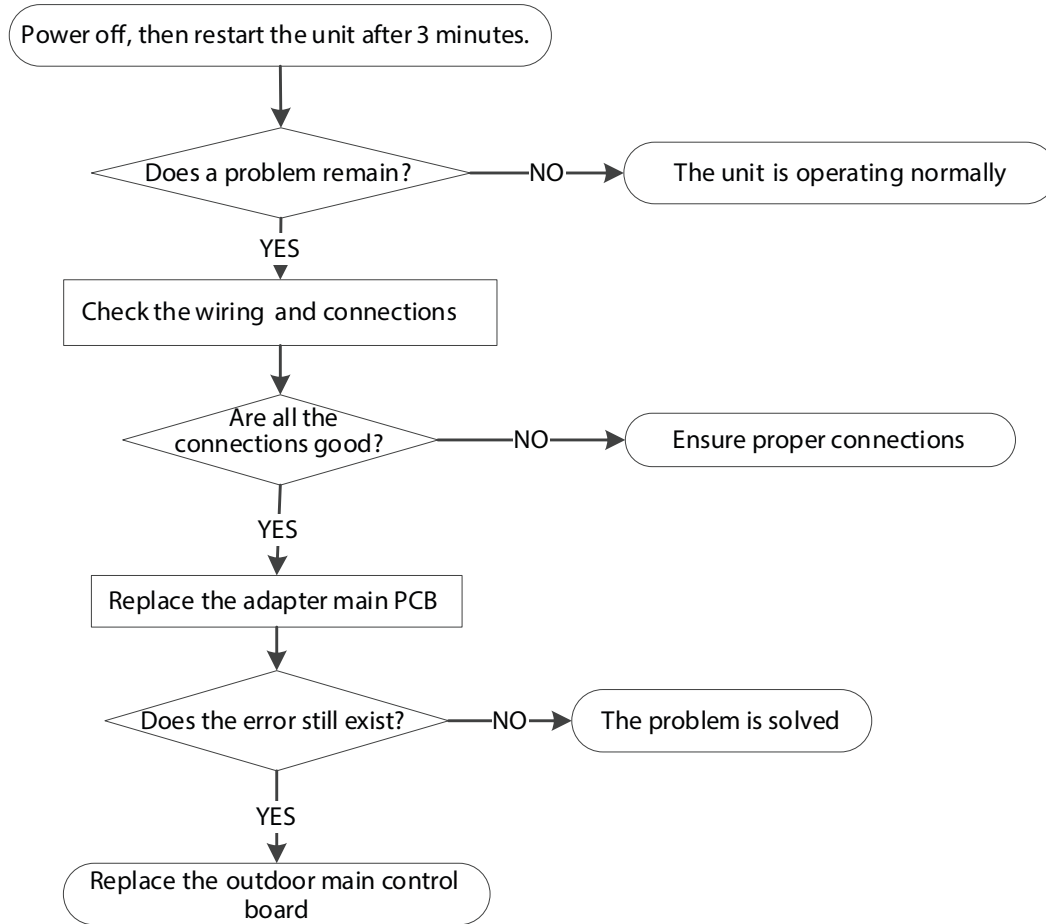


## EL 16 (Communication Malfunction Between Adapter Board and Outdoor Main Board Diagnosis and Solution)

**Description:** The adapter PCB cannot detect the main control board.

**Recommended parts to prepare:** Connection wires, Adapter board, Outdoor main PCB

**Troubleshooting and repair:**



## FL 09 (Mismatch between the new and old platforms diagnosis and solution)

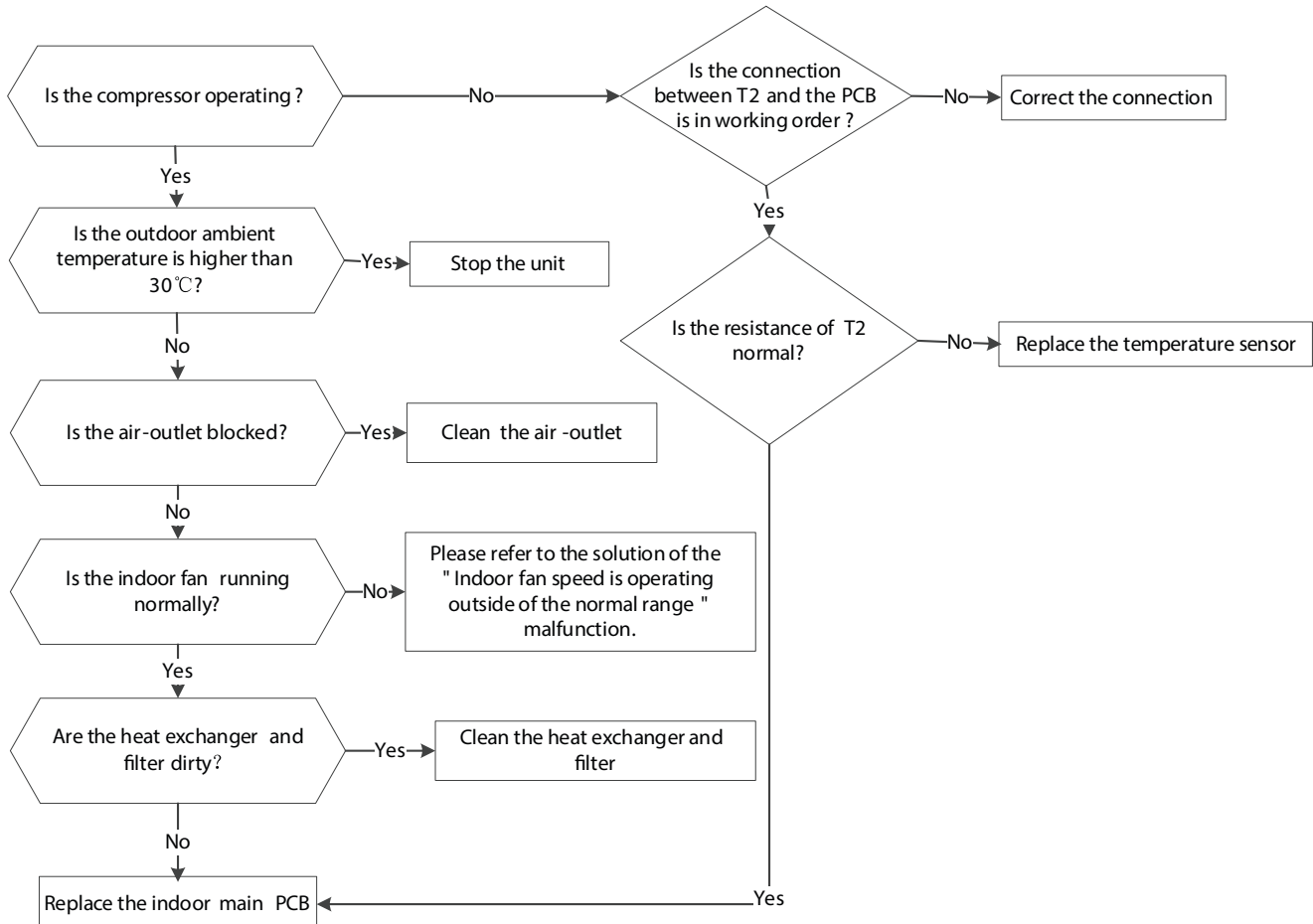
**Description:** Indoor and outdoor units are mismatched, the LED displays this code. Please replace the matching indoor or outdoor unit.

## PH 90 (High temperature protection of evaporator diagnosis and solution)

**Description:** When evaporator coil temperature is more than 60°C in heating mode, the unit stops. It starts again only when the evaporator coil temperature is less than 52°C.

**Recommended parts to prepare:** Connection wires, Evaporator coil temperature sensor (T2), Indoor fan, Indoor main PCB

### Troubleshooting and repair:

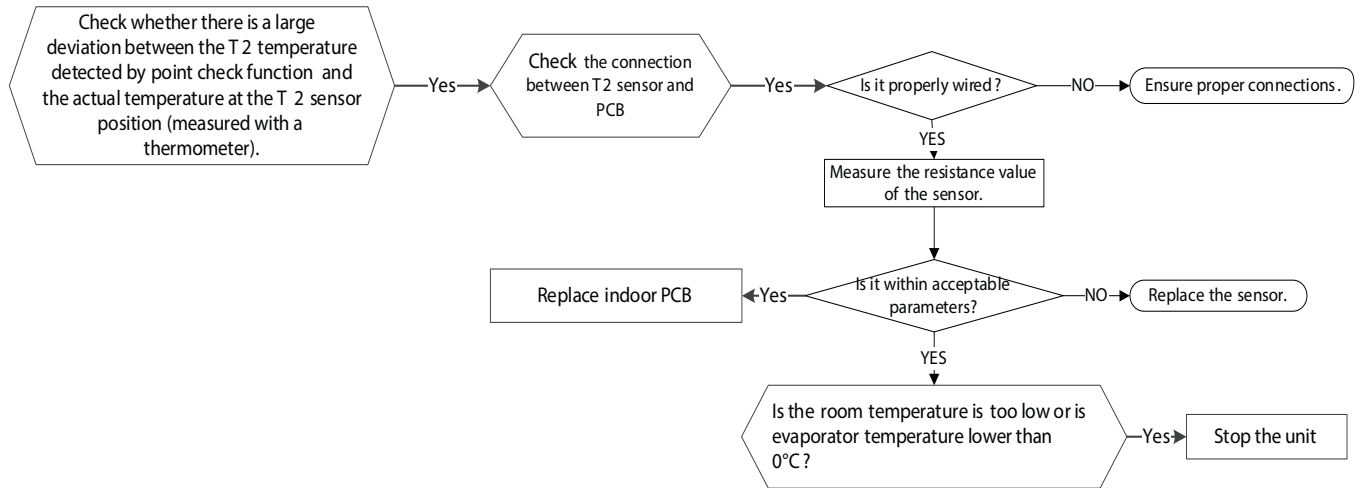


## PH 91 (Low temperature protection of evaporator diagnosis and solution)

**Description:** When evaporator coil temperature is lower than 0°C in cooling mode or drying mode, the unit stops. It starts again only when the evaporator coil temperature is more than 5°C.

**Recommended parts to prepare:** Connection wires, Evaporator coil temperature sensor (T2), Indoor main PCB

### Troubleshooting and repair:



## CHECK PROCEDURES

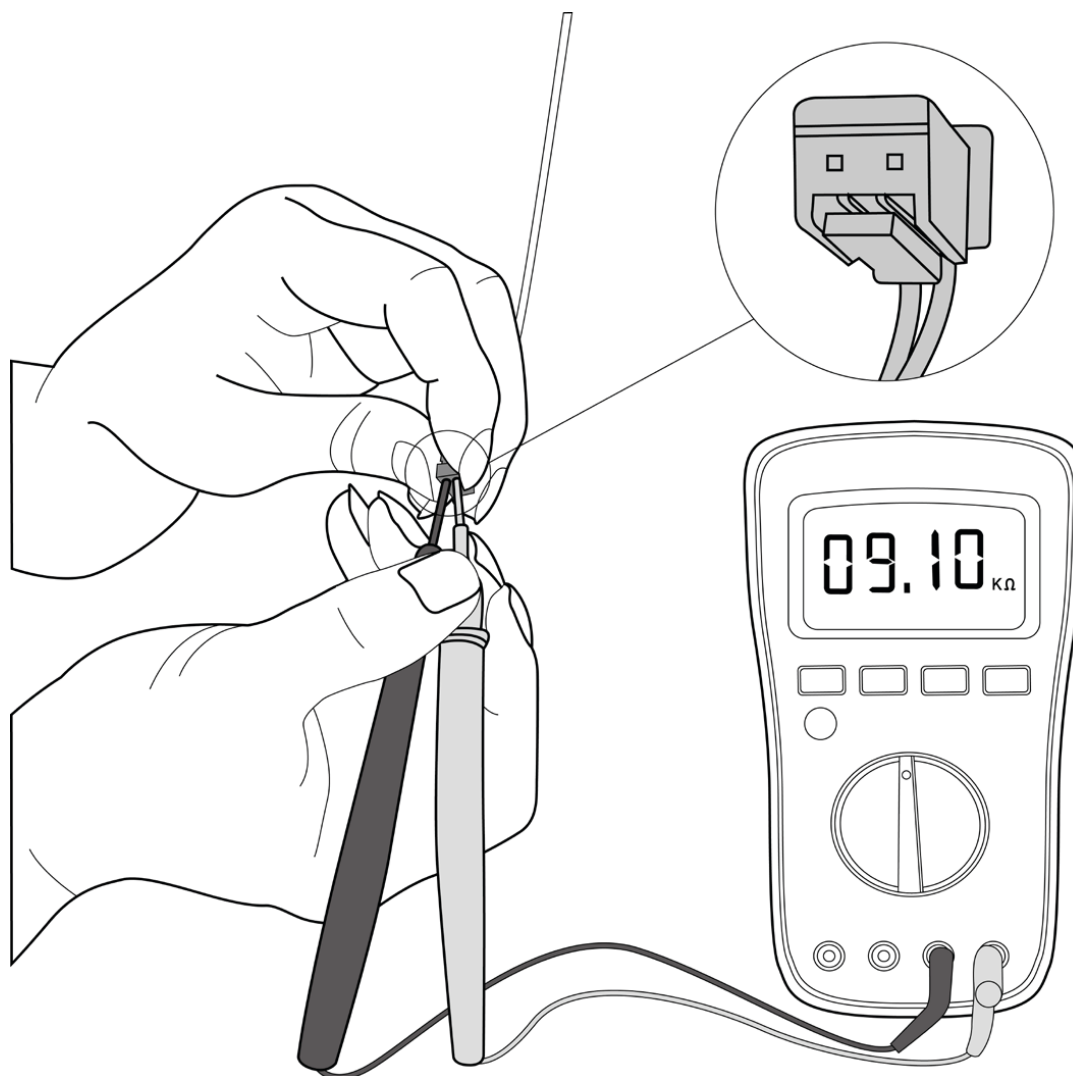
### Temperature Sensor Check

#### WARNING

##### ELECTRICAL SHOCK HAZARD

Be sure to turn off all power supplies or disconnect all wires to avoid electric shock. Operate after compressor and coil have returned to normal temperature in case of injury.

1. Disconnect the temperature sensor from the PCB.
2. Measure the sensor's resistance value with a multi-meter.
3. Check the corresponding temperature sensor resistance value table (see "Temperature Sensor Resistance Value Table for TP (°C - K)" on page 87 and "Other Temperature Sensors Resistance Value Table (°C - K)" on page 88).



**Fig. 33 —Measure the Sensor's Resistance Value**

Compressor Check

- 1. Disconnect the compressor power cord from the outdoor PCB.
- 2. Measure the resistance value of each winding using a multi-meter.
- 3. Check the resistance value of each winding in tables 9 through 12:

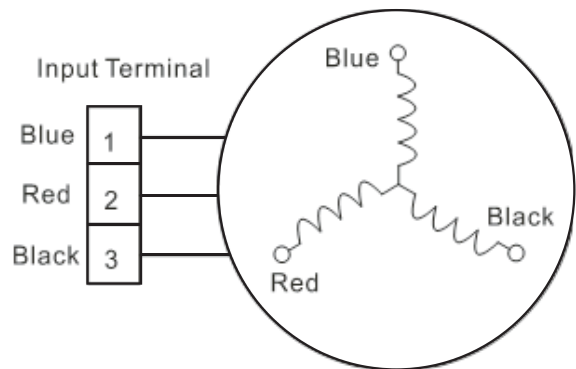


Fig. 34 —Compressor Check

Table 3 – Resistance Values

| Resistance Value | KSN140D58UFZ | KTF250D22UMT | KTM240D46UKT2 | KTF310D43UMT | MTH550UKPC8FU |
|------------------|--------------|--------------|---------------|--------------|---------------|
| Blue-Red         | 1.86Ω        | 0.75Ω        | 1.04Ω         | 0.65Ω        | 0.295Ω        |
| Blue-Black       |              |              |               |              |               |
| Red-Black        |              |              |               |              |               |

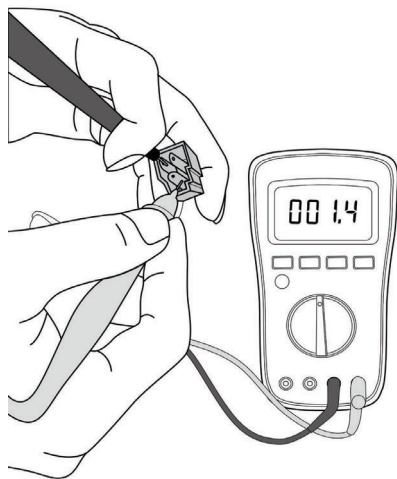


Fig. 35 —Resistance Check

**NOTE:** The picture and the value are only for reference, actual condition and specific value may vary.

IPM Continuity Check

⚠

WARNING

ELECTRICAL SHOCK HAZARD

Electricity remains in capacitors even when the power supply is off.  
Ensure the capacitors are fully discharged before troubleshooting.

1. Turn off outdoor unit and disconnect power supply.
2. Discharge electrolytic capacitors and ensure all energy-storage unit has been discharged.
3. Disassemble outdoor PCB or disassemble IPM board.
4. Measure the resistance value between P and U(V, W, N); U(V, W) and N.

Table 4 – Resistance Value

| Digital Tester |           | Resistance Value          | Digital Tester |           | Resistance Value          |
|----------------|-----------|---------------------------|----------------|-----------|---------------------------|
| (+) Red        | (-) Black | $\infty$<br>(Several MfÇ) | (+) Red        | (-) Black | $\infty$<br>(Several MfÇ) |
| P              | N         |                           | U              | N         |                           |
|                | U         |                           | V              |           |                           |
|                | V         |                           | W              |           |                           |
|                | W         |                           | -              |           |                           |

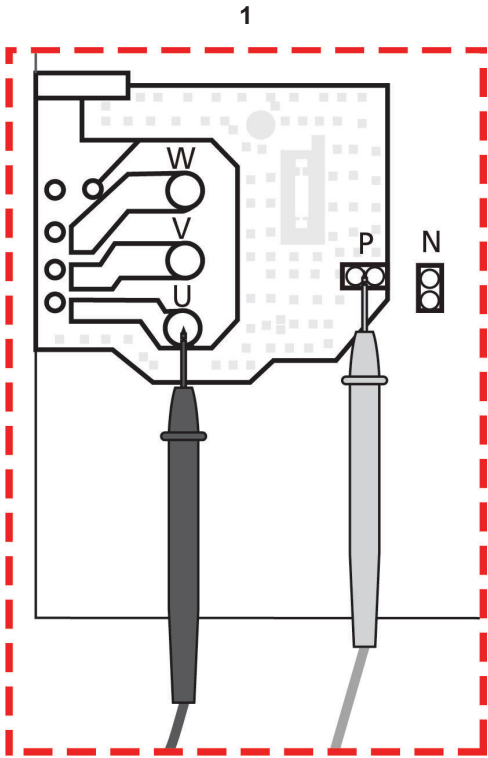


Fig. 36 —Resistance Value

Table 5 – Voltage Range

| 208-240V (1-phase)      |                                |                              |
|-------------------------|--------------------------------|------------------------------|
| In Standby              |                                |                              |
| Around 310VDC           |                                |                              |
| In Operation            |                                |                              |
| With passive PFC module | With partial active PFC module | With fully active PFC module |
| >200VDC                 | >310VDC                        | >370VDC                      |

4-Way Valve Check

- 1. Power on, use a digital tester to measure the voltage, when the unit operates in cooling, it is 0V. When the unit operates in heating, it is about equal to power supply voltage.  
If the value of the voltage is not in the range, the PCB must have problems and need to be replaced.

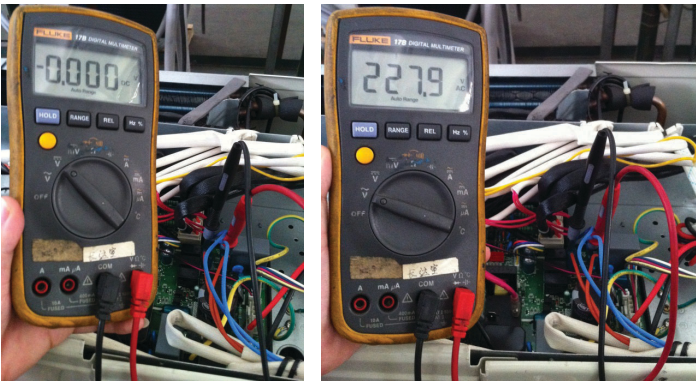


Fig. 37 —Measure the Voltage

- 2. Turn off the power, use a digital tester to measure the resistance. The value should be 1.8~2.5 KΩ.

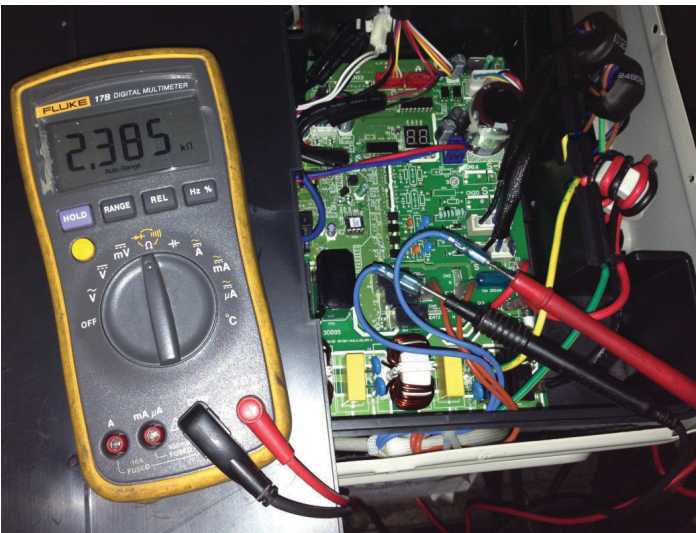


Fig. 38 —Use a Digital Tester to Measure Resistance

EXV Check

**WARNING**

**ELECTRICAL SHOCK HAZARD**  
Electricity remains in the capacitors even when the power is off.  
Ensure the capacitors are fully discharged before troubleshooting.

- 1. Turn off outdoor unit and disconnect power supply.
- 2. Disconnect the connectors of EXV.
- 3. Measure the resistance value between Red and Blue (Yellow); Brown and Orange (White).

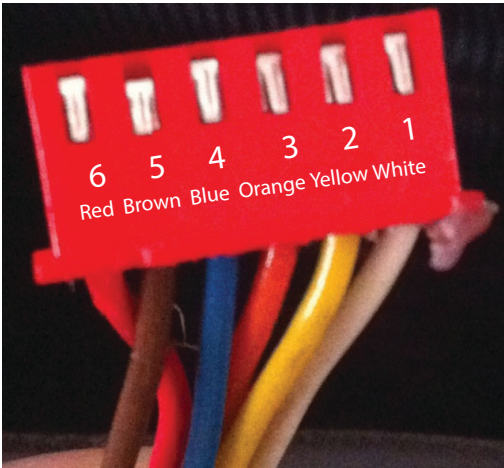


Fig. 39 —EXV Check

Resistance to EXV coil

| Color of Lead Wire | Normal Value |
|--------------------|--------------|
| Red-Blue           | About 500 Ω  |
| Red-Yellow         |              |
| Brown-Orange       |              |
| Brown-White        |              |

Main Parts Check

- 1. Temperature sensor checking  
Disconnect the temperature sensor from PCB, measure the resistance value with a tester.

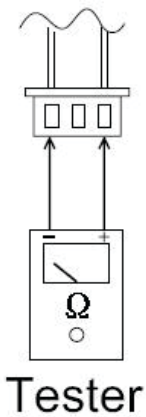
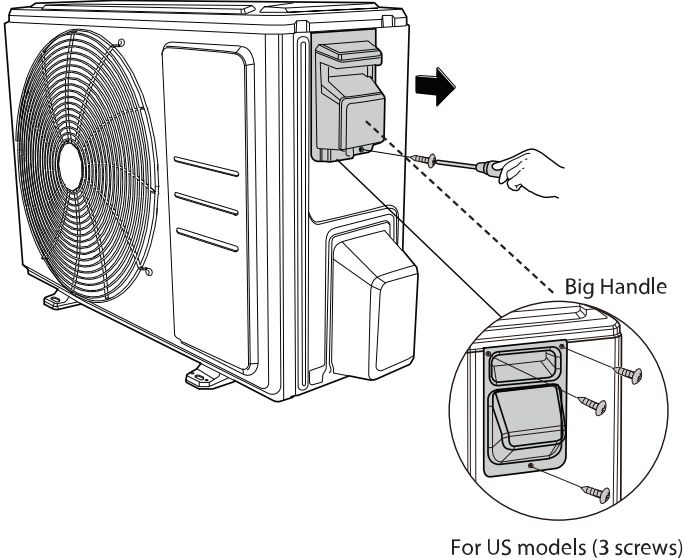
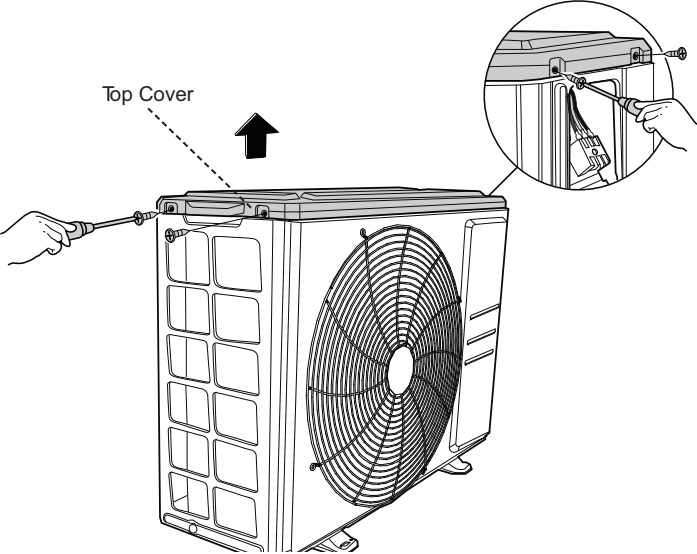
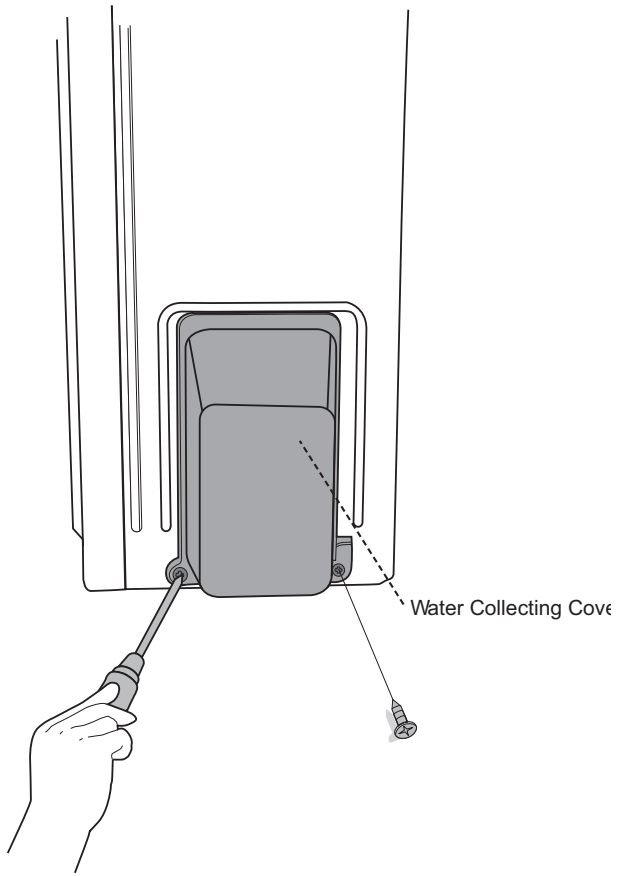
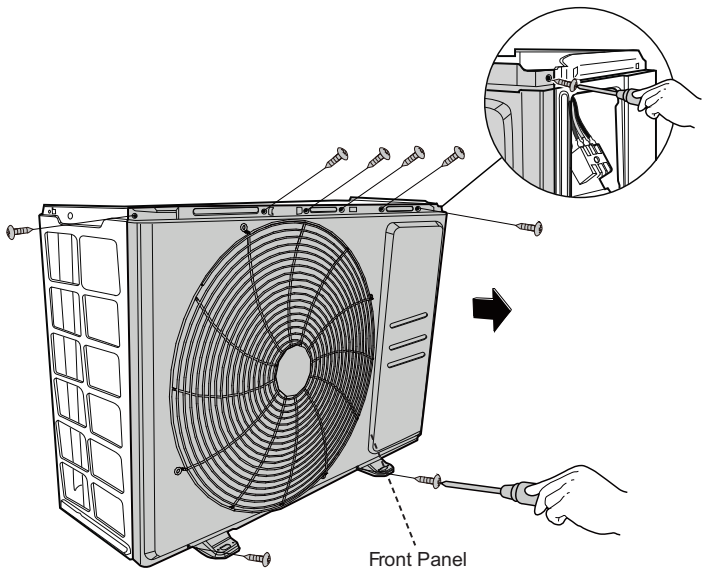


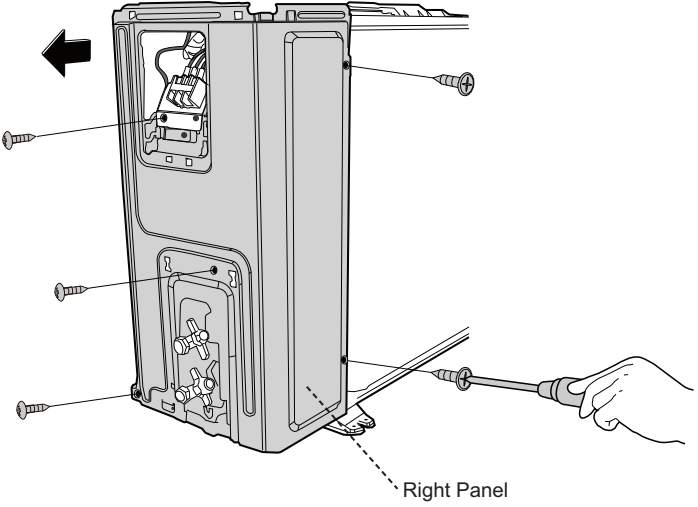
Fig. 40 —Sensor Test

- 2. Temperature sensors
  - a. Room temp. (T1) sensor,
  - b. Indoor coil temp. (T2) sensor,
  - c. Outdoor coil temp. (T3) sensor,
  - d. Outdoor ambient temp. (T4) sensor,
  - e. Compressor discharge temp. (T5) sensor.
  - f. Measure the resistance value of each winding by using the multi-meter.

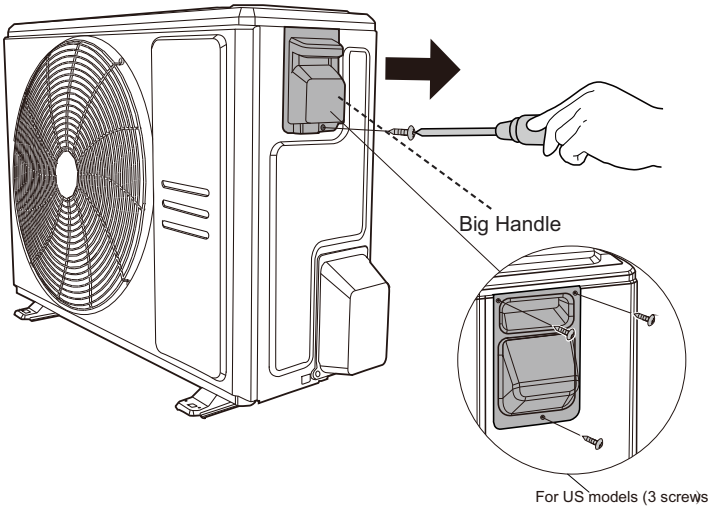
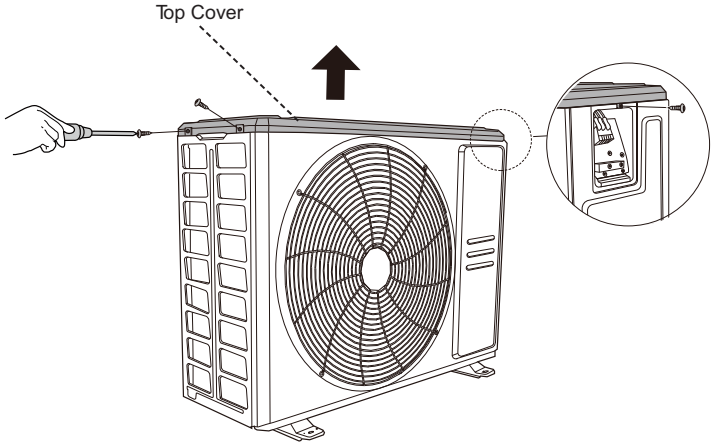
**DISASSEMBLY INSTRUCTIONS****6K - 12K Unit Disassembly - Panel Plate**

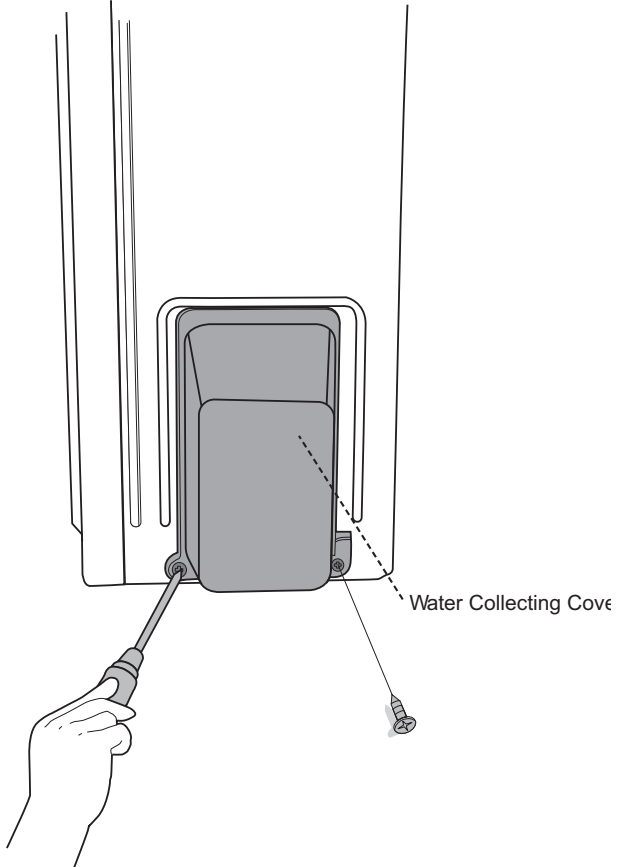
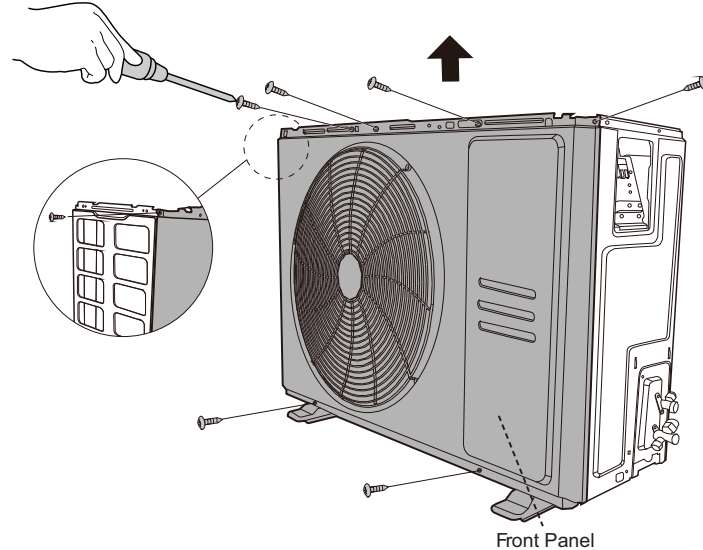
| PROCEDURES                                                                                                                                                             | ILLUSTRATION                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Turn off the air conditioner and the power breaker</p> <p>2. Remove the screw of the big handle and then remove the big handle (3 screws) (see illustration)</p> |  <p>Big Handle</p> <p>For US models (3 screws)</p> |
| <p>3. Remove the screws of the top cover and then remove the top cover (4 screws). One of the screws is located underneath the big handle. (see illustration)</p>      |  <p>Top Cover</p>                                 |

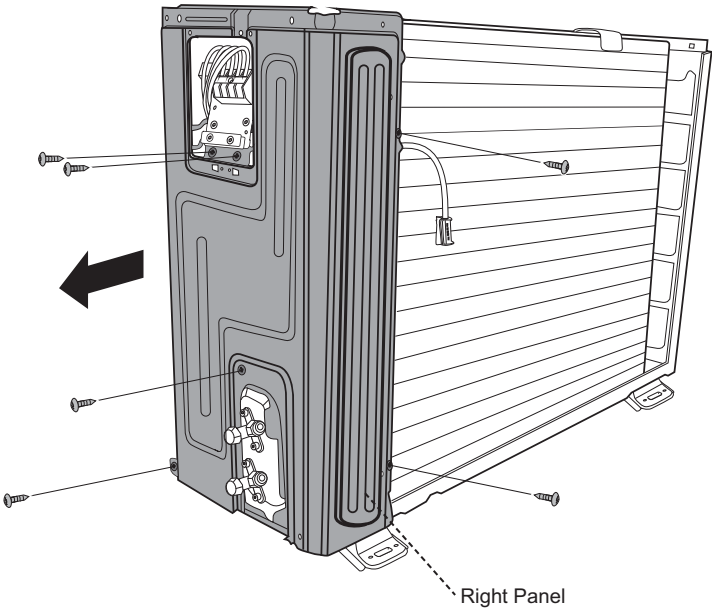
| PROCEDURES                                                                                                                                                  | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>4. Remove the screws of water collecting cover and then remove the water collecting cover (2 screws).(see illustration)</p>                              |  <p>The illustration shows a hand using a screwdriver to remove a screw from the bottom of the water collecting cover. A dashed line points to the cover, which is labeled "Water Collecting Cover".</p>                                                                 |
| <p>5. Remove the screws of the front panel and then remove the front panel (7 screws (on/off models) or 9 screws. (inverter models). (see illustration)</p> |  <p>The illustration shows a hand using a screwdriver to remove a screw from the front panel. A dashed line points to the panel, which is labeled "Front Panel". An inset circular image shows a close-up of the top of the unit where the panel is being detached.</p> |

| PROCEDURES                                                                                                   | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>6. Remove the screws of the right panel and then remove the right panel (5 screws) (see illustration)</p> |  <p>The illustration shows a side view of a rectangular electronic device. A hand is using a screwdriver to remove screws from the right panel. A dashed line indicates the 'Right Panel' being removed. An arrow points to the left, indicating the direction of removal.</p> |

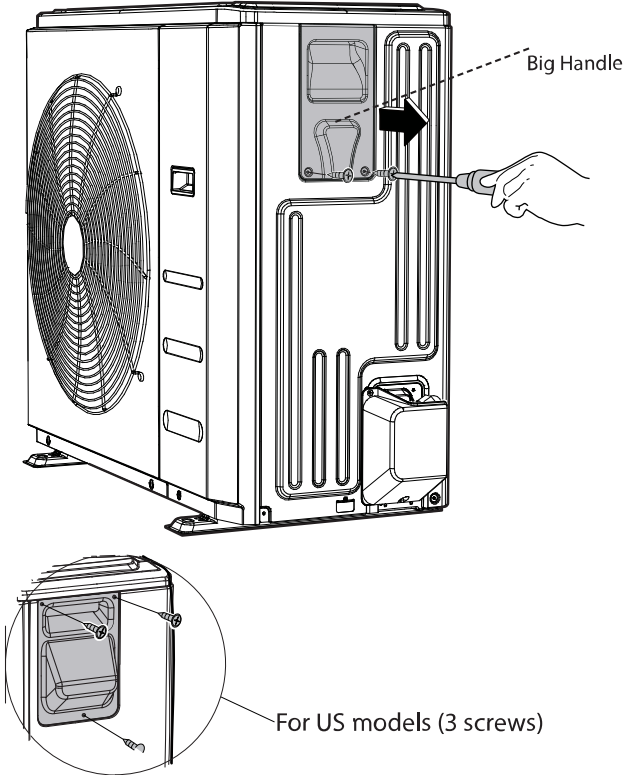
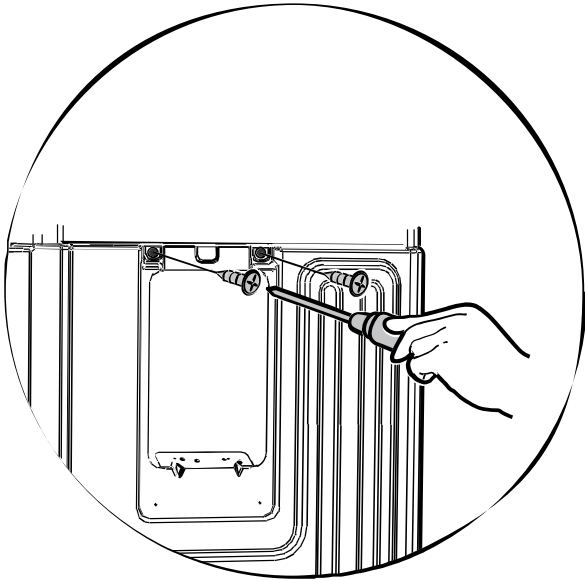
## 18K Unit Disassembly - Panel Plate

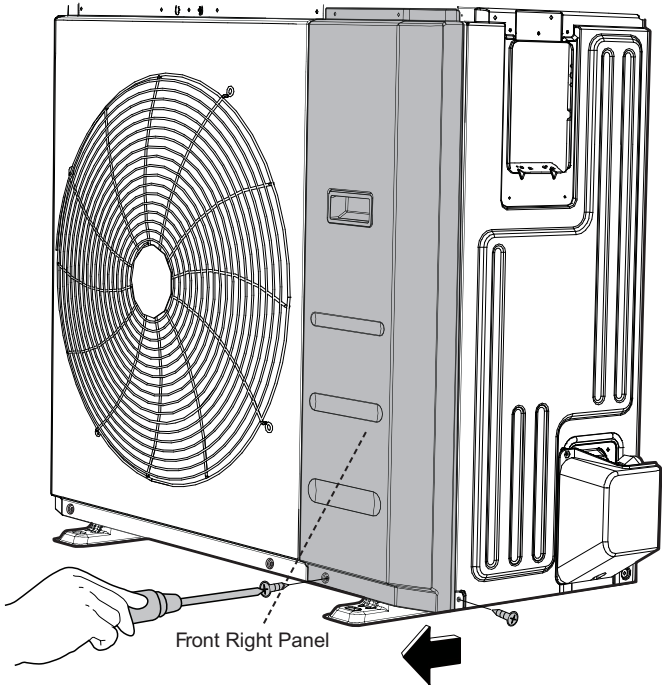
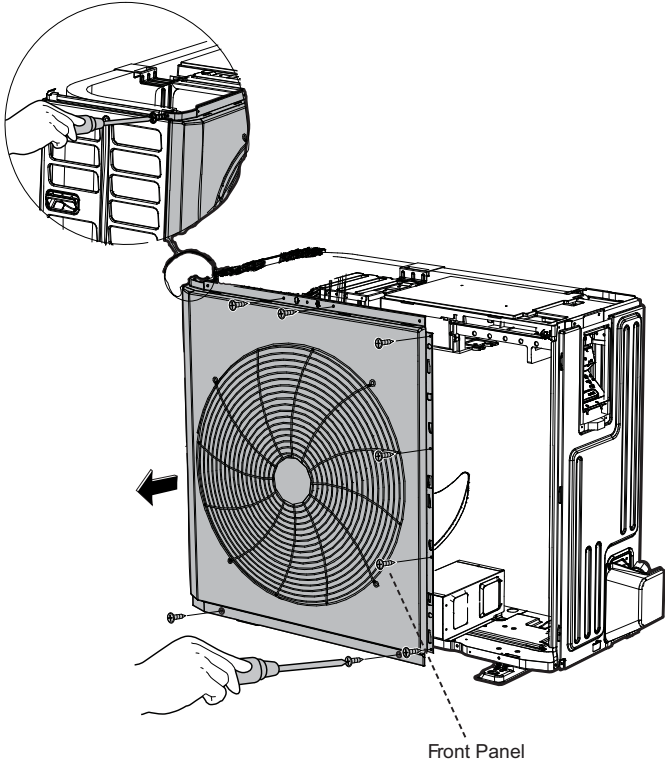
| PROCEDURES                                                                                                                                                             | ILLUSTRATION                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Turn off the air conditioner and the power breaker</p> <p>2. Remove the screw of the big handle and then remove the big handle (3 screws) (see illustration)</p> |  <p>Big Handle</p> <p>For US models (3 screws)</p> |
| <p>3. Remove the screws of the top cover and then remove the top cover (3 screws). One of the screws is located underneath the big handle. (see illustration)</p>      |  <p>Top Cover</p>                                 |

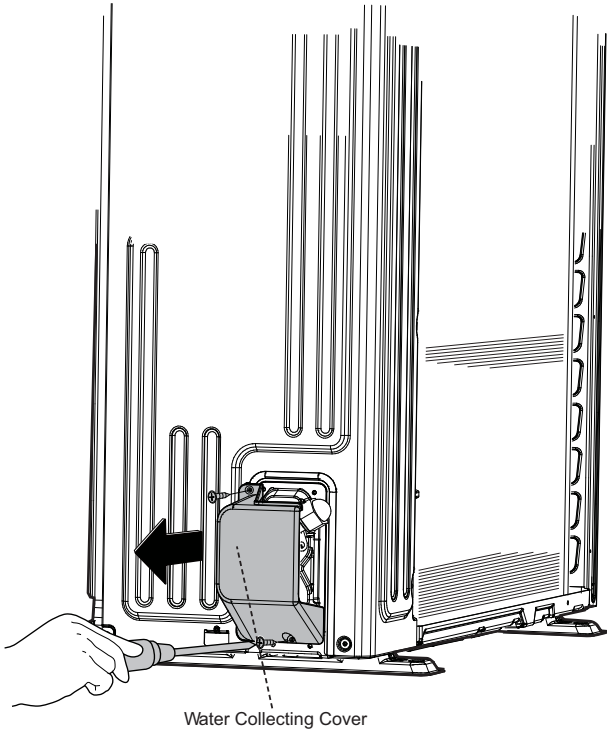
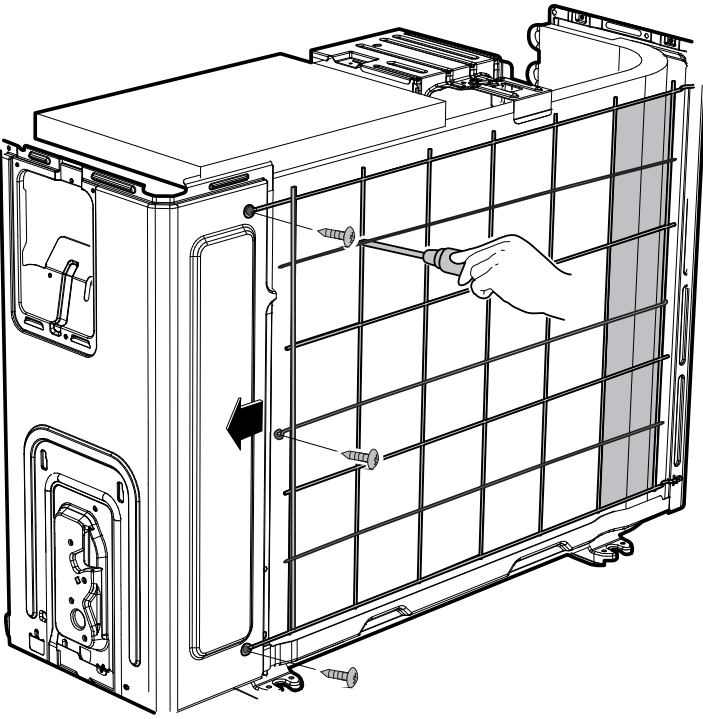
| PROCEDURES                                                                                                                                                  | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>4. Remove the screws of water collecting cover and then remove the water collecting cover (2 screws).(see illustration)</p>                              |  <p>The illustration shows a hand using a screwdriver to remove a screw from the bottom left of a rectangular water collecting cover. The cover is mounted on the side of an outdoor unit. A dashed line points to the cover, and a label 'Water Collecting Cove' is next to it. Another screw is shown being removed from the bottom right.</p> |
| <p>5. Remove the screws of the front panel and then remove the front panel (7 screws (on/off models) or 9 screws. (inverter models). (see illustration)</p> |  <p>The illustration shows a hand using a screwdriver to remove a screw from the top of the front panel of an outdoor unit. A dashed line points to the panel, and a label 'Front Panel' is next to it. An inset shows the front panel with a grid of screws. An upward arrow indicates the direction to lift the panel.</p>                    |

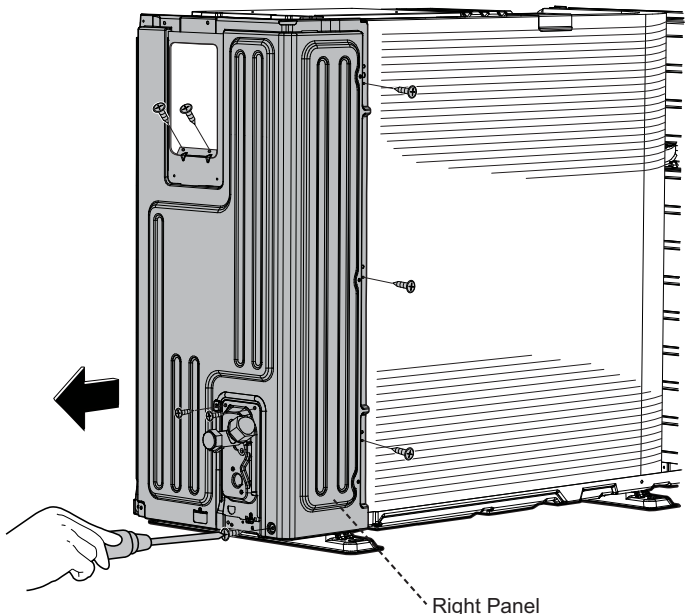
| PROCEDURES                                                                                                                              | ILLUSTRATION                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <p data-bbox="99 457 675 510">6. Remove the screws of the right panel and then remove the right panel (5 screws) (see illustration)</p> |  <p data-bbox="1219 743 1325 766">Right Panel</p> |

**24K - 33K Unit Disassembly - Panel Plate**

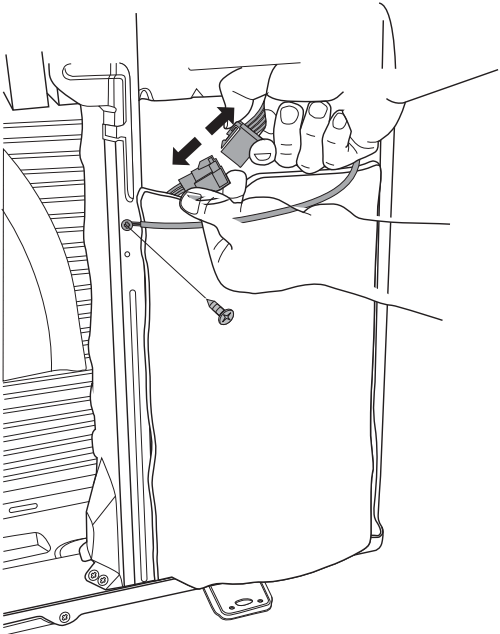
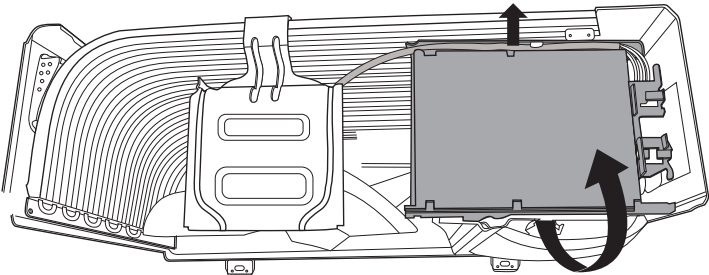
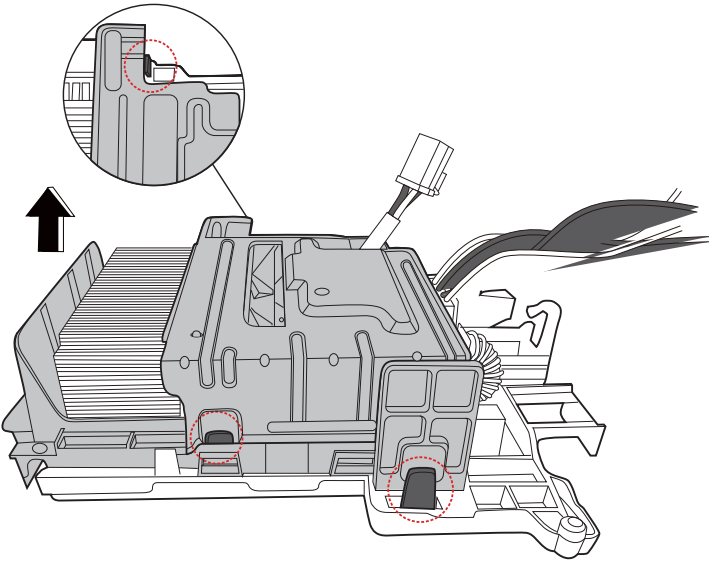
| PROCEDURES                                                                                                                                                             | ILLUSTRATION                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Turn off the air conditioner and the power breaker</p> <p>2. Remove the screw of the big handle and then remove the big handle (2 screws) (see illustration)</p> |  <p>Big Handle</p> <p>For US models (3 screws)</p> |
| <p>3. Remove the screws of the top cover and then remove the top cover (4 screws). One of the screws is located underneath the big handle. (see illustration)</p>      |                                                  |

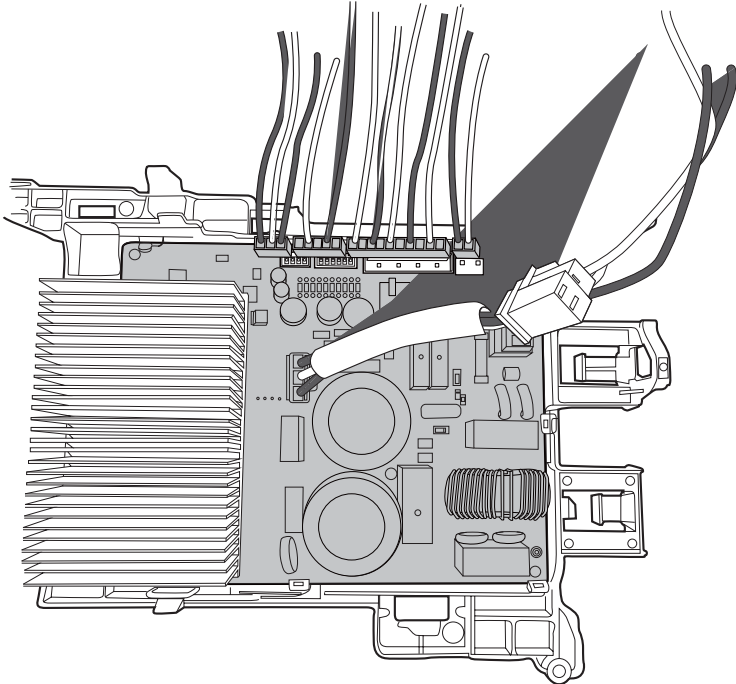
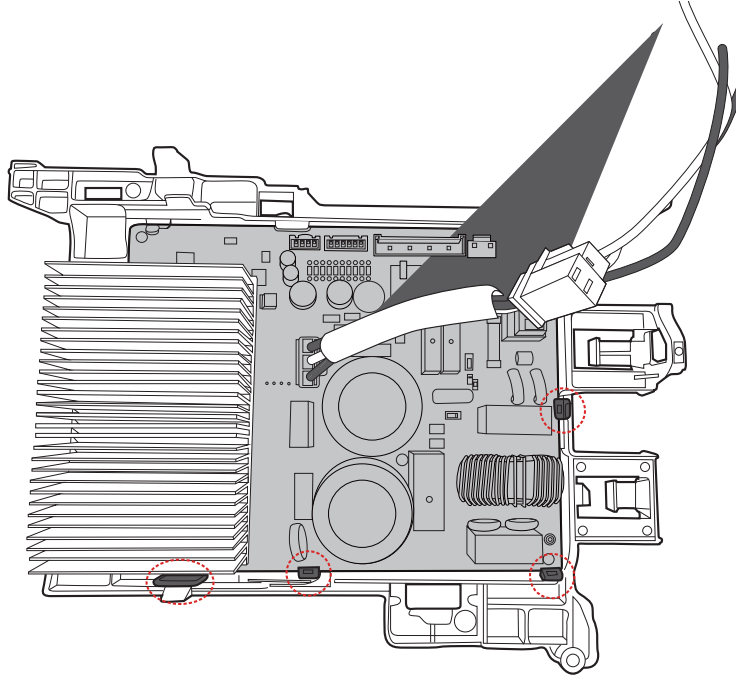
| PROCEDURES                                                                                                               | ILLUSTRATION                                                                         |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p>4. Remove the screws of the front right panel and then remove the front right panel (2 screws).(see illustration)</p> |    |
| <p>5. Remove the screws of the front panel and then remove the front panel (9 screws). (see illustration)</p>            |  |

| PROCEDURES                                                                                                                     | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>6. Remove the screws of water collecting cover and then remove the water collecting cover (2 screws) (see illustration)</p> |  <p>The illustration shows the rear of a vertical unit with a condenser coil. A hand is using a screwdriver to remove screws from a rectangular 'Water Collecting Cover' located below the coil. A dashed line points to the cover, and a large black arrow indicates its removal from the unit.</p> |
| <p>7. For some models, remove the screws of the rear net and then remove the rear net (3 screws) (see illustration)</p>        |  <p>The illustration shows the rear of the unit with a grid-like 'rear net' covering the condenser coils. A hand is using a screwdriver to remove screws from the net. A large black arrow points to the net, indicating its removal from the unit.</p>                                            |

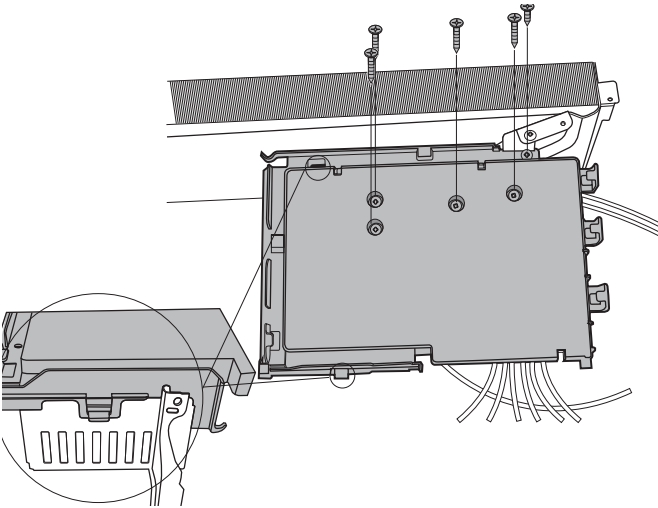
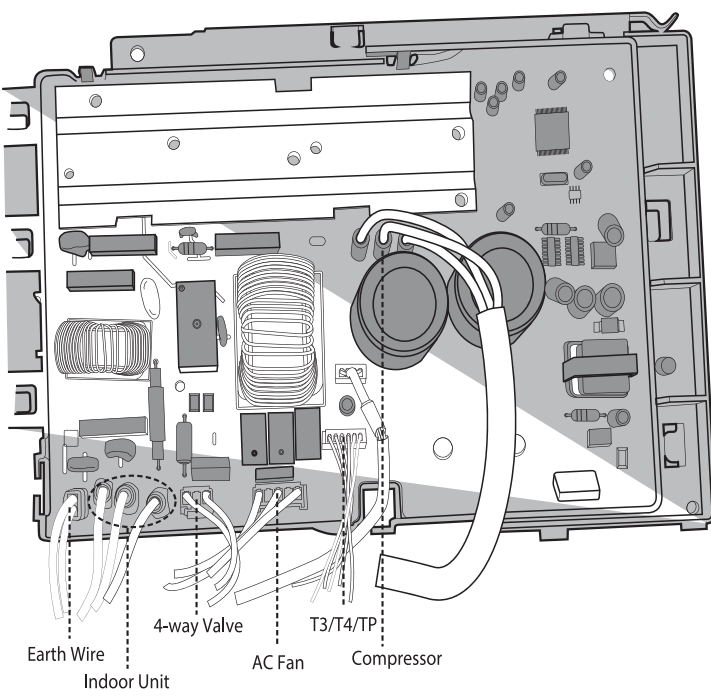
| PROCEDURES                                                                                | ILLUSTRATION                                                                                          |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| <p>8. Remove the screws of the right panel and then remove the right panel (8 screws)</p> |  <p>Right Panel</p> |

12K (115V) Disassembly - Electrical Parts

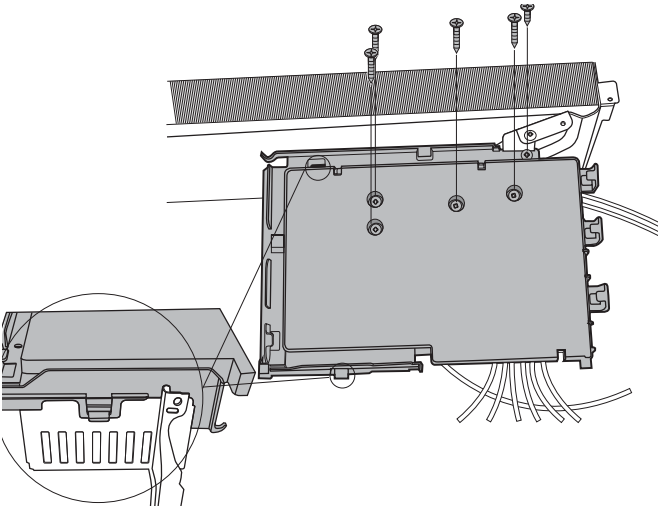
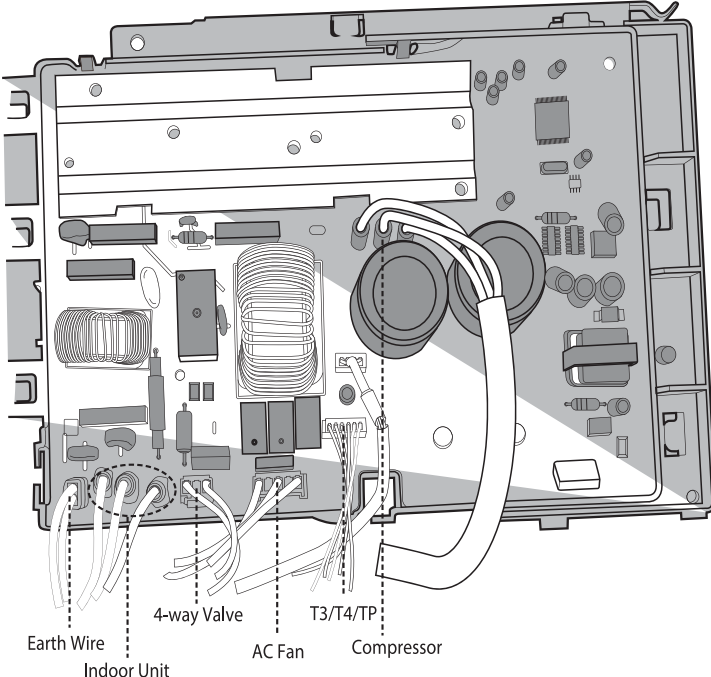
| PROCEDURES                                                                                                                                                                                                                                           | ILLUSTRATION                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p>1. Disconnect the connector for compressor and release the ground wire(1 screw).(.see illustration)</p>                                                                                                                                           |    |
| <p>2. Pull out the wires from electrical supporting plate and turn over the electronic control assembly. (see illustration)</p> <p><b>NOTE:</b> Electric control box cover cannot be removed, so the voltage between P and N cannot be measured.</p> |   |
| <p>3. Remove the electronic installing box subassembly (3 hooks)</p>                                                                                                                                                                                 |  |

| PROCEDURES                                                            | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>4. Disconnect the connectors from the electronic control board</p> |  A technical line drawing of an electronic control board. On the left side, there is a large, vertical, finned heat sink. The board is populated with various electronic components, including two large circular capacitors, several smaller capacitors, and integrated circuits. A bundle of approximately ten black wires is connected to a multi-pin connector on the top edge of the board. A white plastic connector is shown being pulled away from a multi-pin connector on the right side of the board. The board is mounted on a metal chassis, with some mounting points visible. |
| <p>5. Then remove the electronic control board (4 hooks).</p>         |  This illustration shows the same electronic control board as in the previous one, but with four specific removal points highlighted by red dashed circles. These points are located at the bottom of the board: two on the left side near the heat sink and two on the right side. The board is shown from the same perspective, with the heat sink on the left and various components on the main board area. The white plastic connector is still attached to the right side.                                                                                                            |

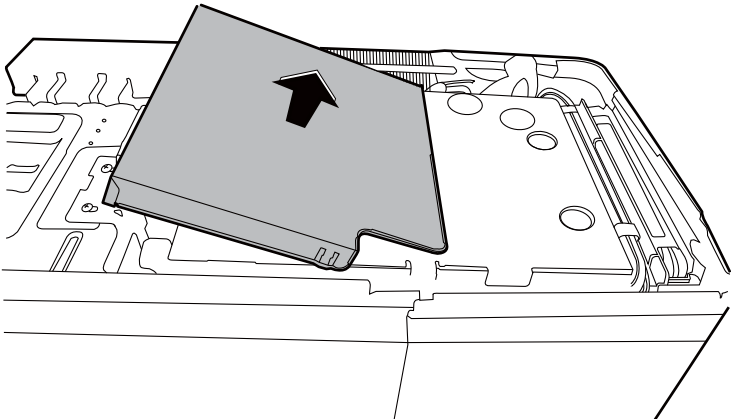
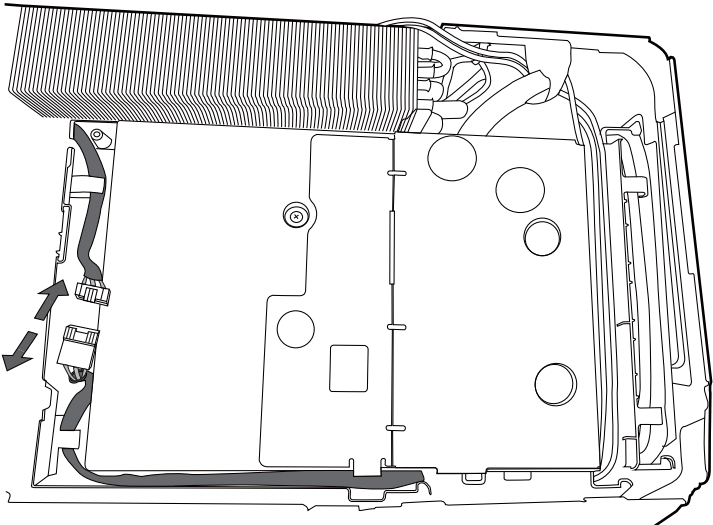
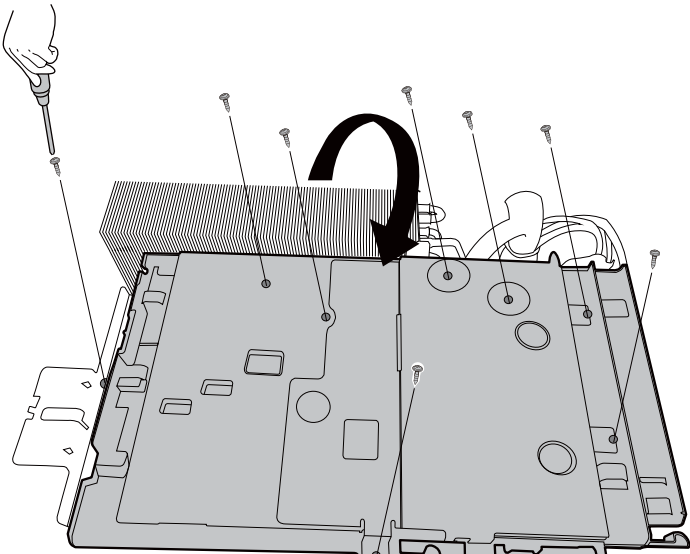
## 18K Disassembly - Electrical Parts

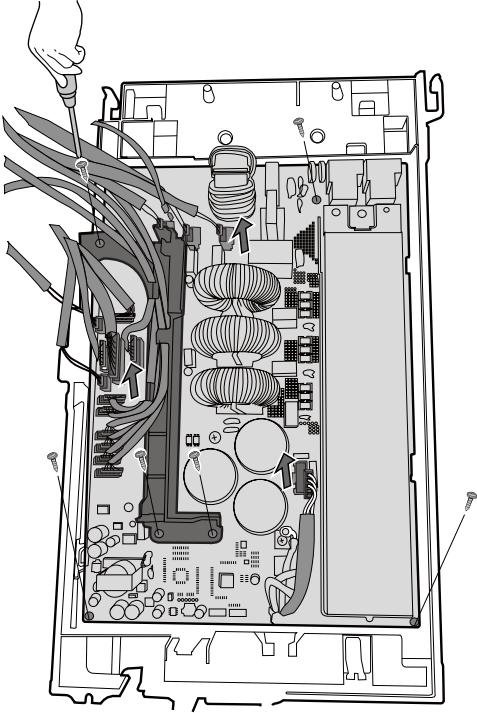
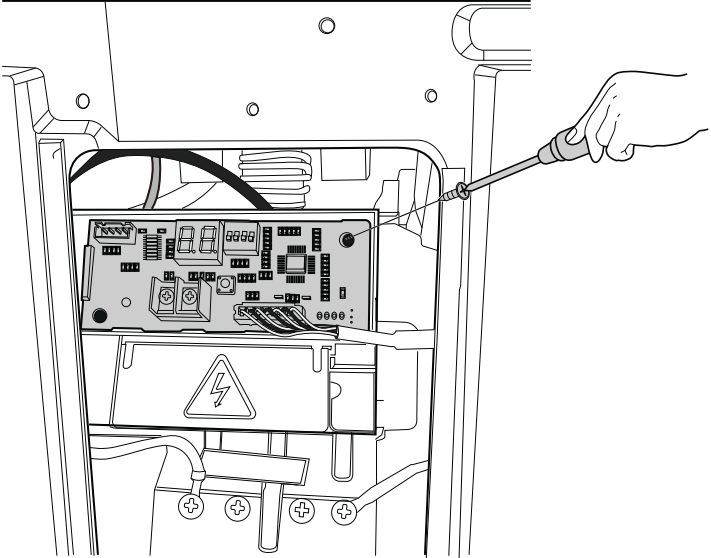
| PROCEDURES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Remove the 5 screws and unfix two hooks. (see illustration)</p> <p><b>NOTE:</b> Electric control box cover cannot be removed, so the voltage between P and N cannot be measured.</p> <p><b>For some models, there might be a wiring diagram covering the screw.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                          |  <p>This illustration shows the electric control box with its cover partially detached. Five screws are being removed from the top of the cover, and two hooks are being unfastened. A separate view shows the cover being lifted away from the main unit, revealing the internal wiring and components.</p>                                                                                                                                   |
| <p>2. Disconnect the connector for fan motor from the electronic control board (see illustration)</p> <p>3. Remove the connector for the compressor</p> <p>4. Pull out the two blue wires connected with the four way valve</p> <p>5. Pull out connectors of the condenser coil temp. sensor(T3), outdoor ambient temp. sensor(T4) and discharge temp. sensor (TP).</p> <p>6. Disconnect the electronic expansion valve wire.</p> <p>7. Remove the connector for the DR and reactor.</p> <p>8. Then remove the electronic control board.</p> <p><b>NOTE:</b> When replacing the electronic control board with a new one, pay attention to applying thermal paste on the heat sink.</p> |  <p>This detailed illustration shows the internal components of the electronic control board. Various wires are connected to different parts of the board. Labels with dashed lines point to specific components: 'Earth Wire' (grounding wire), 'Indoor Unit' (main power input), '4-way Valve' (blue wires), 'AC Fan' (fan motor connector), 'T3/T4/TP' (temperature sensor connectors), and 'Compressor' (compressor motor connector).</p> |

## 18K Disassembly - Electrical Parts

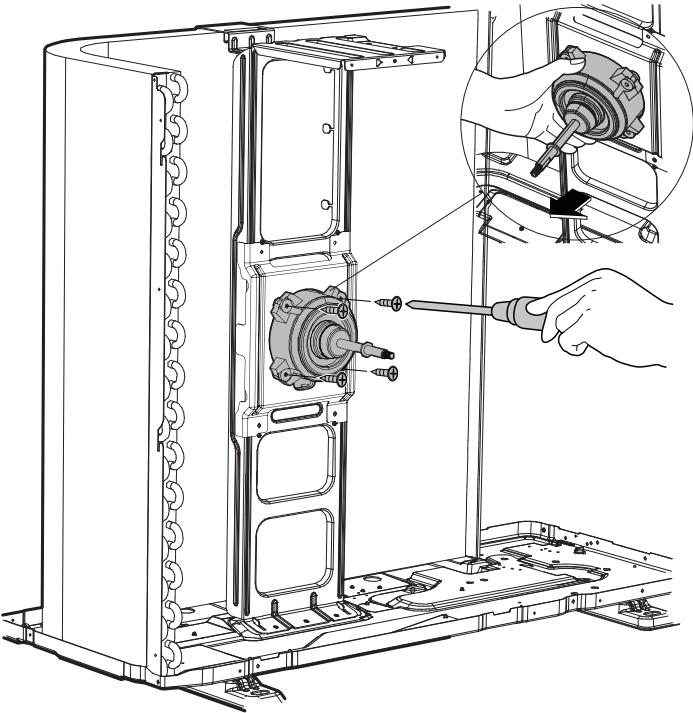
| PROCEDURES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Remove the 5 screws and unfix two hooks. (.see illustration)</p> <p><b>NOTE:</b> Electric control box cover cannot be removed, so the voltage between P and N cannot be measured.</p> <p><b>For some models, there might be a wiring diagram covering the screw.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                        |  <p>This illustration shows the electric control box with its cover partially detached. Five screws are being removed from the top of the cover, and two hooks are being unfastened. A separate view shows the cover being lifted away from the main unit.</p>                                                                                                 |
| <p>2. Disconnect the connector for fan motor from the electronic control board (see illustration)</p> <p>3. Remove the connector for the compressor</p> <p>4. Pull out the two blue wires connected with the four way valve</p> <p>5. Pull out connectors of the condenser coil temp. sensor(T3),outdoor ambient temp. sensor(T4) and discharge temp. sensor (TP).</p> <p>6. Disconnect the electronic expansion valve wire.</p> <p>7. Remove the connector for the DR and reactor.</p> <p>8. Then remove the electronic control board.</p> <p><b>NOTE:</b> When replacing the electronic control board with a new one, pay attention to applying thermal paste on the heat sink.</p> |  <p>This detailed illustration shows the internal components of the electronic control board. Various wires are connected to different parts of the board. Labels with dashed lines point to specific connections: 'Earth Wire' (grounding), 'Indoor Unit', '4-way Valve' (two blue wires), 'AC Fan', 'T3/T4/TP' (temperature sensors), and 'Compressor'.</p> |

24K - 33K, Disassembly -Electrical Parts

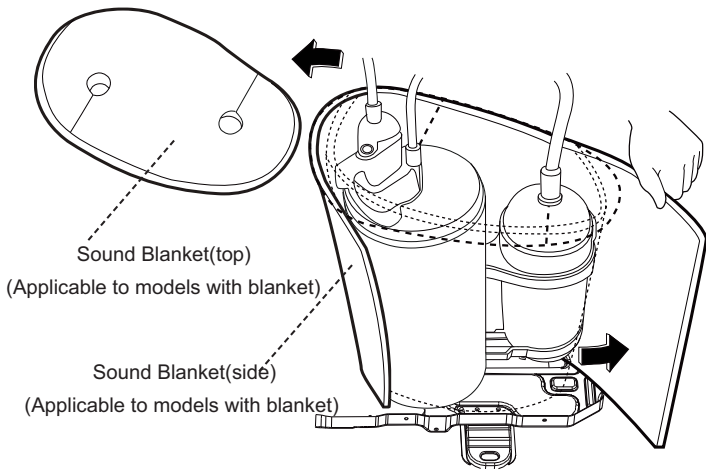
| PROCEDURES                                                                                                | ILLUSTRATION                                                                         |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1. Remove the cover of electrical control box. (see illustration)                                         |    |
| 2. Disconnect the fan motor connector.(see illustration)                                                  |   |
| 3. Remove eight fixing screws.<br>4. Turn over the electronic control box subassembly. (see illustration) |  |

| PROCEDURES                                                                                                                                                                                            | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>5. Remove 3 screws and then remove the bracket.</p> <p>6. Disconnect the connectors from the electronic control board.</p> <p>7. Remove 3 screws and then remove the electronic control board.</p> |  A detailed line drawing of the interior of an appliance chassis. An electronic control board is mounted vertically. A hand is shown using a screwdriver to remove a screw from the top of the board. Various components like capacitors, resistors, and connectors are visible on the board and surrounding area.    |
| <p>8. Pull out the connector, remove one screw and then remove the key board subassembly on terminal board.</p>                                                                                       |  A line drawing showing a hand using a screwdriver to remove a screw from a terminal board. A key board subassembly is being pulled out from the board. A warning symbol (a lightning bolt inside a triangle) is visible on the terminal board. Below the board, there are four circular terminals with plus signs. |

**All Size Units, Disassembly - Fan Motor**

| PROCEDURES                                                                                                        | ILLUSTRATION                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Remove the fixing screws of the fan motor (4 screws)</p> <p>2. Remove the fan motor. (see illustration)</p> |  A technical line drawing of a refrigerator's internal compartment with the door open. A fan motor is mounted on the back wall. A hand is shown using a screwdriver to remove one of four screws securing the motor. An inset circular image provides a magnified view of the screwdriver tip turning a screw on the motor's mounting bracket. |

**All Size Units, Disassembly - Sound Blanket**

| PROCEDURES                                                            | ILLUSTRATION                                                                                                                                                                                                            |
|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>1. Remove the sound blanket (side and top)) (see illustration)</p> |  <p>Sound Blanket(top)<br/>(Applicable to models with blanket)</p> <p>Sound Blanket(side)<br/>(Applicable to models with blanket)</p> |

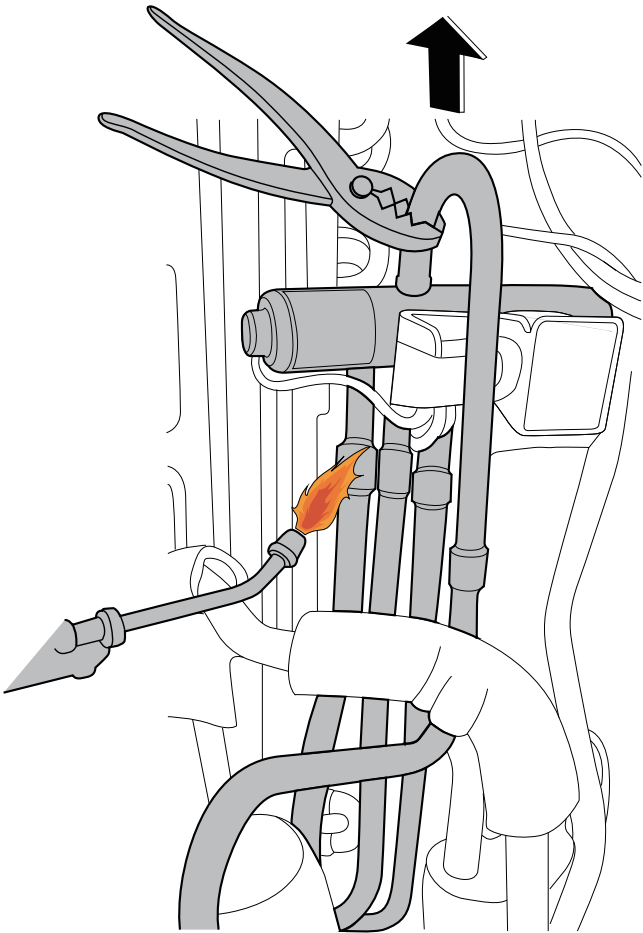
All Size Units, Disassembly - Four-Way Valve (For Heat Pump Models

! WARNING

**FIRE HAZARD**

Evacuate the system and confirm that there is no refrigerant left in the system before removing the four-way valve and the compressor. (For R32 & R290, you should evacuate the system with the vacuum pump; flush the system with nitrogen; then repeat the two steps before heating up the brazed parts. The operations above should be implemented by professionals.)

**NOTE:** Remove the panel plate, connection of four-way valve on PCB (refer to “6K - 12K Unit Disassembly - Panel Plate” on page 66, “18K Unit Disassembly - Panel Plate” on page 69, “24K - 33K Unit Disassembly - Panel Plate” on page 72, or “24K - 33K Unit Disassembly - Panel Plate” on page 72 and “12K (115V) Disassembly - Electrical Parts” on page 76) before disassembling sound blanket.

| PROCEDURES                                                                                                                                                                                                 | ILLUSTRATION                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <div><div>1. Heat up the brazed parts and then detach the the four-way valve and the piper. (see illustration)</div><div>2. Remove the four-way valve assembly with pliers. (see illustration)</div></div> |  |

All Size Units, Disassembly - Compressor

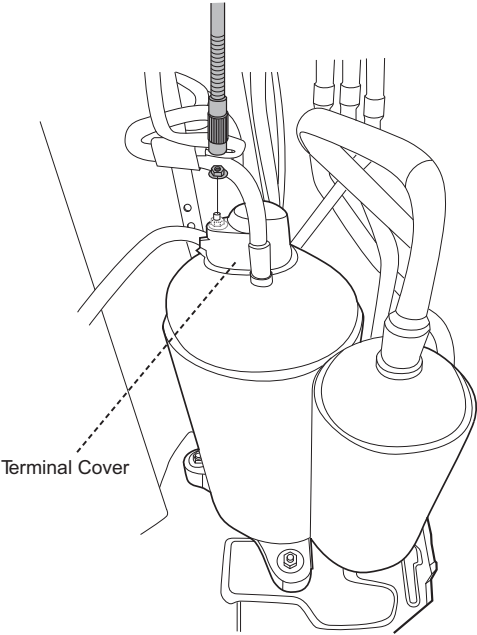
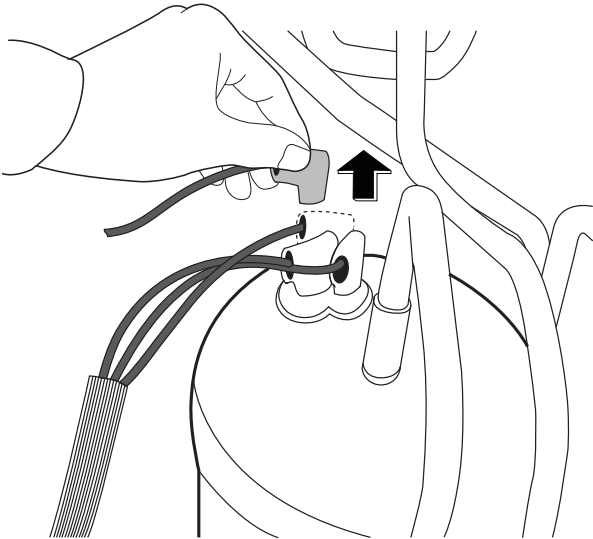


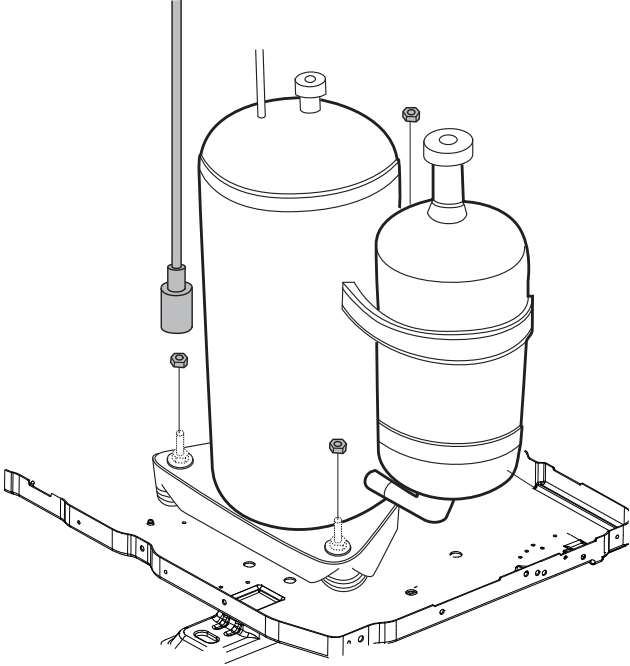
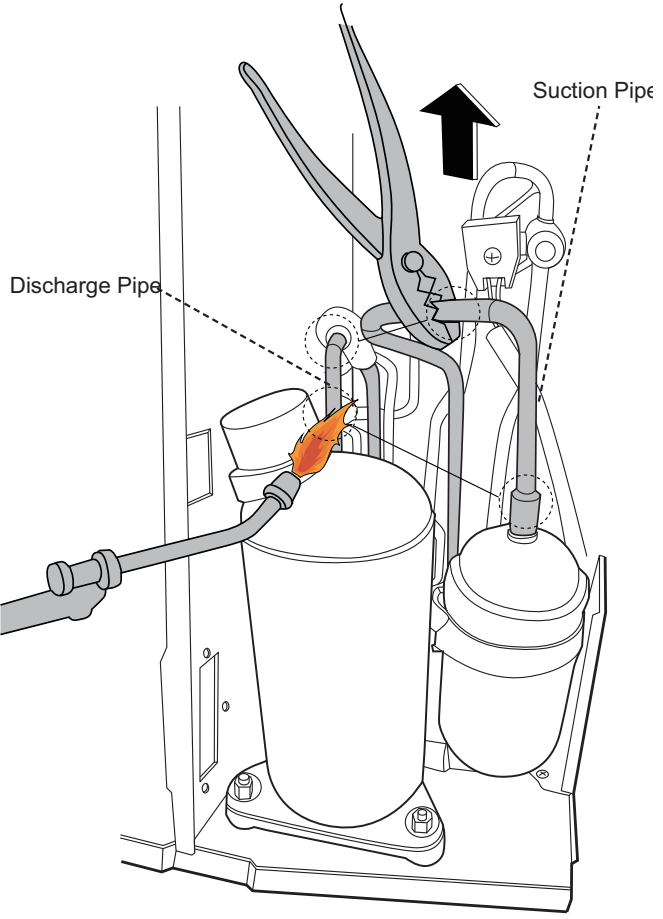
WARNING

EXPLOSION RISK

Evacuate the system and confirm that there is no refrigerant left in the system before removing the four-way valve and the compressor. (For R32 & R290, you should evacuate the system with the vacuum pump; flush the system with nitrogen; then repeat the two steps before heating up the brazed parts. The operations above should be implemented by professionals.)

**NOTE:** Remove the panel plate, connection of four-way valve on PCB (refer to “6K - 12K Unit Disassembly - Panel Plate” on page 66, “18K Unit Disassembly - Panel Plate” on page 69, “24K - 33K Unit Disassembly - Panel Plate” on page 72, or “24K - 33K Unit Disassembly - Panel Plate” on page 72 and “12K (115V) Disassembly - Electrical Parts” on page 76) before disassembling sound blanket.

| PROCEDURES                                                                                  | ILLUSTRATION                                                                         |
|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| 1. Remove the flange nut of terminal cover and remove the terminal cover (see illustration) |   |
| 2. Disconnect the connectors (see illustration)                                             |  |

| PROCEDURES                                                                                                                                                                                              | ILLUSTRATION                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <p>3. Remove the hex nuts and washers securing the compressor, located on the bottom plate. (see illustration)</p>                                                                                      |   |
| <p>4. Heat up the brazed parts and then remove the discharge pipe and the suction pipe. (see illustration)</p> <p>5. Lift the compressor from the base pan assembly with pliers. (see illustration)</p> |  |

**APPENDIX****Temperature Sensor Resistance Value Table for TP (°C - K)**

| °C  | °F | K Ohm | °C | °F  | K Ohm | °C | °F  | K Ohm | °C  | °F  | K Ohm |
|-----|----|-------|----|-----|-------|----|-----|-------|-----|-----|-------|
| -20 | -4 | 542.7 | 20 | 68  | 68.66 | 60 | 140 | 13.59 | 100 | 212 | 3.702 |
| -19 | -2 | 511.9 | 21 | 70  | 65.62 | 61 | 142 | 13.11 | 101 | 214 | 3.595 |
| -18 | 0  | 483   | 22 | 72  | 62.73 | 62 | 144 | 12.65 | 102 | 216 | 3.492 |
| -17 | 1  | 455.9 | 23 | 73  | 59.98 | 63 | 145 | 12.21 | 103 | 217 | 3.392 |
| -16 | 3  | 430.5 | 24 | 75  | 57.37 | 64 | 147 | 11.79 | 104 | 219 | 3.296 |
| -15 | 5  | 406.7 | 25 | 77  | 54.89 | 65 | 149 | 11.38 | 105 | 221 | 3.203 |
| -14 | 7  | 384.3 | 26 | 79  | 52.53 | 66 | 151 | 10.99 | 106 | 223 | 3.113 |
| -13 | 9  | 363.3 | 27 | 81  | 50.28 | 67 | 153 | 10.61 | 107 | 225 | 3.025 |
| -12 | 10 | 343.6 | 28 | 82  | 48.14 | 68 | 154 | 10.25 | 108 | 226 | 2.941 |
| -11 | 12 | 325.1 | 29 | 84  | 46.11 | 69 | 156 | 9.902 | 109 | 228 | 2.86  |
| -10 | 14 | 307.7 | 30 | 86  | 44.17 | 70 | 158 | 9.569 | 110 | 230 | 2.781 |
| -9  | 16 | 291.3 | 31 | 88  | 42.33 | 71 | 160 | 9.248 | 111 | 232 | 2.704 |
| -8  | 18 | 275.9 | 32 | 90  | 40.57 | 72 | 162 | 8.94  | 112 | 234 | 2.63  |
| -7  | 19 | 261.4 | 33 | 91  | 38.89 | 73 | 163 | 8.643 | 113 | 235 | 2.559 |
| -6  | 21 | 247.8 | 34 | 93  | 37.3  | 74 | 165 | 8.358 | 114 | 237 | 2.489 |
| -5  | 23 | 234.9 | 35 | 95  | 35.78 | 75 | 167 | 8.084 | 115 | 239 | 2.422 |
| -4  | 25 | 222.8 | 36 | 97  | 34.32 | 76 | 169 | 7.82  | 116 | 241 | 2.357 |
| -3  | 27 | 211.4 | 37 | 99  | 32.94 | 77 | 171 | 7.566 | 117 | 243 | 2.294 |
| -2  | 28 | 200.7 | 38 | 100 | 31.62 | 78 | 172 | 7.321 | 118 | 244 | 2.233 |
| -1  | 30 | 190.5 | 39 | 102 | 30.36 | 79 | 174 | 7.086 | 119 | 246 | 2.174 |
| 0   | 32 | 180.9 | 40 | 104 | 29.15 | 80 | 176 | 6.859 | 120 | 248 | 2.117 |
| 1   | 34 | 171.9 | 41 | 106 | 28    | 81 | 178 | 6.641 | 121 | 250 | 2.061 |
| 2   | 36 | 163.3 | 42 | 108 | 26.9  | 82 | 180 | 6.43  | 122 | 252 | 2.007 |
| 3   | 37 | 155.2 | 43 | 109 | 25.86 | 83 | 181 | 6.228 | 123 | 253 | 1.955 |
| 4   | 39 | 147.6 | 44 | 111 | 24.85 | 84 | 183 | 6.033 | 124 | 255 | 1.905 |
| 5   | 41 | 140.4 | 45 | 113 | 23.89 | 85 | 185 | 5.844 | 125 | 257 | 1.856 |
| 6   | 43 | 133.5 | 46 | 115 | 22.89 | 86 | 187 | 5.663 | 126 | 259 | 1.808 |
| 7   | 45 | 127.1 | 47 | 117 | 22.1  | 87 | 189 | 5.488 | 127 | 261 | 1.762 |
| 8   | 46 | 121   | 48 | 118 | 21.26 | 88 | 190 | 5.32  | 128 | 262 | 1.717 |
| 9   | 48 | 115.2 | 49 | 120 | 20.46 | 89 | 192 | 5.157 | 129 | 264 | 1.674 |
| 10  | 50 | 109.8 | 50 | 122 | 19.69 | 90 | 194 | 5     | 130 | 266 | 1.632 |
| 11  | 52 | 104.6 | 51 | 124 | 18.96 | 91 | 196 | 4.849 | ?   | ?   | ?     |
| 12  | 54 | 99.69 | 52 | 126 | 18.26 | 92 | 198 | 4.703 | ?   | ?   | ?     |
| 13  | 55 | 95.05 | 53 | 127 | 17.58 | 93 | 199 | 4.562 | ?   | ?   | ?     |
| 14  | 57 | 90.66 | 54 | 129 | 16.94 | 94 | 201 | 4.426 | ?   | ?   | ?     |
| 15  | 59 | 86.49 | 55 | 131 | 16.32 | 95 | 203 | 4.294 | ?   | ?   | ?     |
| 16  | 61 | 82.54 | 56 | 133 | 15.73 | 96 | 205 | 4.167 | ?   | ?   | ?     |
| 17  | 63 | 78.79 | 57 | 135 | 15.16 | 97 | 207 | 4.045 | ?   | ?   | ?     |
| 18  | 64 | 75.24 | 58 | 136 | 14.62 | 98 | 208 | 3.927 | ?   | ?   | ?     |
| 19  | 66 | 71.86 | 59 | 138 | 14.09 | 99 | 210 | 3.812 | ?   | ?   | ?     |

## Other Temperature Sensors Resistance Value Table (°C - K)

| °C  | °F | K Ohm   | °C | °F  | K Ohm   | °C | °F  | K Ohm   | °C  | °F  | K Ohm   |
|-----|----|---------|----|-----|---------|----|-----|---------|-----|-----|---------|
| -20 | -4 | 115.266 | 20 | 68  | 12.6431 | 60 | 140 | 2.35774 | 100 | 212 | 0.62973 |
| -19 | -2 | 108.146 | 21 | 70  | 12.0561 | 61 | 142 | 2.27249 | 101 | 214 | 0.61148 |
| -18 | 0  | 101.517 | 22 | 72  | 11.5    | 62 | 144 | 2.19073 | 102 | 216 | 0.59386 |
| -17 | 1  | 96.3423 | 23 | 73  | 10.9731 | 63 | 145 | 2.11241 | 103 | 217 | 0.57683 |
| -16 | 3  | 89.5865 | 24 | 75  | 10.4736 | 64 | 147 | 2.03732 | 104 | 219 | 0.56038 |
| -15 | 5  | 84.219  | 25 | 77  | 10      | 65 | 149 | 1.96532 | 105 | 221 | 0.54448 |
| -14 | 7  | 79.311  | 26 | 79  | 9.55074 | 66 | 151 | 1.89627 | 106 | 223 | 0.52912 |
| -13 | 9  | 74.536  | 27 | 81  | 9.12445 | 67 | 153 | 1.83003 | 107 | 225 | 0.51426 |
| -12 | 10 | 70.1698 | 28 | 82  | 8.71983 | 68 | 154 | 1.76647 | 108 | 226 | 0.49989 |
| -11 | 12 | 66.0898 | 29 | 84  | 8.33566 | 69 | 156 | 1.70547 | 109 | 228 | 0.486   |
| -10 | 14 | 62.2756 | 30 | 86  | 7.97078 | 70 | 158 | 1.64691 | 110 | 230 | 0.47256 |
| -9  | 16 | 58.7079 | 31 | 88  | 7.62411 | 71 | 160 | 1.59068 | 111 | 232 | 0.45957 |
| -8  | 18 | 56.3694 | 32 | 90  | 7.29464 | 72 | 162 | 1.53668 | 112 | 234 | 0.44699 |
| -7  | 19 | 52.2438 | 33 | 91  | 6.98142 | 73 | 163 | 1.48481 | 113 | 235 | 0.43482 |
| -6  | 21 | 49.3161 | 34 | 93  | 6.68355 | 74 | 165 | 1.43498 | 114 | 237 | 0.42304 |
| -5  | 23 | 46.5725 | 35 | 95  | 6.40021 | 75 | 167 | 1.38703 | 115 | 239 | 0.41164 |
| -4  | 25 | 44      | 36 | 97  | 6.13059 | 76 | 169 | 1.34105 | 116 | 241 | 0.4006  |
| -3  | 27 | 41.5878 | 37 | 99  | 5.87359 | 77 | 171 | 1.29078 | 117 | 243 | 0.38991 |
| -2  | 28 | 39.8239 | 38 | 100 | 5.62961 | 78 | 172 | 1.25423 | 118 | 244 | 0.37956 |
| -1  | 30 | 37.1988 | 39 | 102 | 5.39689 | 79 | 174 | 1.2133  | 119 | 246 | 0.36954 |
| 0   | 32 | 35.2024 | 40 | 104 | 5.17519 | 80 | 176 | 1.17393 | 120 | 248 | 0.35982 |
| 1   | 34 | 33.3269 | 41 | 106 | 4.96392 | 81 | 178 | 1.13604 | 121 | 250 | 0.35042 |
| 2   | 36 | 31.5635 | 42 | 108 | 4.76253 | 82 | 180 | 1.09958 | 122 | 252 | 0.3413  |
| 3   | 37 | 29.9058 | 43 | 109 | 4.5705  | 83 | 181 | 1.06448 | 123 | 253 | 0.33246 |
| 4   | 39 | 28.3459 | 44 | 111 | 4.38736 | 84 | 183 | 1.03069 | 124 | 255 | 0.3239  |
| 5   | 41 | 26.8778 | 45 | 113 | 4.21263 | 85 | 185 | 0.99815 | 125 | 257 | 0.31559 |
| 6   | 43 | 25.4954 | 46 | 115 | 4.04589 | 86 | 187 | 0.96681 | 126 | 259 | 0.30754 |
| 7   | 45 | 24.1932 | 47 | 117 | 3.88673 | 87 | 189 | 0.93662 | 127 | 261 | 0.29974 |
| 8   | 46 | 22.5662 | 48 | 118 | 3.73476 | 88 | 190 | 0.90753 | 128 | 262 | 0.29216 |
| 9   | 48 | 21.8094 | 49 | 120 | 3.58962 | 89 | 192 | 0.8795  | 129 | 264 | 0.28482 |
| 10  | 50 | 20.7184 | 50 | 122 | 3.45097 | 90 | 194 | 0.85248 | 130 | 266 | 0.2777  |
| 11  | 52 | 19.6891 | 51 | 124 | 3.31847 | 91 | 196 | 0.82643 | 131 | 268 | 0.27078 |
| 12  | 54 | 18.7177 | 52 | 126 | 3.19183 | 92 | 198 | 0.80132 | 132 | 270 | 0.26408 |
| 13  | 55 | 17.8005 | 53 | 127 | 3.07075 | 93 | 199 | 0.77709 | 133 | 271 | 0.25757 |
| 14  | 57 | 16.9341 | 54 | 129 | 2.95896 | 94 | 201 | 0.75373 | 134 | 273 | 0.25125 |
| 15  | 59 | 16.1156 | 55 | 131 | 2.84421 | 95 | 203 | 0.73119 | 135 | 275 | 0.24512 |
| 16  | 61 | 15.3418 | 56 | 133 | 2.73823 | 96 | 205 | 0.70944 | 136 | 277 | 0.23916 |
| 17  | 63 | 14.6181 | 57 | 135 | 2.63682 | 97 | 207 | 0.68844 | 137 | 279 | 0.23338 |
| 18  | 64 | 13.918  | 58 | 136 | 2.53973 | 98 | 208 | 0.66818 | 138 | 280 | 0.22776 |
| 19  | 66 | 13.2631 | 59 | 138 | 2.44677 | 99 | 210 | 0.64862 | 139 | 282 | 0.22231 |

**System Pressure Table-R454B**

| Pressure |      |       | Temperature |       | Pressure |       |        | Temperature |       |
|----------|------|-------|-------------|-------|----------|-------|--------|-------------|-------|
| Kpa      | bar  | PSI   | °C          | °F    | Kpa      | bar   | PSI    | °C          | °F    |
| 58.196   | 0.58 | 8.44  | -60         | -76   | 935.23   | 9.35  | 135.64 | 8           | 46.4  |
| 61.517   | 0.62 | 8.92  | -59         | -74.2 | 963.75   | 9.64  | 139.78 | 9           | 48.2  |
| 64.988   | 0.65 | 9.43  | -58         | -72.4 | 992.93   | 9.93  | 144.01 | 10          | 50    |
| 68.615   | 0.69 | 9.95  | -57         | -70.6 | 1022.8   | 10.23 | 148.34 | 11          | 51.8  |
| 72.402   | 0.72 | 10.50 | -56         | -68.8 | 1053.3   | 10.53 | 152.76 | 12          | 53.6  |
| 76.354   | 0.76 | 11.07 | -55         | -67   | 1084.5   | 10.85 | 157.29 | 13          | 55.4  |
| 80.478   | 0.80 | 11.67 | -54         | -65.2 | 1116.4   | 11.16 | 161.91 | 14          | 57.2  |
| 84.776   | 0.85 | 12.30 | -53         | -63.4 | 1149     | 11.49 | 166.64 | 15          | 59    |
| 89.256   | 0.89 | 12.95 | -52         | -61.6 | 1182.3   | 11.82 | 171.47 | 16          | 60.8  |
| 93.923   | 0.94 | 13.62 | -51         | -59.8 | 1216.3   | 12.16 | 176.40 | 17          | 62.6  |
| 98.781   | 0.99 | 14.33 | -50         | -58   | 1251.1   | 12.51 | 181.45 | 18          | 64.4  |
| 103.84   | 1.04 | 15.06 | -49         | -56.2 | 1286.6   | 12.87 | 186.60 | 19          | 66.2  |
| 109.1    | 1.09 | 15.82 | -48         | -54.4 | 1322.8   | 13.23 | 191.85 | 20          | 68    |
| 114.56   | 1.15 | 16.61 | -47         | -52.6 | 1359.9   | 13.60 | 197.23 | 21          | 69.8  |
| 120.25   | 1.20 | 17.44 | -46         | -50.8 | 1397.7   | 13.98 | 202.71 | 22          | 71.6  |
| 126.15   | 1.26 | 18.30 | -45         | -49   | 1436.3   | 14.36 | 208.31 | 23          | 73.4  |
| 132.28   | 1.32 | 19.18 | -44         | -47.2 | 1475.7   | 14.76 | 214.02 | 24          | 75.2  |
| 138.64   | 1.39 | 20.11 | -43         | -45.4 | 1515.9   | 15.16 | 219.85 | 25          | 77    |
| 145.24   | 1.45 | 21.06 | -42         | -43.6 | 1557     | 15.57 | 225.82 | 26          | 78.8  |
| 152.09   | 1.52 | 22.06 | -41         | -41.8 | 1598.9   | 15.99 | 231.89 | 27          | 80.6  |
| 159.18   | 1.59 | 23.09 | -40         | -40   | 1641.6   | 16.42 | 238.09 | 28          | 82.4  |
| 166.54   | 1.67 | 24.15 | -39         | -38.2 | 1685.2   | 16.85 | 244.41 | 29          | 84.2  |
| 174.15   | 1.74 | 25.26 | -38         | -36.4 | 1729.7   | 17.30 | 250.86 | 30          | 86    |
| 182.04   | 1.82 | 26.40 | -37         | -34.6 | 1775     | 17.75 | 257.43 | 31          | 87.8  |
| 190.2    | 1.90 | 27.59 | -36         | -32.8 | 1821.3   | 18.21 | 264.15 | 32          | 89.6  |
| 198.65   | 1.99 | 28.81 | -35         | -31   | 1868.4   | 18.68 | 270.98 | 33          | 91.4  |
| 207.39   | 2.07 | 30.08 | -34         | -29.2 | 1916.5   | 19.17 | 277.95 | 34          | 93.2  |
| 216.42   | 2.16 | 31.39 | -33         | -27.4 | 1965.6   | 19.66 | 285.08 | 35          | 95    |
| 225.76   | 2.26 | 32.74 | -32         | -25.6 | 2015.5   | 20.16 | 292.31 | 36          | 96.8  |
| 235.41   | 2.35 | 34.14 | -31         | -23.8 | 2066.5   | 20.67 | 299.71 | 37          | 98.6  |
| 245.37   | 2.45 | 35.59 | -30         | -22   | 2118.4   | 21.18 | 307.24 | 38          | 100.4 |
| 255.67   | 2.56 | 37.08 | -29         | -20.2 | 2171.3   | 21.71 | 314.91 | 39          | 102.2 |
| 266.29   | 2.66 | 38.62 | -28         | -18.4 | 2225.2   | 22.25 | 322.73 | 40          | 104   |
| 277.25   | 2.77 | 40.21 | -27         | -16.6 | 2280.2   | 22.80 | 330.70 | 41          | 105.8 |
| 288.56   | 2.89 | 41.85 | -26         | -14.8 | 2336.1   | 23.36 | 338.81 | 42          | 107.6 |
| 300.22   | 3.00 | 43.54 | -25         | -13   | 2393.2   | 23.93 | 347.09 | 43          | 109.4 |
| 312.24   | 3.12 | 45.28 | -24         | -11.2 | 2451.3   | 24.51 | 355.52 | 44          | 111.2 |
| 324.63   | 3.25 | 47.08 | -23         | -9.4  | 2510.4   | 25.10 | 364.09 | 45          | 113   |
| 337.39   | 3.37 | 48.93 | -22         | -7.6  | 2570.7   | 25.71 | 372.84 | 46          | 114.8 |
| 350.54   | 3.51 | 50.84 | -21         | -5.8  | 2632.1   | 26.32 | 381.74 | 47          | 116.6 |
| 364.08   | 3.64 | 52.80 | -20         | -4    | 2694.7   | 26.95 | 390.82 | 48          | 118.4 |
| 378.02   | 3.78 | 54.83 | -19         | -2.2  | 2758.3   | 27.58 | 400.04 | 49          | 120.2 |
| 392.37   | 3.92 | 56.91 | -18         | -0.4  | 2823.2   | 28.23 | 409.46 | 50          | 122   |
| 407.13   | 4.07 | 59.05 | -17         | 1.4   | 2889.3   | 28.89 | 419.04 | 51          | 123.8 |

**System Pressure Table-R454B (Continued)**

| Pressure |      |        | Temperature |      | Pressure |       |        | Temperature |       |
|----------|------|--------|-------------|------|----------|-------|--------|-------------|-------|
| Kpa      | bar  | PSI    | °C          | °F   | Kpa      | bar   | PSI    | °C          | °F    |
| 422.31   | 4.22 | 61.25  | -16         | 3.2  | 2956.5   | 29.57 | 428.79 | 52          | 125.6 |
| 437.92   | 4.38 | 63.51  | -15         | 5    | 3025     | 30.25 | 438.72 | 53          | 127.4 |
| 453.98   | 4.54 | 65.84  | -14         | 6.8  | 3094.7   | 30.95 | 448.83 | 54          | 129.2 |
| 470.47   | 4.70 | 68.23  | -13         | 8.6  | 3165.7   | 31.66 | 459.13 | 55          | 131   |
| 487.43   | 4.87 | 70.69  | -12         | 10.4 | 3238.1   | 32.38 | 469.63 | 56          | 132.8 |
| 504.84   | 5.05 | 73.22  | -11         | 12.2 | 3311.7   | 33.12 | 480.30 | 57          | 134.6 |
| 522.73   | 5.23 | 75.81  | -10         | 14   | 3386.7   | 33.87 | 491.18 | 58          | 136.4 |
| 541.1    | 5.41 | 78.48  | -9          | 15.8 | 3463     | 34.63 | 502.25 | 59          | 138.2 |
| 559.95   | 5.60 | 81.21  | -8          | 17.6 | 3540.7   | 35.41 | 513.52 | 60          | 140   |
| 579.31   | 5.79 | 84.02  | -7          | 19.4 | 3619.9   | 36.20 | 525.00 | 61          | 141.8 |
| 599.16   | 5.99 | 86.90  | -6          | 21.2 | 3700.5   | 37.01 | 536.69 | 62          | 143.6 |
| 619.54   | 6.20 | 89.85  | -5          | 23   | 3782.7   | 37.83 | 548.61 | 63          | 145.4 |
| 640.43   | 6.40 | 92.88  | -4          | 24.8 | 3866.3   | 38.66 | 560.74 | 64          | 147.2 |
| 661.86   | 6.62 | 95.99  | -3          | 26.6 | 3951.5   | 39.52 | 573.10 | 65          | 149   |
| 683.82   | 6.84 | 99.18  | -2          | 28.4 | 4038.3   | 40.38 | 585.69 | 66          | 150.8 |
| 706.34   | 7.06 | 102.44 | -1          | 30.2 | 4126.8   | 41.27 | 598.52 | 67          | 152.6 |
| 729.41   | 7.29 | 105.79 | 0           | 32   | 4217     | 42.17 | 611.60 | 68          | 154.4 |
| 753.06   | 7.53 | 109.22 | 1           | 33.8 | 4309     | 43.09 | 624.95 | 69          | 156.2 |
| 777.28   | 7.77 | 112.73 | 2           | 35.6 | 4402.9   | 44.03 | 638.56 | 70          | 158   |
| 802.08   | 8.02 | 116.33 | 3           | 37.4 | 4498.7   | 44.99 | 652.46 | 71          | 159.8 |
| 827.48   | 8.27 | 120.01 | 4           | 39.2 | 4596.5   | 45.97 | 666.64 | 72          | 161.6 |
| 853.49   | 8.53 | 123.78 | 5           | 41   | 4696.5   | 46.97 | 681.15 | 73          | 163.4 |
| 880.11   | 8.80 | 127.64 | 6           | 42.8 | 4798.9   | 47.99 | 696.00 | 74          | 165.2 |
| 907.35   | 9.07 | 131.60 | 7           | 44.6 | 4904.1   | 49.04 | 711.25 | 75          | 167   |